

A PROJECT REPORT ON

**FACE RECOGNITION BASED
ATTENDANCE SYSTEM**

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(Computer Engineering)**

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2023 - 2024**



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Abstract

The integration of face recognition technology into attendance management systems has emerged as a transformative solution for monitoring and optimizing attendance processes in educational institutions and organizations. This report presents an in-depth analysis of a project dedicated to the development and implementation of a Face Recognition Attendance System (FRAS) and examines its efficiency and security in various real-world scenarios. The report begins by discussing the significance of attendance tracking systems in educational and corporate environments, highlighting the shortcomings of traditional methods and the Potential benefits of face recognition technology.

It explores the underlying principles of face recognition, focusing on the algorithms and techniques used to capture and process facial data. The project methodology and implementation details are outlined, covering the hardware and software components used to create the FRAS. Special emphasis is placed on the challenges encountered during the system's development and the strategies employed to overcome them. A comprehensive evaluation of the system's performance and efficiency is presented, including accuracy, speed, and scalability, with comparisons to traditional attendance systems. Moreover, the report addresses security concerns associated with the FRAS, such as data protection, privacy, and vulnerability to spoofing

The report concludes with a discussion of the practical implications and future prospects of implementing face recognition technology in attendance systems. It highlights the system's potential to streamline administrative tasks, reduce errors, and enhance security, while also emphasizing the need for ongoing research and development to address emerging challenges. Overall, this report provides a valuable insight into the evolving landscape of attendance management and the role that face recognition technology can play in revolutionizing these processes, balancing efficiency with security considerations

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CHAPTER 1

SYNOPSIS

1.1 Project Title

Face Recognition Attendance System

1.2 Project Option

The Face Recognition Attendance System is an advanced technology-driven solution designed to automate and streamline the attendance tracking process. Unlike traditional methods involving manual sign-ins or card swipes, this system leverages the power of face recognition to identify and mark the attendance of individuals in a seamless and efficient manner.

1.3 Internal Guide

Prof B.M. Gawale

1.4 Technical Keywords (As per ACM Keywords)

Technical Key Words:

- Face Recognition
- Attendance System
- Computer Vision
- Image Processing
- Machine Learning
- Facial Recognition Algorithms
- Feature Extraction
- Recognition Accuracy
- Face Detection
- Liveness Detection
- Facial Landmarks

- Pattern Recognition
- Security

1.5 Problem Statement

Traditional methods of attendance tracking, such as manual sign-ins or card-based systems, are prone to inaccuracies, time inefficiencies, and the risk of unauthorized access. To address these challenges, there is a need for a modern and automated solution leveraging face recognition technology. The proposed Face Recognition Attendance System aims to streamline the attendance marking process, enhance accuracy, and provide a secure and efficient alternative to conventional methods.

1.6 Abstract

Traditional attendance management systems, often relying on manual roll calls or ID card swipes, are time-consuming, prone to errors, and susceptible to proxy attendance. This inefficiency necessitates exploring alternative solutions. Face recognition technology, with its advancements in accuracy and ease of use, presents a compelling approach for automated attendance systems.

This work explores the design and implementation of a face recognition attendance system (FRAS). The system utilizes a camera to capture images of individuals entering a designated area. A facial detection module locates faces within the captured image. Subsequently, a facial recognition module extracts unique features from the detected faces and compares them against a pre-enrolled database containing authorized individuals' facial data. This comparison process determines whether the individual is recognized and registered in the system.

Upon successful recognition, the FRAS automatically marks the individual's attendance in a digital record. The system offers several advantages over traditional methods. Firstly, it streamlines the attendance recording process, eliminating the need for manual intervention and saving valuable time. Secondly, it enhances the accuracy of attendance data by minimizing the possibility of human error and proxy attendance. Additionally, the contactless nature of facial recognition offers a hygienic attendance recording method, particularly relevant in the current health climate.

The core technical contributions of this work lie in the selection and optimization of facial recognition algorithms. The report delves into the chosen algorithms, including their strengths, limitations, and suitability for the FRAS application. The process of training and fine-tuning the algorithms on a labeled dataset containing facial images of enrolled individuals is also addressed.

Furthermore, the work explores the system's performance evaluation. It details the metrics employed to assess the accuracy, efficiency, and robustness of the FRAS. These metrics may include recognition rate, processing time, and performance under varying lighting conditions or facial occlusions (e.g., wearing glasses or masks). The report discusses the obtained results, analyzing the system's strengths and weaknesses, and identifying potential areas for improvement.

Finally, the work acknowledges the inherent privacy concerns associated with facial recognition technology. It explores strategies to mitigate these concerns, such as anonymizing facial data storage and ensuring user consent for data collection and usage. Additionally, the report discusses the ethical considerations surrounding the implementation of such systems, particularly regarding fairness and bias within the facial recognition algorithms.

This work presents the design, development, and evaluation of a face recognition attendance system. The FRAS offers a promising solution for automating attendance management, improving efficiency, accuracy, and hygiene. The report addresses the technical aspects of the system, its performance evaluation, and the crucial considerations of privacy and ethics surrounding facial recognition technology.

1.7 Goals and Objectives

1.7.1 Goals:

- Develop a system that automates the attendance tracking process.
- Improve the accuracy of attendance records.
- Create a solution that reduces the time spent on attendance tracking.
- Ensure the security and privacy of individuals.
- Design an intuitive and user-friendly graphical interface.

- Develop a system that can scale to meet the growing needs of organizations.
- Provide instant feedback through real-time face recognition.
- Maintain a secure and organized log of attendance records.

1.7.2 Objectives:

- Implement face recognition technology to identify individuals in real-time, eliminating the need for manual attendance marking.
- Utilize machine learning algorithms to train the face recognition model for reliable identification, reducing errors associated with manual methods.
- Implement real-time face recognition to expedite the attendance process, minimizing delays and disruptions.
- Implement robust security measures to prevent unauthorized access and adhere to privacy standards for biometric data.
- Utilize Tkinter to create a GUI that is accessible to both administrators and end-users, facilitating ease of use..
- Design the system architecture to accommodate a large number of users and integrate with existing databases for scalability..
- Implement algorithms that enable quick and seamless recognition, offering immediate attendance updates.
- Develop a database management system to store attendance data securely with timestamps and user information.

1.8 Relevant mathematics associated with the Project

1.8.1 System Description:

This system description outlines the design and implementation of a face recognition attendance system using Haar Cascade and LBPH algorithms. This system combines the efficiency of Haar Cascade for face detection and the robustness of LBPH for face recognition to achieve accurate and reliable

attendance tracking. The system comprises the following hardware and software components:

Hardware:

- **High-resolution camera:** Captures high-quality images of individuals' faces for accurate recognition.
- **Computer with sufficient processing power:** Handles real-time image processing and facial recognition tasks.

Software:

- **Haar Cascade classifier:** Detects faces in captured images using the Haar Cascade algorithm.
- **Local Binary Pattern Histogram (LBPH) algorithm:** Extracts facial features and performs face recognition based on the LBPH algorithm.
- **Attendance management system:** Records attendance data, generates reports, and manages employee information.
- **Enrollment module:** Registers new employees and captures their facial images for enrollment in the system database.

The system operates as follows:

- **Face Detection:** The camera captures a real-time image of the individual. The Haar Cascade classifier is applied to the captured image to detect faces. Faces are marked with bounding boxes to indicate their location in the image.
- **Face Recognition:** For each detected face, the LBPH algorithm extracts facial features from the cropped image. The extracted facial features are compared to the database of enrolled faces using the LBPH algorithm. If a match is found, the individual is identified, and their attendance is recorded. If no match is found, the individual is prompted to enter their ID or contact an administrator for manual enrollment or verification.
- **Attendance Management:** The attendance management system records the attendance data for each identified individual. Attendance reports are generated based on the recorded attendance data. The system maintains an employee database with facial images and attendance records.

1.8.2 Mathematical Model

The mathematical model represents the relationships and dependencies within the system. It uses set notation to define the elements and their interactions.

$S = \{I, O, P, SD, P, T, A, F, HD, D, E, H/w, S/w, Failure, Success\}$
Where

S=System
SD= Student Details
P=Photo Sample
Ad=Admin
T=Train Data
A=Attendance
F=Face Detector
HD= Help Desk
D= Developer
E=Exit

I is Input of system Input $I = \{Input1, Input2, Input3\}$
Where,
Input1=Student Information
Input2=Images of Students
Input3= User Commands
Input4= Time and Date Information
Input5= Hardware Inputs

Procedures $P = \{STD, TP, TD, FD, AD\}$
Where,
STD= Register a Student Details
TP= Take a Photo Sample
TD= Train a Photo using Haar Cascade and LBPH algorithms
FD= Open a Face Detector mark a Attendance
AD= See The Successful Mark Attendance

O is Output of system Output $O = \{Output1, Output2, Output3, Output4, Output5, Output6\}$
Where,
Output1= Student Details successfully Saved
Output2= Successfully Collect Photos Data
Output3= Successfully Trained Photos Data

Output4= Successfully Marked Attendance

Output5= Visual Feedback on the GUI to indicate the recognition status of individual students a green checkmark for attendance success, a red for failures

Output6= The ability to export attendance data and reports to Excel formats

NDD is Non Deterministic Data.

NDD= Facial expressions ,Varying Lighting Conditions, Aging and Appearance, Changes Camera Quality , Angle Lighting conditions, Image quality

DD is Deterministic Data

DD=All the elements of I and O

Core=4 core

H/w is Hardware requirement.

H/W= Laptop,smart phone

S/w is Software requirement.

S/W= VS Code, Python, Tkinter, Database: Mysql

Failure= Student information details are not saved it show a error message and Visual Feedback on the GUI to indicate the recognition status of individual students a red for failures

Success= Student information details are saved it show a success message and Visual Feedback on the GUI to indicate the recognition status of individual students a green for success

1.9 Names of Conferences / Journals where papers can be published

- IEEE/ACM Conference/Journal 1
- Conferences/workshops in IITs
- Central Universities or SPPU Conferences
- IEEE/ACM Conference/Journal 2

1.10 Review of Conference/Journal Papers supporting Project idea

- **R. Hartanto and M. N. Adji, "Face Recognition for Attendance System Detection," 10th International Conference on Information Technology and Electrical Engineering (ICITEE)**

M. N. Adji And R. Hartanto proposed an attendance system by face recognition using the Haar cascade algorithm. Face images are captured using a camera. To determine the location of the face it is performed by a haar algorithm for image detection. It basically draws a box around the faces of the person to whom we want to detect his face. Using different types of the algorithms they have done it. After they have stored the data in the cloud to determine the location of the face. Using Haar Cascade algorithm face detection is done. When the image of the face had been made it would now make or draw a box which is covering the whole face as an Region Of Interest (ROI). Extraction of image features was performed using the LBPH algorithm. Compare the detected faces with all the faces in the database to observe the one closest to the identified face. We have used CSV file format for storing the database. It is used to present the names and directories in the database format.

- **T. A. Kiran, N. D. K. Reddy, A. I. Ninan, P. Krishnan, D. J. Aravindhar and A. Geetha, "PCA based Facial Recognition for Attendance System," 2020 International Conference on Smart Electronics and Communication**

In a paper of authors have suggested a facial recognition for the attendance system which is based on principal component analysis. A single image of each person's face is clicked. After that image is transferred to a grayscale image and stored in the database. After entering class, the camera is placed over the whole face and it will change the face into gray scale and after that it is submitted to the next part. For extracting features we are using fisher face method. Face recognition is accomplished by comparing all received variables to previously recorded values in the database. If the quality is compatible, the face is recognized and the name connected to that face is presented. The face data.xml file is filled with information when the face is recognized.

- **A Real-Time Face Detection Method Based on Blink Detection :**

Face anti-spoofing refers to the computer determining whether the face

detected is a real face or a forged face. In user authentication scenarios, photo fraud attacks are easy to occur, where an illegal user logs into the system using a legitimate user's picture. The method first extracts the image texture features through the LBP algorithm, which eliminates the problem of illumination changes to a certain extent. Then the extracted features are input into the ResNet network, and the facial feature extraction is enhanced by adding an attention mechanism is added to enhance the face feature extraction. Meanwhile, the BiLSTM method is used to extract the temporal characteristics of images from different angles or at different times to obtain more facial details. In addition, the fusion of local and global features is realized by SPP pooling, which enriches the expression ability of feature maps and improves detection accuracy. Finally, the eye EAR value is calculated by the face key point detection technology to achieve face anti-spoofing, and then the real-time face recognition against fraud is realized. The experimental results show that the algorithm proposed in this paper has good accuracy on NUAA, CASIASURF and CASIA-FASD datasets, which can reach 99.48, 98.65 and 99.17, respectively

- **Surveillance System for Real-Time High-Precision Recognition of Criminal Faces From Wild Videos :**

As violent criminals, such as child sex offenders, tend to have high recidivism rates in modern society, there is a need to prevent such offenders from approaching socially disadvantaged and crimeprone areas, such as schools or childcare centers. Accordingly, national governments and related institutions have installed surveillance cameras and provided additional personnel to manage and monitor them via video surveillance equipment. However, naked-eye monitoring by guards and manual image processing cannot properly evaluate the video captured by surveillance cameras. To address the various problems of conventional systems that simply store and retrieve image data, a system is needed that can actively classify captured images in real-time, in addition to assisting surveillance personnel. The proposed system detects the faces of criminals in real time from videos captured by a surveillance camera and notifies relevant institutions of the appearance of criminals. The reliability of the identification results is complemented by an identification score accumulation method. In this paper, we prototyped the proposed system and experimented with the recognition model, achieving an accuracy of 0.900 and an F-1 score of 0.943. We also experimentally confirmed that the models proposed in other studies have higher performance when using the tracked instance-level face identification

method proposed in this paper. It is expected that the proposed system can be used to locate criminals and protect national facilities, and such responses can quickly prevent accidents/incidents. The dataset and code are available at <https://github.com/aengoo/focusface>

- **When Age-Invariant Face Recognition Meets Face Age Synthesis: A Multi-Task Learning Framework and a New Benchmark**

To minimize the impact of age variation on face recognition, age-invariant face recognition (AIFR) extracts identity-related discriminative features by minimizing the correlation between identity- and age-related features while face age synthesis (FAS) eliminates age variation by converting the faces in different age groups to the same group. However, AIFR lacks visual results for model interpretation and FAS compromises downstream recognition due to artifacts. Therefore, we propose a unified, multi-task framework to jointly handle these two tasks, termed MTLFace, which can learn the age-invariant identity-related representation for face recognition while achieving pleasing face synthesis for model interpretation. Specifically, we propose an attention-based feature decomposition to decompose the mixed face features into two uncorrelated components—identity- and age-related features—in a spatially constrained way. Unlike the conventional one-hot encoding that achieves group-level FAS, we propose a novel identity conditional module to achieve identity-level FAS, which can improve the age smoothness of synthesized faces through a weight-sharing strategy. Benefiting from the proposed multi-task framework, we then leverage those high-quality synthesized faces from FAS to further boost AIFR via a novel selective fine-tuning strategy. Furthermore, to advance both AIFR and FAS, we collect and release a large cross-age face dataset with age and gender annotations, and a new benchmark specifically designed for tracing long-missing children. Extensive experimental results on five benchmark cross-age datasets demonstrate that MTLFace yields superior performance than state-of-the-art methods for both AIFR and FAS. We further validate MTLFace on two popular general face recognition datasets, obtaining competitive performance on face recognition in the wild. The source code and datasets are available at <http://hzzone.github.io/MTLFace>.

- **Invisible Adversarial Attacks on Deep Learning-Based Face Recognition Models**

Deep learning technology has grown rapidly in recent years and achieved

tremendous success in the field of computer vision. At present, many deep learning technologies have been applied in daily life, such as face recognition systems. However, as human life increasingly relies on deep neural networks, the potential harms of neural networks are being revealed, particularly in terms of deep neural network security. More and more studies have shown that existing deep learning-based face recognition models are vulnerable to attacks by adversarial samples, resulting in misjudgments that could have serious consequences. However, existing adversarial face images are rather easy to identify with the naked eye, so it is difficult for attackers to carry out attacks on face recognition systems in practice. This paper proposes a method for generating adversarial face images that are indistinguishable from the source images based on facial landmark detection and superpixel segmentation. First, the eyebrows, eyes, nose, and mouth regions are extracted from the face image using a facial landmark detection algorithm. Next, the superpixel segmentation algorithm is used to include the pixels neighboring the extracted facial landmarks with similar pixel values. Lastly, the segmented regions are used as masks to guide existing attack methods to insert adversarial noise within the masked areas. Experimental results show that our method can generate adversarial samples with high Structural Similarity Index Measure (SSIM) values at the cost of a small percentage of attack success rate. In addition, to simulate real-time physical attacks, printouts of the adversarial images generated by the proposed method are presented to the face recognition system via a camera and are still able to fool the face recognition model. Experimental results indicated that the proposed method can successfully perform adversarial attacks on face recognition systems in real-world scenarios.

- **Smart Security System For Door Access Based On Unique Authentication :**

Security is a prime aspect of concern in order to maintain confidentiality of our home, work places and to avoid intrusion of unauthorized persons. In voice password and biometric based authentication door locking system, authentication using the unique identification like biometric and voice recognition plays a vital role to provide high level of security. The finger ridges of individual do not match with any other finger ridges and an individual's voice cannot be impersonated with accuracy. This paper proposes a smart voice password and biometric based security system for door locking in smart homes. To enhance the security, in place of conventional door locking system, a finger print

sensor along with a micro phone are used to authenticate, i.e. to lock and unlock the doors [1]. The data base will maintain the data of the persons who tried to access the door. The entire system is controlled by the Raspberry pi 3 B+ processor. PIWHO software is used for the purpose of voice recognition. Door access may be provided to the registered users based on his voice pass word and thumb impression. Door will be opened only when both the factors are satisfied, otherwise buzzer will be activated and the authorized person will receive an SMS alert message

- **”Face Recognition Based Smart Attendance System,” 2020 International Conference on Intelligent Engineering and Management (ICIEM)**

They have proposed a system for face recognition using the LBPH algorithm. First, a database is created by enrolling students. They are taking several images of students using cameras. Then the images are converted into grayscale by using the LBP algorithm and after that a histogram is generated. A video stream is given to the system by which the LBPH algorithm matches the live image which is taken by using camera streams along the images present in the database. When a face is detected successfully, attendance is marked

1.11 Plan of Project Execution

[illegible]

Figure 1.1: Plan Timeline

CHAPTER 2

TECHNICAL KEYWORDS

2.1 Area of Project

The area of a project involving a face recognition attendance system typically falls within the broader domain of computer vision and image processing

2.2 Technical Keywords

Technical Key Words:

- **Face Detection:** The ability to locate and identify faces in images and videos.
- **Feature Extraction:** Extracting unique facial features, such as the distance between eyes, nose, and mouth, to create a facial template.
- **Face Recognition:** Comparing extracted facial features to a database of known faces to identify individuals.
- **Machine Learning:** Utilizing machine learning algorithms to train facial recognition models and improve their accuracy.
- **Attendance Management:** Integrating face recognition technology with attendance management systems to automate attendance tracking.
- **Facial Recognition Algorithms**
- **Recognition Accuracy**
- **Liveness Detection**
- **Facial Landmarks**
- **Pattern Recognition**
- **Security**

CHAPTER 3

INTRODUCTION

3.1 Need

The implementation of a face recognition attendance system addresses several needs and challenges in traditional attendance tracking methods. Here are some key reasons why a face recognition attendance system may be needed:

The need for such a system arises from:

- **Accuracy:** Face recognition systems offer a high level of accuracy in identifying individuals. Traditional methods such as manual attendance tracking or barcode scanning may be prone to errors, either due to intentional manipulation or unintentional mistakes.
- **Efficiency:** Automated face recognition significantly improves the efficiency of the attendance tracking process. It eliminates the need for manual data entry and reduces the time spent on attendance-related administrative tasks.
- **Contactless Technology:** Especially relevant in situations where hygiene and health are priorities, face recognition systems are contactless. Users can mark their attendance without physical contact with any device, reducing the risk of the spread of germs or viruses.
- **Security:** Face recognition enhances security by ensuring that the individual marking attendance is the authorized person. This is particularly important in environments where attendance is linked to access control or sensitive areas within an organization.
- **Scalability:** With the growth of organizations, the traditional methods of attendance tracking may become impractical. Face recognition systems are scalable and can easily handle a large number of users, making them suitable for organizations of various sizes.
- **Reduction of Fraud:** Face recognition reduces the possibility of proxy attendance, where one person marks attendance on behalf of another. The unique facial features of each individual make it difficult for unauthorized individuals to manipulate the system.

3.2 Motivation of the Project

There are several reasons might be motivated to create a face recognition attendance system:

- **Improved Accuracy and Reduced Errors:** Compared to traditional methods like sign-in sheets or fingerprint scanners, facial recognition can offer a more accurate way to track attendance. Fingerprint scanners can be fooled by wounds or dirt, and sign-in sheets are vulnerable to buddy punching someone signing in for another person.
- **Increased Efficiency and Automation:** Facial recognition systems can automate the attendance tracking process, saving time and resources for both employees and administrators. This can free up HR staff to focus on other tasks.
- **Enhanced Security:** Facial recognition attendance systems can also be used to improve security by ensuring that only authorized personnel are entering a building. This can be especially useful in restricted areas.
- **Convenience for Employees:** Facial recognition systems can be faster and more convenient for employees to use than traditional methods. Employees don't need to remember a badge or punch a specific key.
- **Potential Cost Savings:** While there may be upfront costs associated with setting up the system, facial recognition can potentially lead to cost savings in the long run by reducing administrative time and payroll errors.

3.3 Basic Concepts

- **Face Detection:**
This involves identifying faces in images or videos. Face detection algorithms typically use machine learning techniques to locate and mark facial regions in captured data.
- **Feature Extraction:**
Once a face is detected, the system extracts unique facial features that serve as identifiers. Common feature extraction methods include Local Binary Patterns (LBP).
- **Face Recognition:**
The extracted features are compared to a database of known faces to determine a match. Face recognition algorithms

- **Haar Cascade Algorithm:**

An object detection algorithm used for identifying objects or features in images or video. In face recognition, the Haar Cascade algorithm is often used for face detection, helping to locate faces within images or video frames.

- **LBPH Algorithm:**

An algorithm for facial recognition that considers the local patterns of pixel intensities in an image. LBPH is commonly used for feature extraction in face recognition systems. It is robust against variations in lighting conditions and facial expressions.

- **Attendance Recording:**

Once an individual is identified, their attendance is recorded in the system's database. This information can be used to generate attendance reports and track employee attendance patterns.

CHAPTER 4

LITERATURE SURVEY

4.1 Comparative Study

Table 4.1: Literature Survey

Paper	Title	Hardware	Algorithm	Application	Advantage And Disadvantage
Paper [1]	Face Recognition Based Attendance System	Camera	Haarcascade Object Detection, Local Binary Pattern	To Recognize The Face Of a Person	a] they are very accurate and difficult to cheat b] expensive to implement
Paper [2]	Surveillance System For Real-Time High-Precision Recognition Of Criminal Faces From Wild Videos	Camera	a] Haar Feature-Based Cascade Classifiers b] Local Binary Patterns Histograms (LBPH)	a] Facial Recognition b] Robotics	a] Prediction Results For Facial Object Ids Are Never Reversed. b] Calculation Speed Is Faster
Paper [3]	When Age-Invariant Face Recognition Meets Face Age Synthesis: A Multi-Task Learning Framework And a New Benchmark	Camera	MTLFace, Face Age Synthesis (FAS), age-invariant face recognition (AIFR)	Continuous Face Age Synthesis, Comparison With Ground-Truth Faces	Improve The Age Accuracy Due To The Multi-Level Age Conditions.
Paper [4]	Real-time eye state detection system using haar cascade classifier and circular Hough transform	Camera Image Input Device ,CPU, GPU	a] Haar Cascade Classifier b] Circular Hough Transform (CHT)	Eye state detection, Drowsiness detection	Accurate, Real-time

Paper	Title	Hardware	Algorithm	Application	Advantage And Disadvantage
Paper [5]	Invisible Adversarial Attacks On Deep Learning-Based Face Recognition Models	Camera	FaceNet , IR152, IRSE50, and ArcFace face recognition models	Mask Generation Method Based On Facial Landmark Detection And Superpixel Segmentation	Improving Existing Adversarial Attack Methods On Face Recognition Models, Both Digital And Physical.
Paper [6]	A real-time face recognition system based on the improved LBPH algorithm	Computer CPU GPUs	LBPH	Face recognition	Tolerant Of Gaps In Feature Boundary Descriptions And Is Relatively Unaffected By Image Noise.
Paper [7]	Face Recognition for Attendance System Detection	Camera ,	a] Haar cascade classifier Local [b] Binary Pattern Histogram (LBPH) algorithm	Attendance system	a] Simple to implement, efficient, robust to illumination changes, expression variations, and attitude deflections b] Sensitive to occlusion and other factors
Paper [8]	Class Attendance Management System Using Face Recognition	Camera Raspberry Pi RaspberrPi Camera 5-inch screen	HaarCascade Classifier Local Binary Patterns (LBP)	Class attendance system	Accurate Efficient Robust to illumination changes, expression variations, and attitude deflections .

Paper	Title	Hardware	Algorithm	Application	Advantage And Disadvantage
Paper [9]	Face Recognition based Automated Attendance System in - built with “Two -Tier Authentication” Method	2 cameras Computer with GPU	Faster R-CNN Two -tier authentication	Automatic attendance system	Accurate Efficient Two -tier authentication for increased security.
Paper [10]	Face Recognition based Class Management and Attendance System	Webcam CPU Memory Storage	HaarCascade LBPH	Class attendance	Promotes contactless attendance taking

CHAPTER 5

PROBLEM DEFINITION AND SCOPE

5.1 Problem Statement

In traditional attendance systems, manual methods such as paper registers or card swiping are commonly employed, leading to inefficiencies, inaccuracies, and time-consuming processes. To address these challenges, there is a growing demand for automated attendance systems that leverage face recognition technology. The objective is to create a Face Recognition Attendance System that enhances accuracy, efficiency, and security in recording and managing attendance data.

5.1.1 Goals and objectives

5.1.2 Goals:

1. Improve the efficiency and accuracy of attendance tracking in educational settings.
2. Reduce the time and effort spent on manual attendance marking.
3. Provide a more reliable and secure method of attendance verification

5.1.3 Objectives:

- Implement a secure and compliant face recognition attendance system.
- Develop and deploy face recognition algorithms that are accurate and reliable.
- Establish clear policies and procedures for the use of the system.
- Educate students and staff about the system and its benefits.
- Monitor and evaluate the system's effectiveness and make adjustments as needed.
- Overall goals and objectives of software, input and output description with necessary syntax, format etc are described

5.1.4 Statement of scope

- **System Description**

This document outlines the scope of a face recognition attendance system designed to automate the process of recording employee or student attendance.

- **Inputs**

Size of Input: Image data captured by a high-definition camera. The size will vary depending on the camera resolution but will typically be several 2-3 megapixels.

Bounds on Input: The system expects a well-lit image containing one or more human faces within the frame.

Input Validation: The system will validate the input image for: Presence of a face (using facial detection algorithms) Image quality (e.g., clarity, lighting)

Input Dependency: The system relies on a pre-populated database of user facial images for comparison.

Major Inputs: User enrollment: New user facial images and corresponding identification information (e.g., employee ID, name). Daily attendance capture: Live camera feed capturing faces for recognition.

- **Outputs**

Attendance records: Timestamped log of identified users marked as present.

User identification: Name or ID associated with the recognized face.

System notifications: Alerts for errors (e.g., unrecognizable face, poor image quality).

Optional: Reports summarizing attendance data for a specified period.

- **Inclusions**

Facial recognition algorithms for accurate user identification.

Secure user database management for storing facial images and associated information.

User interface for system administration (e.g., user enrollment, data export).

Error handling and notification mechanisms.

- **Exclusions**

Integration with existing access control systems (separate development required).

Liveness detection (verifying a real person is present, not a photograph).

Advanced features like emotion recognition or activity tracking.

- **Future Considerations**

Integration with access control systems for seamless entry/exit management.

Liveness detection for enhanced security. Mobile app integration for user self-service attendance recording.

5.2 Major Constraints

The major constraints include:

- **Privacy Concerns** The collection and storage of facial data raise concerns about data breaches, unauthorized access, and potential misuse
- **Accuracy and Bias** Face recognition technology can be inaccurate due to factors like lighting, facial expressions, and accessories, and may exhibit bias towards certain racial or ethnic groups..
- **Technical Challenges** Implementation requires technical expertise, infrastructure, and integration with existing attendance systems.
- **Cost:** Hardware, software, installation, and maintenance costs can be significantly higher than traditional methods
- **Data Integrity and Security:** Facial data must be securely stored and transmitted to prevent unauthorized access or misuse.

5.3 Methodologies of Problem solving and efficiency issues

The single problem can be solved by different solutions. This considers the performance parameters for each approach. Thus considers the efficiency issues.

Methodologies for Problem Solving

- **Data Quality**
and Preprocessing: Ensure high-quality images in the facial database. Preprocess images to address lighting variations, occlusions (like glasses or masks), and facial expressions. Techniques like image normalization and histogram equalization can improve accuracy.
- **Algorithm Selection and Training:**
Choose a robust facial recognition algorithm that can handle variations and achieve high accuracy. Train the algorithm with a diverse dataset that reflects the real-world scenarios the system will encounter.

- **Liveness Detection:**
Implement liveness detection to prevent spoofing attacks (using photos or videos to impersonate someone). Liveness detection can involve blinking detection, facial landmark movement tracking, or depth sensing cameras.
- **Security and Privacy:**
Store facial data securely using encryption and access controls. Define a clear data retention policy and obtain user consent for data collection and usage.
- **System Monitoring and Maintenance:**
Regularly monitor system performance and accuracy. Address any degradation in performance by retraining the algorithm or improving data quality.

Efficiency Issues in Face Recognition Attendance Systems

- **Computational Cost:**
Facial recognition algorithms can be computationally expensive, impacting processing speed, especially with large databases.
- **Lighting Variations:**
Poor lighting conditions or sudden changes in lighting can affect the accuracy of facial recognition.
- **Facial Occlusions:**
Items like hats, glasses, or masks can obscure facial features and hinder recognition.
- **Scalability:**
Large-scale deployments with a high number of users might require additional hardware or cloud-based solutions to handle the processing load.
- **Cost:**
Implementing and maintaining a face recognition system can be expensive, depending on the chosen hardware and software solutions.

5.4 Outcome

Face recognition attendance systems have the potential to revolutionize the way we track attendance and manage access control. They offer several advantages over traditional methods, such as:

- **Increased accuracy:** Face recognition technology is highly accurate and can identify individuals with minimal error. Reduced fraud: Face recognition can help to prevent buddy punching and other forms of attendance fraud.
- **Enhanced security:** Face recognition can be used to control access to restricted areas and improve overall security.
- **Improved efficiency:** Face recognition can automate attendance tracking and save time for administrators.
- **Convenience:** Users do not need to carry ID cards or badges, they can simply sign in using their faces.

5.5 Applications

The Applications of the project include:

- **Education Institutions:** Schools and Universities: Streamline attendance tracking for students and staff, reducing administrative workload. Libraries: Improve access control and track attendance of library patrons.
- **Corporate Settings:** Office Attendance: Replace traditional attendance systems with a more secure and efficient face recognition system. Conferences and Meetings: Simplify attendance tracking for events and conferences. Access Control: Enhance security by using face recognition for access to secure areas.
- **Healthcare Facilities:** Hospitals and Clinics: Monitor attendance of healthcare professionals, ensuring staff presence during critical hours. Restricted Areas: Use face recognition for secure access to sensitive areas within healthcare facilities.
- **Government Institutions:** Government Offices: Improve attendance tracking for government employees. Secure Facilities: Enhance security in government facilities by implementing face recognition for access control.
- **Retail and Hospitality:** Employee Attendance: Streamline attendance tracking for retail and hospitality staff. Access Control: Use face recognition for secure areas within retail establishments.

- **Transportation Hubs:** Airports and Train Stations: Track employee attendance and enhance security at restricted areas. Customs and Immigration: Implement face recognition for identity verification.

5.6 Hardware Resources Required

Sr. No.	Parameter	Minimum Requirement	Justification
1	Processor (CPU)	Multi-core, 2.5 GHz or Intel i3-10th generation	Face recognition algorithms require significant computational power for image processing and feature extraction. A multi-core CPU with a clock speed of at least 2.5 GHz ensures efficient processing.
2	Camera	720p resolution (HD)	Captures clear facial images for accurate recognition. Lower resolutions might struggle with identifying individuals, especially in less-than-ideal lighting conditions.
3	RAM	8 GB	Sufficient RAM is necessary to store and process face images, feature vectors, and intermediate results during recognition tasks, ensuring smooth and efficient operation.

Table 5.1: Hardware Requirements

Sr. No.	Parameter	Minimum Requirement	Justification
4	Storage	256GB SSD	Adequate storage space, preferably on a Solid State Drive (SSD), is required to store face images, training data, model parameters, and system logs for fast data access and retrieval.
5	Network	Reliable internet connection	connection provides high-speed and reliable network connectivity, essential for communication between system components and data transfer.
6	GPU	Dedicated graphics card (for large deployments or complex algorithms)	Offloads some processing tasks from the CPU, potentially improving performance. Beneficial for large-scale deployments with many users or systems using computationally expensive algorithms.
7	Cooling System	Fans	Adequate cooling mechanisms, such as fans and heat sinks, are necessary to prevent overheating and maintain optimal performance, especially in systems with high computational demands.

5.7 Software Resources Required

Platform :

1. **Operating System:**

Windows 10 or 11 for development

2. **IDE:**

VS-Code is suitable for development

3. **Programming Language**

Python 3.7(version)

4. **Libraries and Frameworks**

OpenCV: Provides extensive support for image processing tasks, including face detection, recognition, and feature extraction.

Tkinter: For Designing the Graphical User Interface

5. **Algorithms**

Haar Cascade Classifier: The Haar cascade classifier is a machine learning-based object detection algorithm used to identify objects in images or video streams.

LBPH (Local Binary Patterns Histograms): LBPH is a texture-based face recognition algorithm that extracts local binary patterns from facial images and creates histograms of these patterns for classification.

6. **Database Management System (DBMS):**

MySQL is Used to Stores and manages facial data and attendance records securely.

CHAPTER 6

PROJECT PLAN

6.1 Project Estimates

Here's a breakdown of the project estimates using the Waterfall model:

- **Requirements Analysis:**

Objective:

Gather and analyze requirements for the face recognition attendance system.

Activities:

Gather requirements, define system functionalities, create use cases, identify hardware and software resources, establish project timeline and budget.

Duration: 2 weeks

- **Design and Architecture:**

Objective:

Design the system architecture, database schema, and user interface.

Activities:

Develop system architecture diagrams and flowcharts.

Design database schema and data models.

Create wireframes or mockups for the user interface.

Duration: 3 weeks

- **Implementation :**

Objective:

Develop the face recognition attendance system based on the design specifications.

Activities:

Implement face detection using Haar cascade algorithm.

Implement LBPH algorithm for face recognition.

Develop user authentication, attendance recording, and reporting modules.

Duration: 8 weeks

- **Testing:**

Objective:

Verify and validate the system functionalities through testing.

Activities:

Perform unit testing for individual modules and components.

Conduct integration testing to ensure modules work together seamlessly.

Execute system testing to validate user requirements and performance.

Duration: 4 weeks

- **Deployment and Maintenance :**

Objective:

Deploy the system to production and ensure ongoing maintenance and support..

Activities:

Set up deployment environments

Deploy the system and perform user acceptance testing (UAT).

Establish maintenance procedures and provide user training and support.

Allocate additional time for bug fixes, future enhancements, and system updates based on user feedback or changing requirements.

Duration: Ongoing

6.1.1 Reconciled Estimates

6.1.1.1 Cost Estimate

- Software Costs:
Licensing fees for any proprietary software or development frameworks.
Total Software Cost: *Rs.500* (open source / free software will used)
- Communication Costs:
Internet, phone bills, and other communication
- Hardware Costs:
Camera and Storage
- Miscellaneous Costs:
Contingency budget for unforeseen expenses.
Total Miscellaneous Cost: *Rs.1,000* expenses

6.1.1.2 Time Estimates

- Requirements Analysis: 2 weeks
- Design: 3 weeks
- Implementation: 8 weeks
- Testing: 4 weeks
- Deployment and Maintenance: Ongoing

6.1.2 Project Resources

- **People:**
All group members.
- **Hardware:**
Camera: 720p or higher resolution USB webcam (for single user) or multiple high-definition cameras (for multi-user systems).
Laptop: A mid-range computer (e.g., Intel Core i3 or equivalent) with at least 8GB RAM (for single user development). A more powerful server with multiple cores, 8GB+ RAM (for multi-user deployment).
Storage: Adequate storage space(256GB SSD) on the computer(s) to store facial images and attendance data.
- **Software:**
Operating System: Windows 10/11
Programming Language: Python with OpenCV library for computer vision tasks (face detection, recognition).
Database Management System (DBMS): MySQL ,to store facial data and attendance records securely, especially for larger deployments.
- **Development Tools:**
Integrated Development Environments (IDEs) like Visual Studio Code,for code editing and debugging.
- **Version Control System (VCS):**
Git for version control and collaboration among developer

6.2 Risk Management w.r.t. NP Hard analysis

This section discusses Project risks and the approach to managing them.

6.2.1 Risk Identification

1. Technology Risks:

- **Algorithm Complexity:** Face recognition algorithms, especially those based on deep learning or NP-hard problems, may be complex and challenging to implement efficiently.

- **Integration Issues:** Integrating face recognition algorithms with existing systems or databases can pose compatibility and integration challenges.
- **Performance Optimization:** Ensuring real-time performance and accuracy of face recognition under varying environmental conditions (lighting, pose, occlusions) can be a risk.
- **Data Security:** Risks related to biometric data security, including unauthorized access, data breaches, and privacy concerns.

2. Project Management Risks:

- **Scope Creep:** Expansion of project scope beyond initial requirements, leading to timeline delays and resource over utilization.
- **Resource Allocation:** Inadequate resources (human, financial, hardware) allocated to the project, impacting its success and timely completion.
- **Communication Breakdown:** Poor communication among project stakeholders, team members, and external vendors, leading to misunderstandings and delays.

3. Operational Risks:

- **System Downtime:** Unplanned system outages or downtime affecting attendance tracking and system availability.
- **Data Integrity:** Risks of data corruption, loss, or inconsistency within the attendance records or database.
- **User Training:** Inadequate training or user adoption issues leading to errors in system usage or data entry.

4. Cost and Performance:

- The cost and performance of face recognition technology can be a challenge, particularly for organizations with limited resources or large-scale attendance tracking needs. Ensuring the cost-effectiveness and efficiency of the system is crucial to its success.

5. User Acceptance:

- Involve users in the development process and address their concerns.
- Provide clear instructions and training on system usage.

- Ensure the system is user-friendly and reliable.
- Address privacy concerns transparently and demonstrate adherence to regulations.

6.2.2 Risk Analysis

- **Risk:**

Delays in project milestones, such as requirement gathering, design, implementation, testing, and deployment, can impact the overall project timeline.

Impact:

Reduced time for thorough testing and optimization, rushed implementation leading to quality issues, and missed deadlines.

Mitigation:

Break down the project into manageable phases with clear milestones and deadlines. Allocate sufficient time for each phase, considering dependencies and potential delays. Regularly monitor progress and adjust timelines as needed. Implement agile practices for iterative development and flexibility in responding to changes.

- **Risk:**

Insufficient focus on quality assurance (QA) processes, including testing, validation, and verification, can lead to defects, errors, and system failures.

Impact:

Reduced system reliability, accuracy, and user satisfaction. Increased post-deployment issues and maintenance efforts.

Mitigation:

Implement comprehensive QA strategies, including unit testing, integration testing, system testing, and user acceptance testing (UAT). Define clear quality criteria and acceptance criteria for each deliverable. Conduct regular code reviews, peer testing, and quality audits. Invest in automated testing tools and continuous integration/continuous deployment (CI/CD) pipelines.

- **Risk:**

Limited resources (human, financial, hardware) allocated to the project can hinder progress, compromise quality, and increase the risk of project failure.

Impact:

Inadequate staffing, insufficient budget for necessary tools or technologies, and hardware limitations affecting system performance.

Mitigation:

Conduct resource planning and allocation based on project requirements and priorities. Ensure adequate training and skill development for team members. Explore cost-effective alternatives for software licenses, hardware procurement, and cloud services. Optimize resource utilization and prioritize critical tasks.

- **Risk:**

Complex technical requirements, such as implementing advanced face recognition algorithms, integrating with existing systems, and ensuring compatibility with diverse hardware and software environments, can lead to challenges and delays.

Impact:

Increased development effort, potential technical issues, and compatibility issues affecting system functionality..

Mitigation:

Conduct thorough technical analysis and feasibility studies before project initiation. Break down complex tasks into smaller, manageable components. Collaborate with domain experts, leverage existing libraries and frameworks, and use proven methodologies and best practices. Implement robust error handling and fallback mechanisms.

- **Risk:**

Poor quality input data or inadequate preprocessing can lead to inaccurate face recognition results and attendance records..

Mitigation:

Implement data quality checks, image preprocessing techniques (e.g., normalization, denoising), and face alignment methods to improve the accuracy and reliability of recognition.

6.2.3 Overview of Risk Mitigation, Monitoring, Management

Following are the details for each risk.

ID	Risk Description	Probability	Impact		
			Schedule	Quality	Overall
1	Inaccurate Recognition	Moderate	High	High	High
2	System Performance Issues	High	High	Moderate	High

Table 6.1: Risk Table

ID	Risk Description	Probability	Impact		
			Schedule	Quality	Overall
3	Data Security Breach	Low	High	High	High
4	Project Delays	Moderate	High	High	High
5	User Resistance	Low	Moderate	Low	Moderate

Probability	Value	Description
High	Probability of occurrence is	$> 75\%$
Medium	Probability of occurrence is	$26 - 75\%$
Low	Probability of occurrence is	$< 25\%$

Table 6.2: Risk Probability definitions [?]

Impact	Value	Description
Very high	> 10%	Schedule impact or Unacceptable quality
High	5 – 10%	Schedule impact or Some parts of the project have low quality
Medium	< 5%	Schedule impact or Barely noticeable degradation in quality Low Impact on schedule or Quality can be incorporated

Table 6.3: Risk Impact definitions [?]

Risk ID	1
Risk Description	Face Identification
Category	Scanned Image
Source	Database
Probability	Medium
Impact	High
Response	Matching with person's face
Strategy	Accuracy checking
Risk Status	Observed and identified

Risk ID	2
Risk Description	Integration issues with existing systems
Category	Integration
Source	Identified during system design and integration planning.
Probability	Medium
Impact	High
Response	by conducting thorough integration testing, implementing robust data handling protocols, and ensuring compatibility checks
Strategy	Develop clear integration interfaces and protocols .
Risk Status	Identified

Risk ID	3
Risk Description	Insufficient data quality
Category	Data Management
Source	Identified during data collection and preprocessing phases.
Probability	High
Impact	High
Response	by implementing data quality checks, preprocessing techniques, and data cleansing procedures
Strategy	Develop data quality standards and validation processes.
Risk Status	Identified

Risk ID	4
Risk Description	Insufficient computational resources may lead to performance bottlenecks during face recognition processing, impacting system responsiveness and accuracy.
Category	Infrastructure
Source	identified during system architecture design and hardware planning.
Probability	Medium
Impact	Medium
Response	by optimizing resource usage, upgrading hardware if necessary, and implementing performance monitoring
Strategy	Conduct load testing, performance profiling, and resource utilization analysis.
Risk Status	Identified

6.3 Project Schedule

6.3.1 Project task set

Major Tasks in the Project stages are:

- **Task 1: Project Initiation:**
Define project objectives, scope.
Conduct feasibility studies and risk assessments.
Establish project governance and team roles.
- **Task 2: Requirements Gathering and Analysis:**
Collect and analyze user requirements for the attendance system.
Define functional and non-functional requirements.
Create use cases, user stories, and system specifications.
- **Task 3: System Design:**
Design the system architecture, including database schemas, backend logic, and user interfaces.
Select face recognition algorithms and integration with attendance track-

ing functionalities.

Develop system workflows and data flow diagrams.

- **Task 4: Implementation:**

Develop and integrate face recognition algorithms into the system.

Implement user interfaces for registration, attendance tracking, and reporting.

Integrate with hardware components such as cameras .

- **Task 5: Testing and Quality Assurance:**

Conduct unit testing for individual components and modules.

Perform integration testing to ensure system functionality and data consistency.

Execute system testing, including performance testing and user acceptance testing (UAT).

Implement quality assurance processes and bug tracking mechanisms.

- **Task 6: Deployment:**

Prepare the system for deployment in the production environment.

Configure hardware, software, and networking components.

Conduct system validation and user training.

- **Task 7: Monitoring and Maintenance:**

Monitor system performance, security, and data integrity.

Address user feedback and system enhancements.

Perform regular maintenance, updates, and backups.

- **Task 8: Documentation and Reporting:**

Create project documentation, including requirements documents, design specifications, user manuals, and technical guides.

Generate reports on system usage, attendance statistics, and performance metrics.

6.3.2 Timeline Chart

[illegible]

Figure 6.1: Timeline Chart

Above points should be covered in Project Planner as Annex C and you can mention here Please refer Annex C for the planner

6.4 Team Organization

The manner in which staff is organized and the mechanisms for reporting are noted.

6.4.1 Team structure

- **Project Guide:**
Prof. B.M. Gawale
- **Team Members:**
Sujit Kanhaiyalal Yadav
Om Rajendra Jain
Vivek Sunil Sangale
Arjala Sajendra Patil
- **Project Guide :**
Provides guidance and mentorship to the team throughout the project life cycle. Offers technical expertise and assistance in problem-solving. Reviews project deliverables and provides feedback for improvement. Assists in resolving any issues or challenges faced by the team. Approves project milestones and final deliverables.
- **Team Members:**
Collaborate with the team leader and project guide to execute project tasks. Contribute to the design, development, testing, and deployment phases of the project. Take ownership of assigned tasks and ensure timely completion. Participate in team meetings, brainstorming sessions, and progress reviews. Provide input and suggestions for improving project efficiency and quality.

6.4.2 Management reporting and communication

Management reporting and communication are crucial aspects of project management, ensuring that stakeholders are kept informed about the project's progress, challenges, and achievements. Here's a suggested approach for management reporting and communication for your messenger application development project:

- **Weekly Progress Meetings:**
Conduct weekly meetings with the project team to review progress, discuss any challenges, and plan upcoming tasks. Team leader should lead these meetings, with all team members providing updates on their respective tasks. Use these meetings to identify any bottlenecks or issues that need resolution.
- **Bi-weekly Status Reports:**
Prepare bi-weekly status reports outlining the project's progress, key milestones achieved, and upcoming tasks. Include updates on budget utilization, resource allocation, and any risks or issues encountered. Distribute these reports to the project guide, team members, and any other relevant stakeholders.
- **Monthly Review Meetings with Project Guide:**
Schedule monthly review meetings with the project guide to provide a comprehensive overview of the project's status. Discuss progress against the project plan, any deviations, and mitigation strategies for addressing risks. Seek guidance and advice from the project guide on critical decisions or challenges faced by the team.
- **Change Control Reporting:**
Implement a change control process to manage any changes to project scope, schedule, or requirements. Document change requests, including the rationale for the change and its potential impact on the project. Report on approved changes and their implications in management reports.
- **Documentation and Version Control:**
Maintain comprehensive documentation of project plans, requirements, design documents, and test cases. Use version control systems like Git to track changes to project artifacts and ensure consistency across team members.

CHAPTER 7

**SOFTWARE REQUIREMENT
SPECIFICATION**

7.1 Introduction

To develop a comprehensive and efficient attendance management system that utilizes facial recognition technology for tracking and recording attendance in various settings, such as educational institutions, businesses, or organizations. To reduce the time and effort required for manual attendance tracking and improve accuracy by automating the process. To enhance security and accountability by ensuring that only authorized individuals can mark their attendance.

7.1.1 Purpose and Scope of Document

- **Face detection:**
The system should effectively detect faces in images or video streams captured from a webcam.
- **Face recognition:**
The system should accurately identify individual students based on their facial features.
- **Attendance recording:**
The system should automatically record the attendance of each student in the classroom.
- **Reporting:**
The system should generate comprehensive attendance reports, including student attendance records and overall attendance statistics.
- **Administration:**
The system should provide an administrative interface for managing student information, attendance rules, and system settings.

7.1.2 Overview of responsibilities of Developer

- **Requirement Analysis:**
Collaborate with stakeholders to understand and analyze system requirements related to face recognition, attendance tracking, and user interface functionalities. Translate functional and non-functional requirements into technical specifications and design documents.
- **System Design and Architecture**
Design and architect the system components, including face recognition

algorithms, data storage, user interfaces, and integration with hardware devices (e.g. cameras). Ensure scalability, performance, and security considerations are addressed in the system design.

- **Algorithm Development:**

Develop or integrate face recognition algorithms, such as Haar cascade, LBPH (Local Binary Patterns Histograms), to accurately identify and recognize faces. Optimize algorithms for speed, accuracy, and robustness under varying environmental conditions (e.g., lighting, pose, occlusions).

- **Software Development:**

Implement software components using programming languages (e.g., Python) and development frameworks (e.g., OpenCV) to support face recognition and attendance tracking functionalities. Develop user interfaces for registration, attendance marking, reporting, and system administration.

- **Testing and Quality Assurance:**

Conduct unit testing, integration testing, and system testing to validate the functionality, performance, and reliability of the face recognition system. Identify and fix bugs, errors, and performance bottlenecks through debugging and troubleshooting.

- **Documentation:**

Create technical documentation, including system architecture diagrams, code documentation, API specifications, and user manuals. Maintain documentation for future reference, maintenance, and knowledge transfer.

- **Collaboration and Communication:**

Communicate progress, challenges, and technical solutions effectively to team members and stakeholders.

- **Continuous Learning and Improvement:**

Stay updated with advancements in face recognition technology, machine learning algorithms, and software development best practices. Participate in training, workshops, and knowledge-sharing sessions to enhance skills and contribute to project success.

7.2 Usage Scenario

- **Enrolls in the system:**
The employee's facial image is captured and stored in the system's database.
- **Face is recognized:**
The system's camera captures the employee's face and compares it to the images stored in the database.
- **Client attendance is recorded:**
If the employee's face is recognized, their attendance is automatically recorded
- **Receives confirmation:**
The employee receives a confirmation that their attendance has been recorded.
- **Data is reported::**
The system automatically generates attendance reports for managers.

7.2.1 User profiles

- **Students:**
Description: Individuals enrolled in the program/institution using the system.Enroll in the system by providing facial image and identification information.Have their face recognized for attendance marking upon entering/leaving designated areas.
- **Teachers/Faculty:**
Description: Individuals who conduct classes and manage student attendance.Access the system using a secure login process (computer)View attendance reports for their classes, including student names, ID numbers, and attendance status (present, absent).Download or print attendance reports for record-keeping.
- **System Administrators:**
Description: Individuals responsible for system setup, configuration, and maintenance Manage user accounts (add, edit, delete) for students, staff, and potentially other users. Configure system settingsMonitor system performance and address any technical issues. Back up and

restore system data to ensure data security. Manage user access levels and permissions within the system

7.2.2 Use-cases

Sr No.	Use Case	Description	Actors	Assumptions
1	Register Employee or Student	Allows administrators to register new employees or students into the system	Administrator, Employee or Student	Assumes employees or students have valid credentials for registration.
2	Mark Attendance	Capture a user's face and mark their attendance based on recognition.	Student	The system has cameras positioned for attendance capture.
3	Attendance Report Generation	Generate attendance reports for specific classes, dates, or date ranges.	Teacher or Faculty	The system stores attendance data in a structured format.
4	User Profile Management	View and manage user information	System Administrator	The system stores user profiles in a secure database
5	Configuration Management	Adjust system settings such as time thresholds, camera sensitivity, access control policies.	System Administrator	The system provides a user interface for configuration management.
6	Data Backup and Restore	Back up and restore system data for security and recovery purposes.	System Administrator	The system has a backup and restore functionality.

Table 7.1: Use Cases

7.2.3 Use Case View

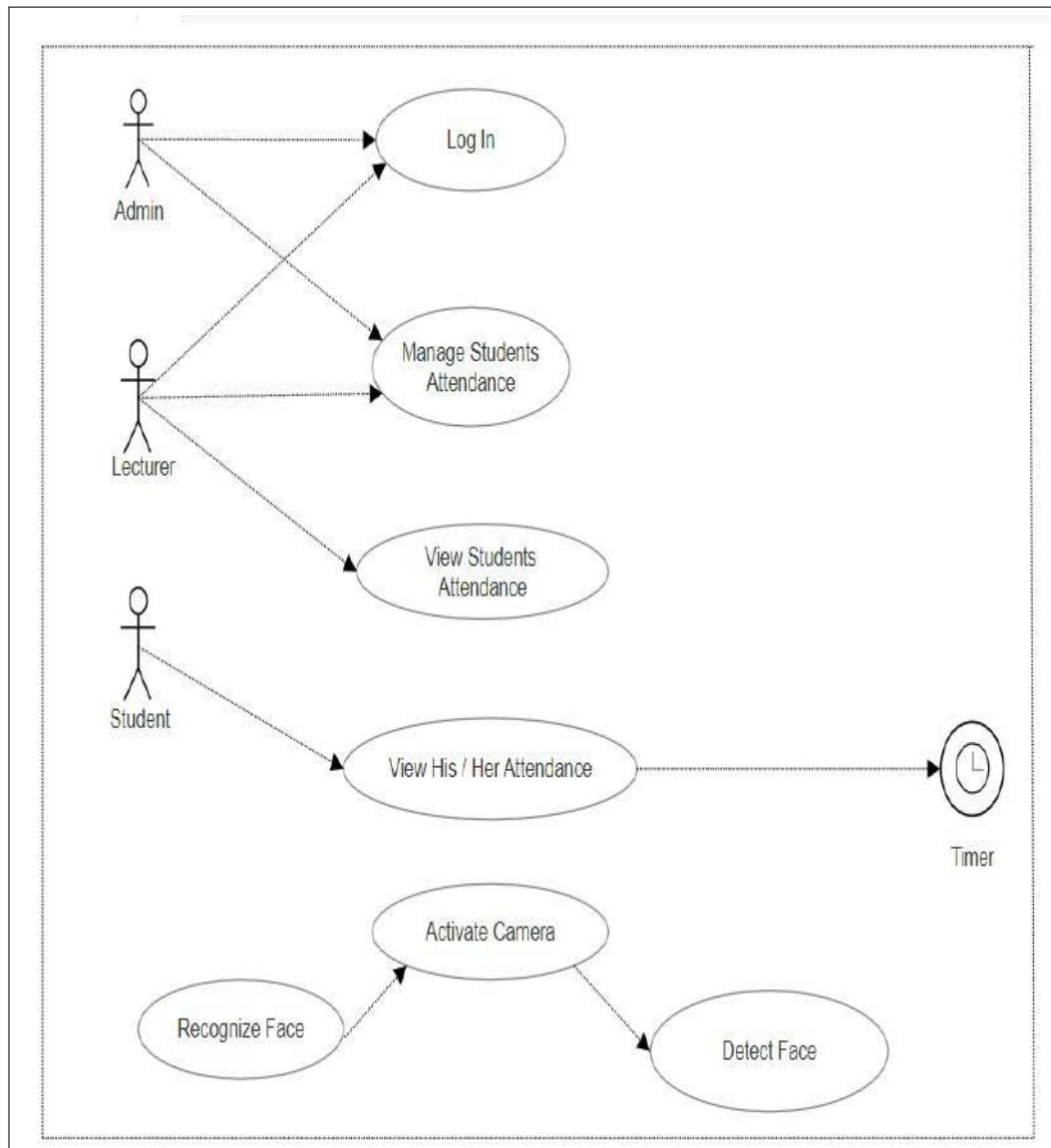


Figure 7.1: Use case diagram

7.3 Data Model and Description

7.3.1 Data Description

Data objects that will be managed/manipulated by the software are described in this section. The database entities or files or data structures required to be described. For data objects details can be given as below

7.3.2 Data objects and Relationships

- **Entities:**

User: Attributes: User ID (primary key), Name, Identification Number (e.g., Student ID), Email , User Type (e.g., Student, Teacher, Administrator)

Image: Attributes: Image ID (primary key), User ID (foreign key references User.UserID), Image Data (stores facial image data securely), Capture Date and Time

Attendance Record: Attributes: Attendance Record ID (primary key), User ID (foreign key references User.UserID), Date, Time, Location (optional)

- **Relationships:**

One User can have Many Images: A user can have multiple facial images captured during enrollment or re-enrollment. (1:N relationship between User and Image)

One User can have Many Attendance Records: A user can have numerous attendance records for different days and times. (1:N relationship between User and Attendance Record)

7.4 Functional Model and Description

This functional model outlines the key functionalities of a face recognition attendance system, using a combination of data flow and class descriptions:

7.4.1 Data Flow Diagram

- DFDs illustrate how data moves through the system, Depicting processes, Data stores, Data Flows, & External units.
- Create a Level-0 DFD, gives overview of the system's main processes and data flows.

- Create Level–1 and Level–2 DFDs to break down processes into more detail.

7.4.1.1 Level 0 Data Flow Diagram

The level 0 DFD represents the input and output that will be generated by the system.

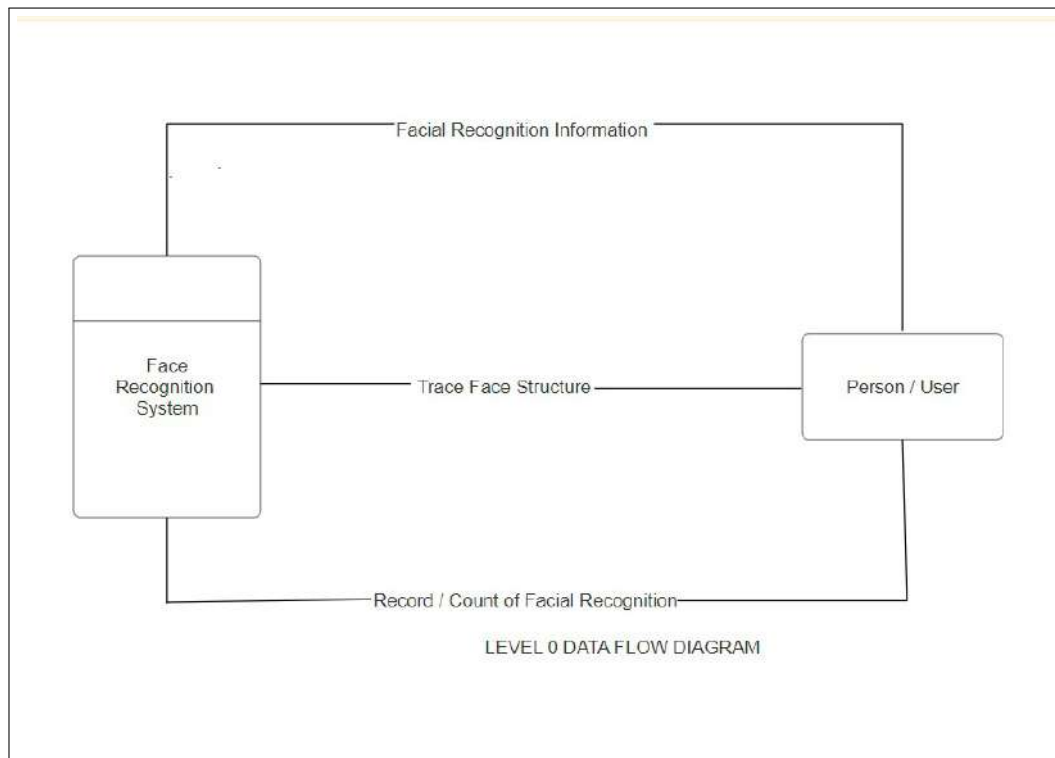


Figure 7.2: Level 0 Data Flow Diagram

7.4.1.2 Level 1 Data Flow Diagram

The system appears to be designed for capturing student facial images, likely for enrollment purposes, and marking attendance when a student is recognized by the camera. The system leverages OpenCV and NumPy libraries for image processing tasks.

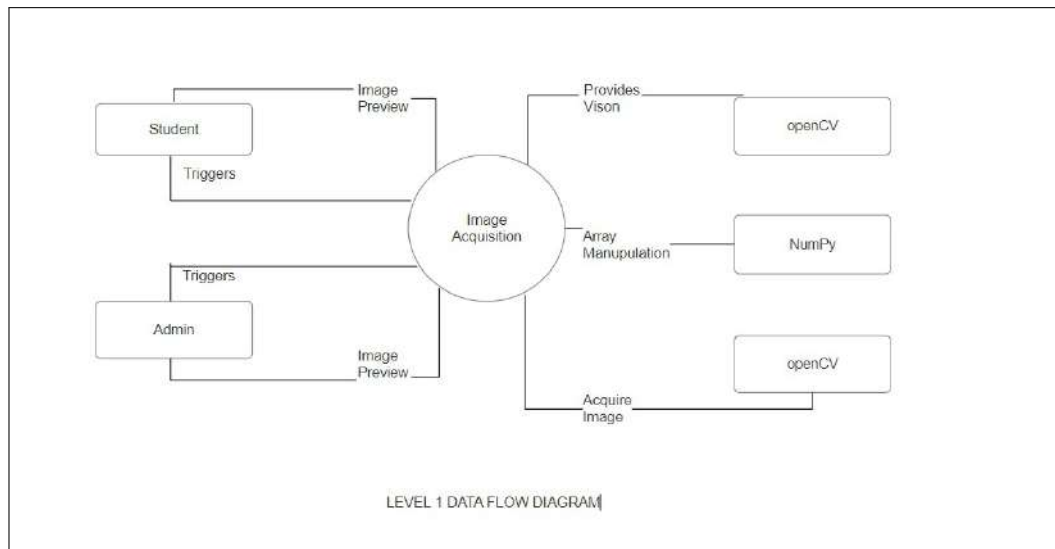


Figure 7.3: Level 1 Data Flow Diagram

7.4.1.3 Level 2 Data Flow Diagram

The DFD level-2 represents the various operations performed by the System

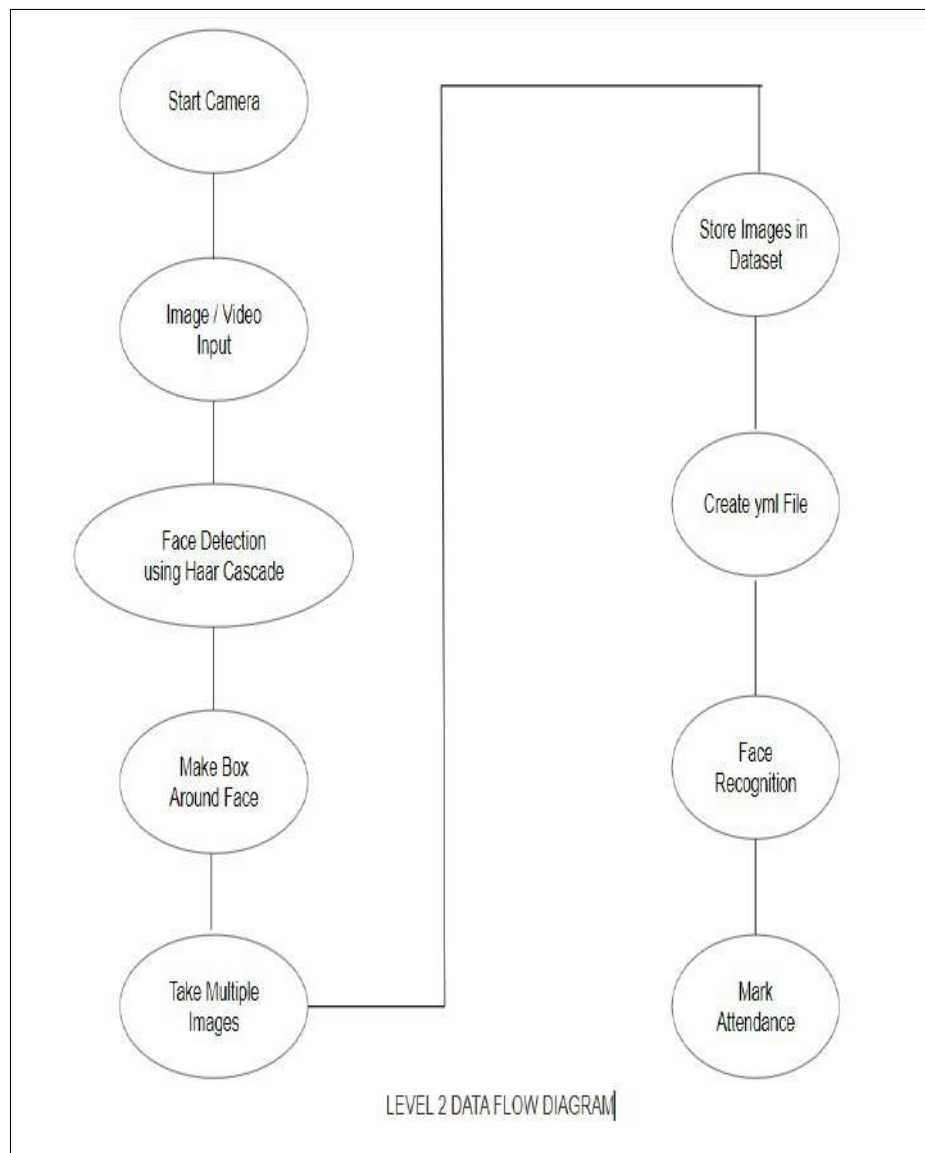


Figure 7.4: Level 2 Data Flow Diagram

7.4.2 Activity Diagram:

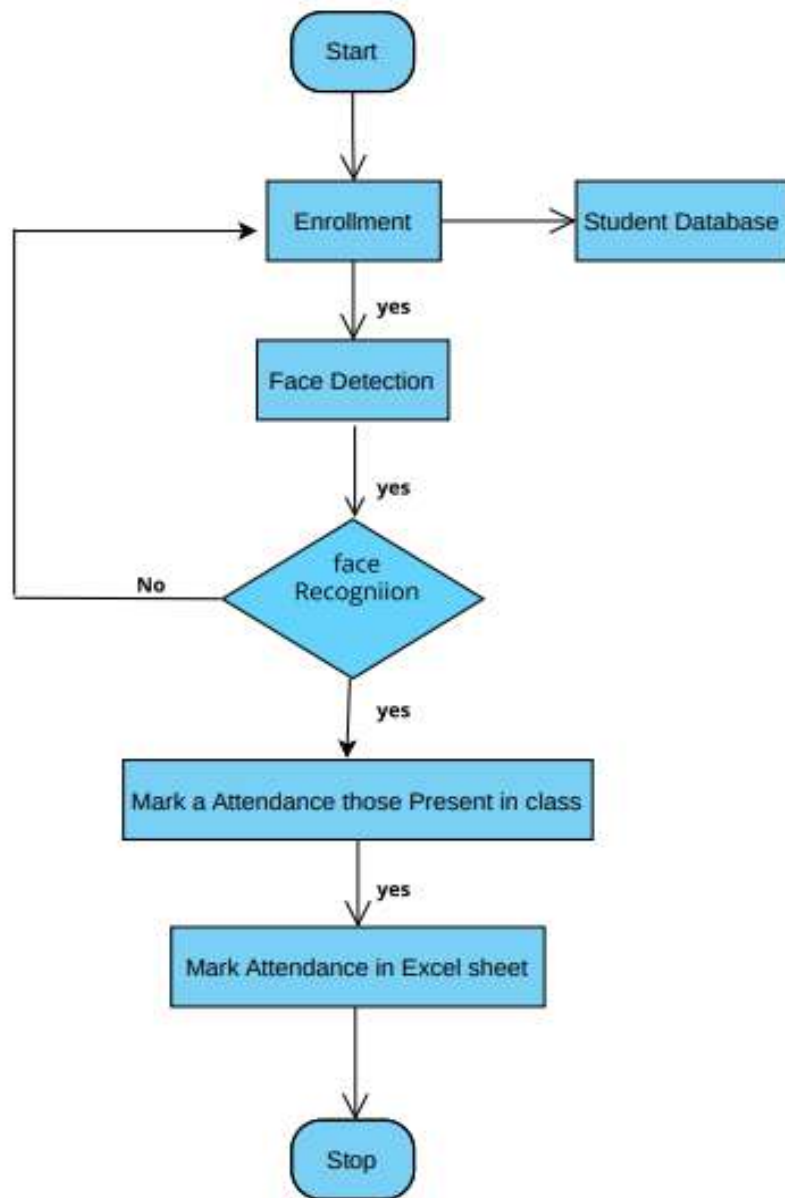


Figure 7.5: Activity Diagram

7.4.3 Non Functional Requirements:

- **Interface Requirements:**

Administrator Interface: The administrator interface should have functionalities for user management, including adding, editing, and deleting user profiles. It should provide options for configuring access control rules, setting up attendance policies, and generating reports.

Employee/Student Interface:

The employee/student interface should allow users to mark their attendance easily using face recognition technology. It should display attendance status, historical records, and notifications related to attendance policies or updates.

- **Performance Requirements**

Response Time: The system should have low latency in recognizing faces and marking attendance, aiming for response times within milliseconds. Response time for generating attendance reports and accessing user profiles should be fast, even with a large number of concurrent users.

Scalability:

The system should be able to handle increasing user loads and data volumes without compromising performance.

Recognition accuracy:

The system should achieve a high degree of accuracy in recognizing faces under various lighting conditions and poses.

- **Software Quality Attributes:**

Availability The system should be highly available and reliable, with minimal downtime or disruptions. It should have failover mechanisms and redundancy to ensure continuous operation.

Modifiable:

The system should be easily modifiable to accommodate changes in business rules, user requirements, or regulatory standards. It should exhibit portability across different platforms and environments, allowing deployment flexibility.

Security:

The system should adhere to industry-standard security practices to protect sensitive data, user privacy, and system integrity.

Test ability: The system should be designed for easy testing, including unit testing, integration testing, and performance testing. Test ability features should enable developers to identify and fix issues quickly dur-

ing development and maintenance.

Usability: The system should be user-friendly and accessible to users with varying technical backgrounds. It should support self-adaptability, allowing users to customize settings and preferences. User adaptability features should provide options for adjusting system behavior based on user feedback and usage patterns.

7.4.4 Design Constraints

- **Hardware Constraints:**

1. **Camera quality and resolution:**

The system's accuracy relies on the camera's ability to capture clear facial images under various lighting conditions. Higher resolution cameras generally provide better image detail for recognition.

2. **Processing power**

Real-time face detection and recognition require sufficient processing power. This may limit the number of cameras or users the system can handle on a single machine

- **Software Constraints:**

1. **Algorithm performance:**

The chosen face recognition algorithms should achieve high accuracy within acceptable processing time constraints.

2. **Resource usage:**

The software should be optimized to minimize memory and CPU usage, especially for resource-constrained devices.

3. **Scalability**

The system architecture should be designed to scale efficiently to accommodate a growing number of users and attendance records.

- **Environmental Constraints:**

1. **Lighting conditions:**

The system should function reliably under varying lighting conditions (bright light, low light, shadows). Techniques like facial normalization or illumination compensation may be required.

2. **Background clutter:**

Minimize distractions in the background where faces are captured to avoid recognition errors.

3. **Viewing angle:**

The system's performance may be affected by the angle at which faces are presented to the camera. Ideally, users should be positioned within a specific range.

- **Security Constraints:**

1. **Data security:**

facial image data and user information should be encrypted at rest and in transit to prevent unauthorized access.

2. **System access control:**

Implement strong access control mechanisms to restrict unauthorized users from modifying settings or accessing sensitive data.

3. **Privacy considerations:**

Adhere to data privacy regulations regarding the collection, storage, and use of facial image data. Users should be informed about how their data is used and have the option to opt out.

- **Additional Considerations:**

1. **Cost:**

The overall system cost should be balanced with desired functionality and performance requirements.

2. **Deployment environment:**

Consider the physical environment where the system will be deployed (classroom, office, outdoor setting) when choosing hardware and software components.

3. **User acceptance:**

The system design should be user-friendly and minimize disruption to existing workflows..

7.4.5 Software Interface Description

- **Administrator Interface:**

Allows administrators to manage user profiles, configure access control settings, and generate attendance reports. Features include forms for user registration, access control configuration panels

- **Employee/Student Interface:**
Enables employees or students to mark their attendance using face recognition technology. Includes a simple interface with a "Mark Attendance" button and feedback messages for successful or failed recognition.
- **Integration API:**
Facilitates integration with external systems such as HRMS, student databases, and third-party applications. Provides endpoints for data synchronization, user management, access control configuration, and report generation.
- **Camera Interface:**
Interfaces with cameras and facial scanners for capturing facial images during attendance marking and user registration.
- **Database Interface:**
Integrates with databases (MySQL) for storing user profiles, attendance records, access logs, and configuration settings.

CHAPTER 8

DETAILED SYSTEM DESIGN

8.1 Introduction

This document specifies the design that is used to solve the problem of Product.

8.2 Architectural Design

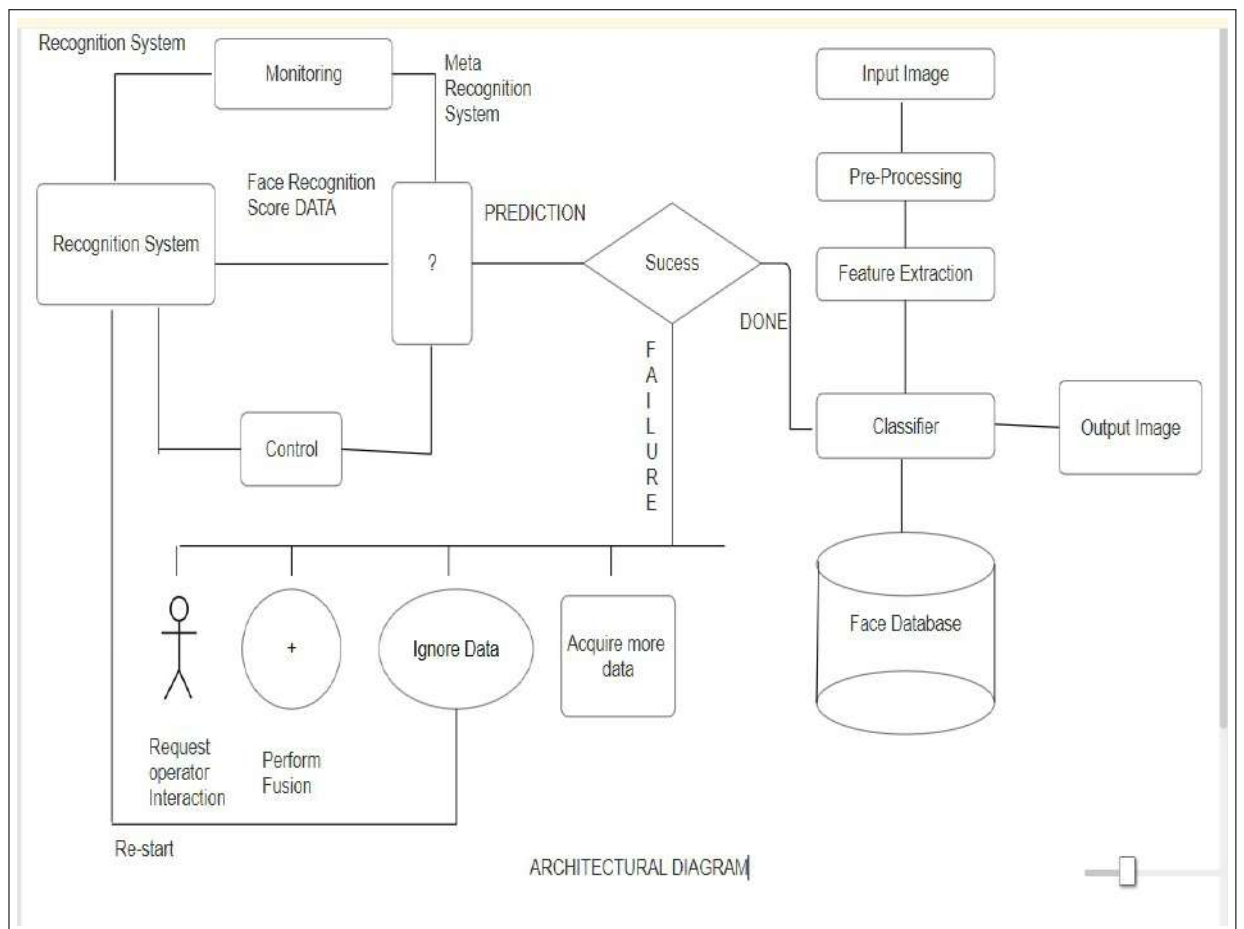


Figure 8.1: Architecture diagram

8.3 Data design (using Annexures A and B)

8.3.1 Internal software data structure

The internal software data structure refers to data structures that are passed among components within the software. In a face recognition attendance system, these data structures may include:

1. **User Profile Data Structure**

Fields: User ID, Name, Department, Position, Email, Phone, Face Template.

Purpose: Stores information about registered users for face recognition and attendance tracking.

2. **Attendance Record Data Structure:**

Fields: User ID, Timestamp, Status (Present/Absent).

Purpose: Records attendance events including user identification and timestamp.

8.3.2 Global data structure

Global data structures are available to major portions of the architecture. In this context, global data structures in the face recognition attendance system may include:

1. **System Configuration Data Structure:**

Fields: Recognition Threshold, Camera Settings, System Preferences.

Purpose: Stores configuration settings that affect the overall behavior of the system.

2. **Access Logs Data Structure:**

Fields: User ID, Timestamp, Access Type (Login/Logout), Access Result (Success/Failure).

Purpose: Logs access control events for security monitoring and audit purposes.

8.3.3 Temporary data structure

Temporary data structures are files created for interim use. In the face recognition attendance system, temporary data structures may include

1. **Temporary Face Image Storage:**

Format: Image files (e.g., JPEG, PNG).

Purpose: Stores temporary facial images captured during face recognition processes.

2. Temporary Data Cache:

Format: In-memory data structure (e.g., cache object).

Purpose: Caches frequently accessed data to improve system performance and response times.

8.3.4 Database description

The database(s) used as part of the application can include tables related to user management, attendance tracking, access control, and system configuration.

1. User Table:

Fields: User ID (Primary Key), Name, Department, Position, Email, Phone, Face Template.

2. Attendance Table:

Fields: Attendance-ID (Primary Key), User ID (Foreign Key), Timestamp, Status.

3. Access Control Table

Fields: Access-ID (Primary Key), User ID (Foreign Key), Access Level, Access Right.

8.4 Component Design

8.4.1 Class Diagram

A class diagram is a type of UML (Unified Modeling Language) diagram that represents the static structure of a system. It's particularly useful for visualizing the classes and their relationships in an object-oriented system. Here's an overview of the key components of a class diagram

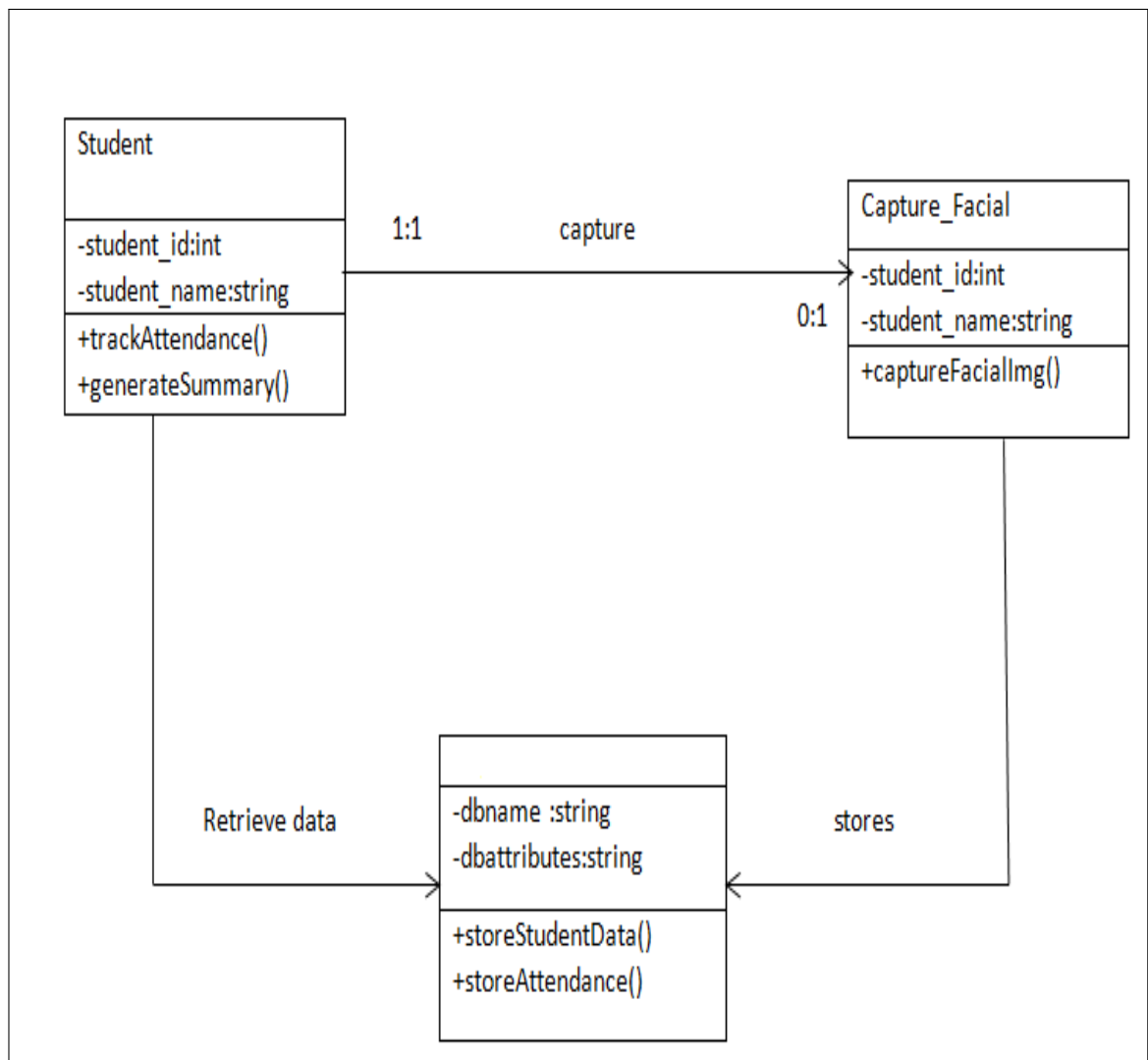


Figure 8.2: Class Diagram

CHAPTER 9

PROJECT IMPLEMENTATION

9.1 Introduction

9.1.1 Motivation

A face recognition attendance system offers several advantages and motivations for its implementation in various settings. Here are some key motivations for using face recognition technology in attendance systems:

- **Efficiency and Accuracy:**

Time-saving: Face recognition systems can quickly and accurately identify individuals, reducing the time it takes to record attendance compared to traditional methods like manual entry or card scanning.

Reduced errors: Automated systems are less prone to errors associated with manual data entry, leading to more accurate attendance records.

- **Security:**

Prevention of proxy attendance: Face recognition helps prevent instances of proxy attendance, where someone else might attempt to clock in on behalf of another individual.

Enhanced security: Facial features are unique to individuals, making it difficult for unauthorized individuals to gain access.

- **Contactless Technology:**

Hygiene and health considerations: Especially in the context of global health concerns, contactless systems reduce the risk of spreading germs compared to traditional methods involving physical touch or proximity.

- **Automation and Integration:** Integration with other systems: Face recognition attendance systems can be seamlessly integrated with other software and databases, streamlining administrative processes and providing real-time data.

Automation of attendance tracking: Eliminates the need for manual data entry and paperwork, reducing administrative burden and potential errors.

- **Cost-Effective:** Reduced administrative costs: Automation of attendance tracking can lead to cost savings by reducing the need for manual record-keeping and associated administrative tasks

- **User Convenience:**

User-friendly: Face recognition systems are generally user-friendly, requiring minimal effort from individuals to register their attendance.

No need for physical tokens: Eliminates the need for physical ID cards or keycards, providing a more convenient experience for users.

9.1.2 Problem Definition

Traditional attendance tracking methods in educational institutions, such as manual sign-in sheets and roll call, are inefficient, inaccurate, and prone to fraud. These methods can waste time, distract from instruction, and make it difficult to track student progress accurately.

Face recognition attendance systems (FRAS) offer a number of advantages over traditional attendance tracking methods. FRAS are accurate, efficient, and convenient. They can also help to improve security and reduce cheating. However, FRAS are a relatively new technology, and there are still some challenges associated with their implementation in educational institutions, including

- **Accuracy and Reliability:**

The face recognition algorithm must be accurate and reliable, ensuring minimal false positives and false negatives. The system should be able to handle variations in lighting conditions, facial expressions, and minor changes in appearance over time.

- **Real-time Processing:**

The system should operate in real-time, providing quick and efficient recognition without significant delays. It must be able to process a large number of faces in a short time, making it suitable for attendance tracking in various scenarios such as classrooms, offices, or events.

- **User-Friendly Interface:**

The system should have an intuitive and user-friendly interface for both administrators and end-users. Administrators should be able to easily manage the system, add or remove users, and view attendance records. End-users (individuals being recognized) should experience a seamless and non-intrusive attendance process.

- **Security and Privacy:**

The system must prioritize the security and privacy of the stored facial data and attendance records. It should comply with relevant privacy regulations and standards to ensure the protection of personal information.

9.2 Tools and Technologies Used

1. Python Tkinter for GUI:

- Tkinter is a standard GUI toolkit in Python used for creating graphical user interfaces.
- It provides various widgets like buttons, labels, entry fields, etc., for building interactive interfaces.
- Tkinter is easy to use, platform-independent, and integrates well with other Python libraries.

2. Open CV Library:

- Open CV (Open Source Computer Vision Library) is a popular open-source library for computer vision and image processing tasks.
- It provides functions for image manipulation, object detection, face recognition, and more.
- Open CV supports multiple programming languages, including Python, C++, and Java.

3. MySQL for Database:

- MySQL is a widely used open-source relational database management system (RDBMS).
- It allows storing and managing structured data in tables with support for SQL queries.
- MySQL is known for its reliability, scalability, and performance, making it suitable for various applications, including web and enterprise systems.

4. Haar Cascade Algorithm:

- The Haar Cascade algorithm is used for object detection, including face detection, in images or video streams.
- It works by detecting features based on Haar-like features and cascading classifiers.
- Haar Cascade classifiers are trained to recognize specific patterns, such as faces, eyes, or smiles.

5. LBPH (Local Binary Patterns Histograms) Algorithm:

- LBPH is a face recognition algorithm used for identifying faces based on local texture patterns.
- It creates histograms of local binary patterns (LBP) in face images and uses them for recognition.
- LBPH is robust to variations in lighting conditions and facial expressions, making it suitable for real-world face recognition systems.

9.3 Methodologies/Algorithm Details

9.3.1 Haar Cascade Face Detection Algorithm

1. **Step 1:** Load the pre-trained Haar Cascade face detection model
2. **Step 2:** Initialize the webcam or image source for capturing frames.
3. **Step 3:** Repeat for each frame:
 - Read the frame from the webcam or image source.
 - Convert the frame to gray scale for efficient processing.
 - Use the Haar Cascade model to detect faces in the gray scale frame.
4. **Step 4:** For each detected face:
 - Draw a bounding box around the face on the original color frame.
 - Extract the face region from the frame for further processing.
5. **Step 5:** Display the processed frame with detected faces.
6. **Step 6:** If required, save or export the processed frames with bounding boxes.

9.3.2 LBPH Face Recognition Algorithm

1. **Step 1:** Load the pre-trained LBPH face recognition model.
2. **Step 2:** Initialize the database or data structure to store user profiles with face templates.
3. **Step 3:** Initialize the webcam or image source for capturing frames.
4. **Step 4:** Repeat for each frame:
 - Read the frame from the webcam or image source.
 - Convert the frame to grayscale for efficient processing.
 - Use the LBPH model to recognize faces in the grayscale frame.
5. **Step 5:** For each recognized face:
 - Compare the face template with stored user profiles in the database.
 - If a match is found with sufficient confidence:
 - (a) Mark attendance for the corresponding user.
 - (b) Display the user's name or ID on the frame.
6. **Step 6:** If no match is found or confidence is low:
 - Display "Unknown" or handle unrecognized faces based on system requirements.
7. **Step 7:** Display the processed frame with recognized faces and attendance status.
8. **Step 8:** If required, save or export attendance records and processed frames.

9.4 Verification and Validation for Acceptance

Verification focuses on ensuring that the software meets its specified requirements and that each component functions correctly

1. **Requirements Verification:**
 - Review and validate the initial requirements document to ensure they are clear, complete, and consistent.

- Verify that the system's functionalities, such as face detection, recognition, attendance marking, and database management, align with the specified requirements.

2. Code Review:

- Conduct code reviews to check for adherence to coding standards, readability, and maintainability.
- Perform static code analysis to identify potential bugs, code smells, and security vulnerabilities.

3. Unit Testing:

- Develop and execute unit tests for individual modules and components (e.g., face detection algorithm, database interactions) to verify their correctness.
- Use testing frameworks like unittest (Python) to automate and document test cases.

4. System Testing:

- Conduct system-level testing to evaluate the overall functionality, performance, and usability of the face recognition attendance system.
- Test boundary cases, error handling, and system resilience under varying conditions..

5. User Acceptance Testing :

- Collaborate with stakeholders, including end users, to define acceptance criteria and scenarios for UAT.
- Collaborate with stakeholders, including end users, to define acceptance criteria and scenarios for UAT.

CHAPTER 10

TESTING

10.1 Type of Testing Used

10.1.1 Unit Testing:

Unit Testing is a type of software testing where individual units or components of software are tested. The purpose is to validate that each unit of the software code performs as expected. Unit Testing is done during the development (coding phase) of an application by the developers. Unit testing is used to check the execution path of the module, function, and procedure of the system. The test is conducted with the help of normal data and abnormal data. This testing includes different factors like statement coverage, branch coverage, loop processing, abnormality, circulation, etc. With the help of this Unit testing system check that all the statement in the code is executed or not so it avoids the dead code statement. It checks all the branches and execution paths of the code. It ensures that all the internal methods of the program are executed and properly integrated with the program.

10.1.2 Integration Testing

In integrated testing, all the modules are checked together to ensure that all the modules are executing together according to the program specification. Once all the modules have been tested individually, the most legitimate question that can be asked is when all the modules are working properly, and why there is a need for integrated testing. The answer is, though all modules are working properly problems may occur while interfacing individual modules. Data can lose across an interface one module can have an adverse on another.

10.1.3 System Testing:

System testing for a face recognition attendance system involves evaluating the system as a whole to ensure it meets the specified requirements, functions correctly, and delivers the expected outcomes.

- **Functional Testing:** Verify that the system's functions, such as face detection, recognition, attendance marking, and database operations, work as intended.
- **Non-functional Testing:** Evaluate the system's non-functional aspects, including performance, security, usability, reliability, and compliance.

10.2 Test Cases and Test Results

TC ID	TC Name	Description	Expected	Actual	Result
TC 01	Open Application	Open application on PC.	Application should get opened.	Application is getting opened	Pass
TC 02	Registration page	Click on Registration page	Application should show Registration page	Application is showing Registration form to fill.	Pass
TC 03	Registration verification	Verify the registration Details of user	Application should show Successful registration of user	Application is showing Successful registration of user.	Pass
TC 04	Login with valid credentials	Enter a valid username and password. Click the login button.	User is successfully logged in and redirected to the appropriate page	User is successfully logged in and redirected to the appropriate page	Pass
TC 05	Student or Client Details Saving	Enter valid student/client details (e.g., name, ID, contact information). Save the details to the database.	Student/client details should be stored accurately in the database with unique identifiers.	Details saved successfully, and unique IDs assigned in the database	Pass
TC 06	Face Sample Collection	Select a client from the database to collect face samples. Capture multiple face	Face samples should be captured and associated with the respective student	Face samples captured successfully and linked to the correct student/client IDs	Pass

Table 10.1: Test Cases

TC ID	TC Name	Description	Expected	Actual	Result
TC 07	Training with Haar Cascade Algorithm	Trigger training using the Haar Cascade algorithm for a selected student/client. Verify training completion and accuracy metrics	Training should complete without errors, and accuracy metrics should indicate successful learning	Training completed successfully, and accuracy metrics meet expected standards	Pass
TC 08	Training with LBPH Algorithm	Initiate training using the LBPH algorithm for a selected student/client. Validate training results and accuracy.	LBPH training should yield accurate recognition results and meet defined accuracy thresholds	LBPH training successful, and recognition accuracy meets expectations.	Pass
TC 09	Face Recognition and Attendance Marking	Present a registered student/client face for recognition. Verify if the system correctly recognizes the face and marks attendance.	System should accurately recognize faces and mark attendance for recognized individuals	Face recognition successful, and attendance marked for recognized individuals	Pass
TC 10	Attendance Logging - Excel Output	Mark attendance for multiple individuals with recognized faces. Check the generated Excel attendance log for accuracy	Excel log should accurately reflect the attendance marked by the system..	Excel log generated accurately, reflecting marked attendance	Pass

Table 10.2: Test Cases

CHAPTER 11

RESULT

11.1 Screen shots

11.1.1 Login Page

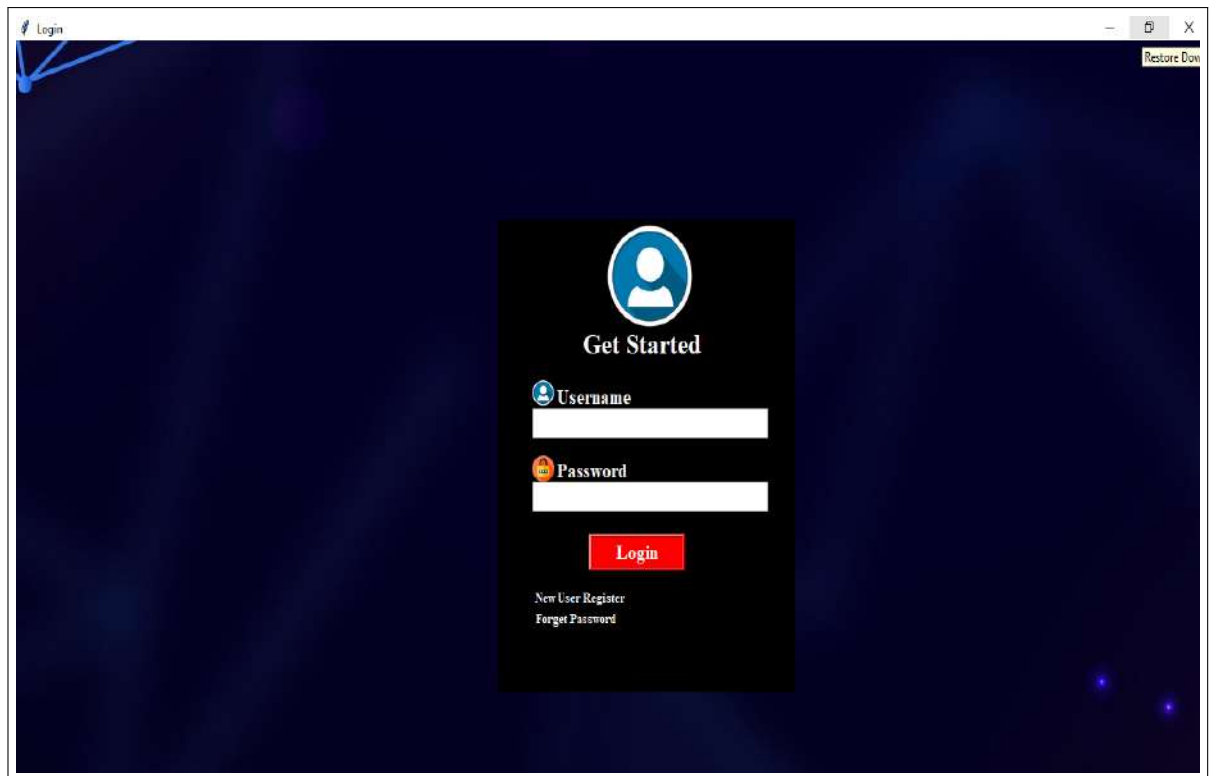
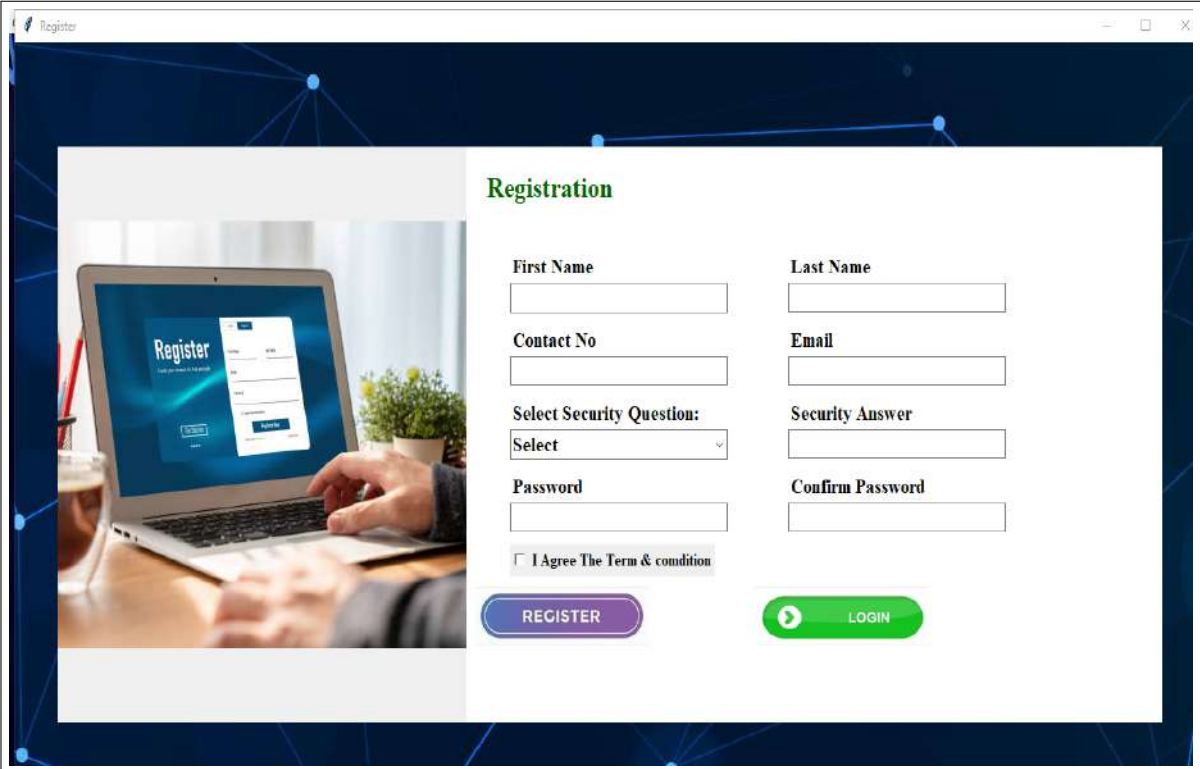


Figure 11.1: Login Page

1. Login This is the title at the top of the page.
2. Username This is a field where the user enters their username.
3. Password This is a field where the user enters their password.
4. Login This is a button that the user clicks to log in.
5. New User Register This is a link that the user clicks to create a new account.
6. Forget Password This is a link that the user clicks if they forget their password.

11.1.2 Registration page

1. First Name This is a field where the user enters their first name.
 2. Last Name This is a field where the user enters their last name.
 3. Contact No This is a field where the user enters their contact number.
 4. Email This is a field where the user enters their email address.
 5. Select Security Question This is a dropdown menu where the user selects a security question.
 6. Security Answer This is a field where the user enters their answer to the security question.
- Password This is a field where the user enters their password.



Registration

First Name

Last Name

Contact No

Email

Select Security Question:

Security Answer

Password

Confirm Password

☐ I Agree The Term & condition

REGISTER **LOGIN**

Figure 11.2: Registration page

11.1.3 Home Page

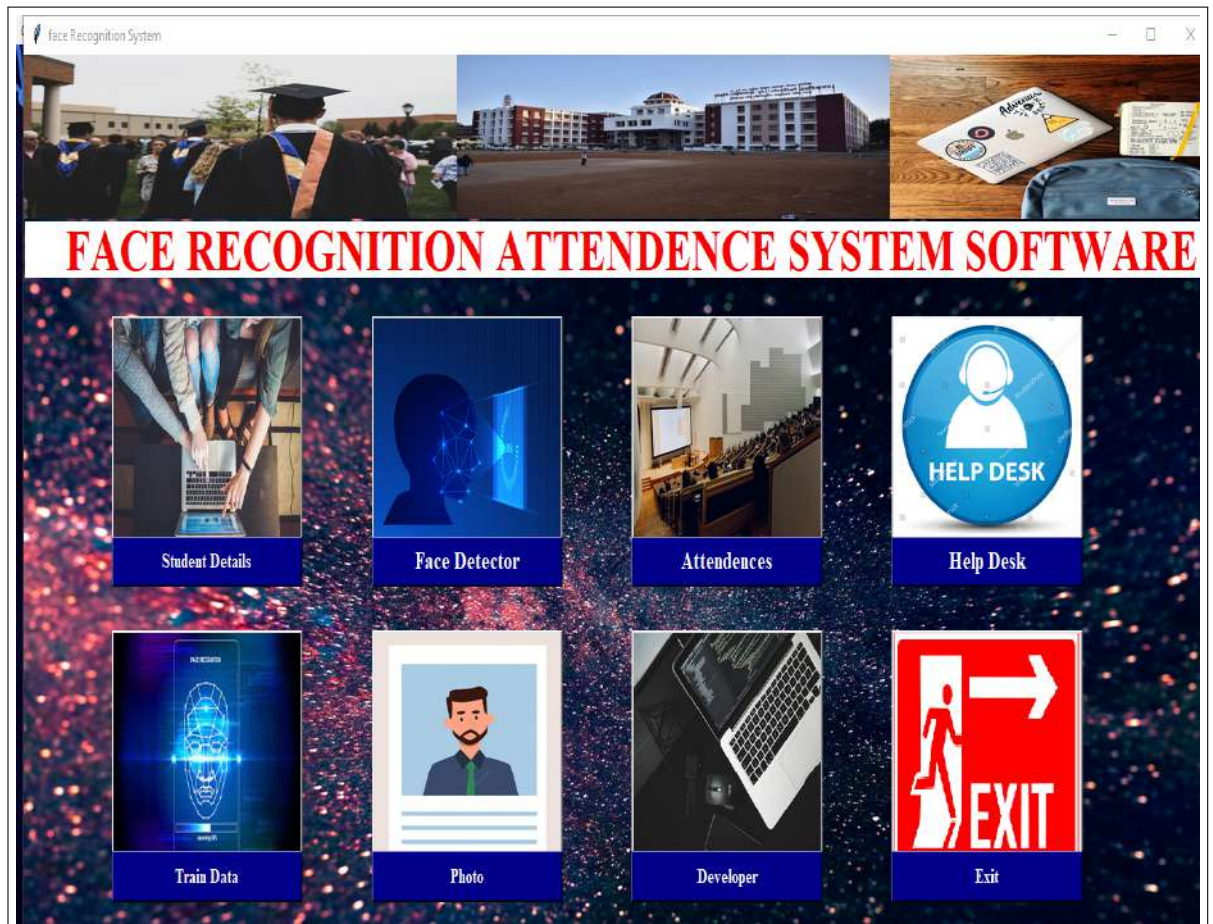


Figure 11.3: Home Page

11.1.4 Student

Student Information Filled and Saved in Database

Face Recognition System

STUDENT MANAGEMENT SYSTEM

Student Detail

Student Detail

Department
Select Department
Course
Select Course

Year
Select Yaer
Semester
Select Semester

Class Student information

StudentID

Student Name

Class Division

Roll No

Gender

DOB

Email

Phone No

Address

Teacher Name

☐ take Photo sample
☐ No Photo Sample

Save

Update

Delete

Reset

Take Photo Sample

Student Detail

Search System

Search By

Select

Search

Show All

Department	Course	year	Semester	StudentId	Name	
Computer	BE	2021-22	Semester-1	1	sujit	A
IT	TE	2022-23	Semester-2	2	Dhoni	A
IT	BE	2021-22	Semester-1	3	Rohit	A
Civil	SE	2022-23	Select Semester	4	rmodi	A
Civil	TE	Select Yaer	Select Semester	5	selman	A
Computer	BE	2023-24	Semester-2	6	om jain	A
Computer	SE	2021-22	Semester-1	7	akshay	A

Figure 11.4: Home Page

11.1.5 Attendance Record

The primary purpose of an attendance record is to keep track of when individuals are present or absent at a specified location or event. It helps in monitoring attendance patterns, identifying trends, and ensuring compliance with attendance requirements.

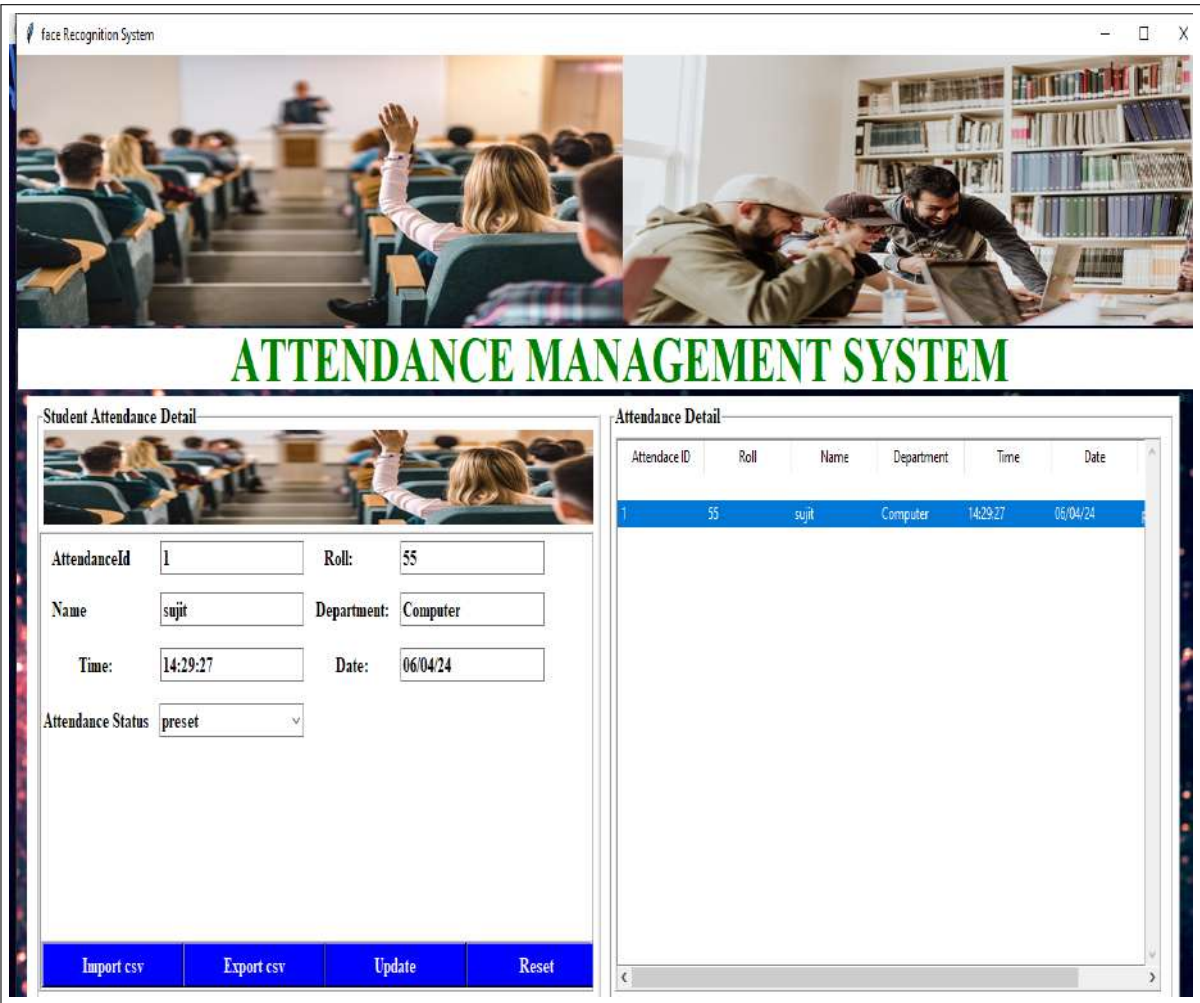


Figure 11.5: Attendance Record

11.2 Outputs

11.2.1 Student Detail saved

Saved Student Detail in Database

Student Detail



Search System

Search By

Select

Search

Show All

Department	Course	year	Semester	StudentId	Name	
Computer	BE	2021-22	Semester-1	1	sujit	A
IT	TE	2022-23	Semester-2	2	Dhoni	A
IT	BE	2021-22	Semester-1	3	Rohit	A
Civil	SE	2022-23	Select Semester	4	modi	A
Civil	TE	Select Yaer	Select Semester	5	salman	A
Computer	BE	2023-24	Semester-2	6	om jain	A
Computer	SE	2021-22	Semester-1	7	akshay	A

Figure 11.6: Student Detail saved

11.2.2 Collect Photo Sample

To collect photo samples for a Face Recognition Attendance System (FRAS),



Figure 11.7: Collect Photo Sample

11.2.3 Face Detection for Attendance

the importance of accurate face detection in ensuring the reliability and efficiency of attendance tracking. he methodology used for face detection, including the algorithms and techniques employed.

- Haar Cascade Algorithm: Explain how Haar Cascade is used for object detection, specifically faces, by analyzing features such as edges, lines, and textures.

- Local Binary Patterns Histograms (LBPH) Algorithm: Describe the LBPH algorithm's approach to recognizing faces based on local texture patterns.

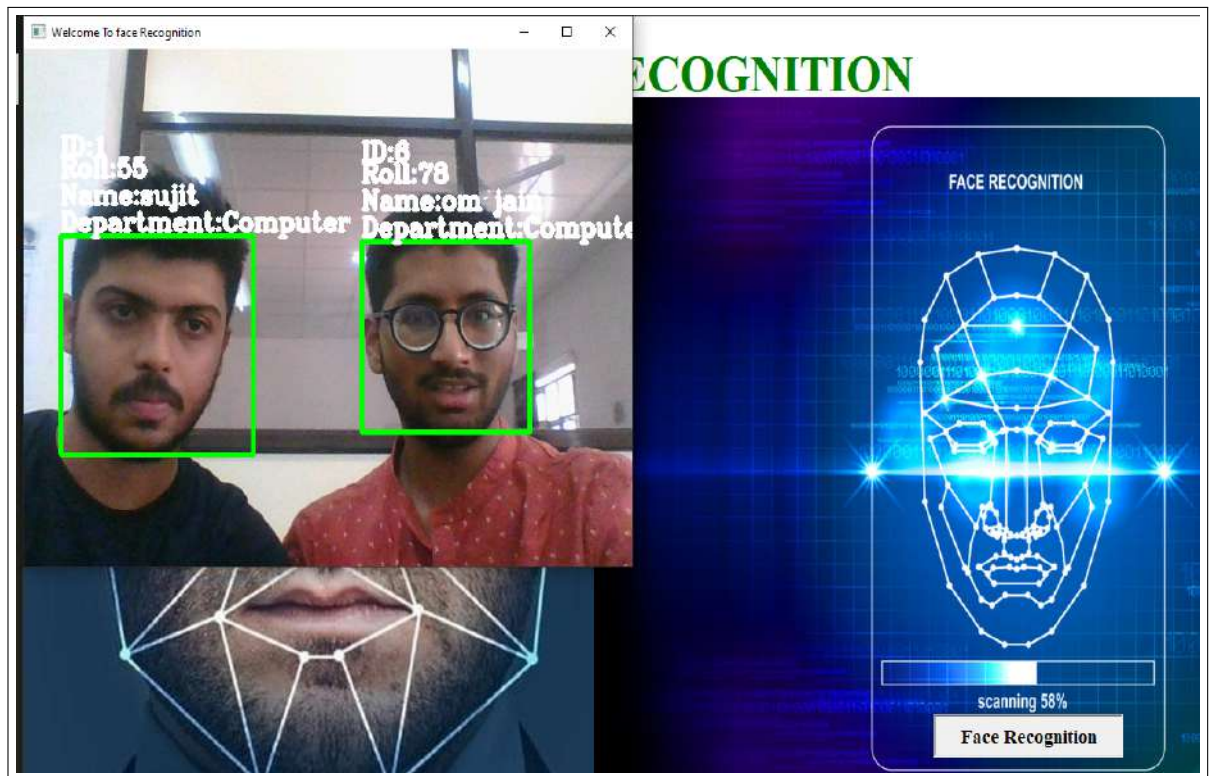


Figure 11.8: Face Detection for Attendance

11.2.4 Attendance record

Once a face is detected, the system attempts to match it against a pre-existing database of authorized individuals. This database typically contains student IDs, employee IDs, or other unique identifiers linked to facial features

CHAPTER 12

**DEPLOYMENT AND
MAINTENANCE**

12.1 Installation and un-installation

This system utilizes Python with Tkinter for the GUI, Haar cascade and LBPH algorithms for face recognition, and MySQL for database storage. Here's a breakdown of installation and uninstallation

12.1.1 Installation

- Python: Ensure you have Python 3.x installed. Download it from the official website if needed: <https://www.python.org/downloads/>.
- Libraries: Open a terminal or command prompt and use pip to install the required libraries:
`pip install opencv-python numpy pillow mysql-connector-python tk`
 1. OpenCV-python: Provides computer vision functionalities.
 2. NumPy: Enables numerical computations.
 3. Pillow (PIL Fork): Used for image processing.
 4. mysql-connector-python: Connects Python to MySQL databases.
 5. Tk: The core Tkinter library for building GUIs.
- Haar Cascade Classifier: Download a pre-trained Haar cascade classifier for frontal face detection from OpenCV's GitHub repository: <https://github.com/opencv-java/facedetection> Place it in your project directory.
- MySQL Database:
Set up a MySQL server if you don't have one already.
Create a database for your attendance system using tools like MySQL Workbench or command-line tools.
Design tables to store student information (ID, Name, etc.) and attendance records (Date, Time, Student ID, etc.).

12.1.2 un-installation

- Python Packages: want to uninstall Python packages used in your FRAS project, you can use pip.
`pip uninstall numpy opencv-python mysql-connector-python tkinter`

- **Remove MySQL Database:** Drop or delete the MySQL database and associated tables created for your FRAS application. Use MySQL commands to drop the database:
DROP DATABASE yourdatabasename;
- **Remove face recognition attendance system Project:** Delete the face recognition attendance system project directory from your system if you no longer need it.
- **Clean Up:** Optionally, you can clean up any additional files or directories created during the installation and usage of your FRAS application.

12.2 User help

- **Logging In:**
Start the application and log in using your credentials. Ensure that you have the necessary permissions to access attendance data.
- **Capture Your Face**
 1. When prompted, position yourself in front of the camera.
 2. Ensure good lighting and avoid glare or shadows on your face.
 3. Look directly at the camera for the best recognition results.
- **Face Recognition:**
 1. The system will analyze your facial features using algorithms like Haar Cascade or LBPH.
 2. Wait for the system to recognize and verify your face.
- **Attendance Marking:**
 1. Once your face is successfully recognized, the system will mark your attendance automatically.
 2. You may receive a confirmation message or see your attendance status on the screen.
- **Viewing Attendance Records:**
You can usually view your attendance records within the application.
- **Feedback and Support:**
Provide feedback about your experience with the system to help improve its functionality.

CHAPTER 13

**CONCLUSION AND FUTURE
SCOPE**

13.1 Conclusion

Face recognition attendance systems are a modern and efficient way to track attendance in a variety of settings. They offer a number of advantages over traditional attendance systems, such as accuracy, convenience, security, automation, and integration. Face recognition attendance systems are becoming increasingly popular as the technology becomes more accurate and affordable. These systems offer a number of advantages over traditional attendance systems, making them a valuable tool for organizations of all sizes. While face recognition attendance systems do have some limitations, such as accuracy bias, privacy concerns, cost, and complexity, the benefits outweigh the risks for many organization

13.2 Future Work

- **Improving Accuracy and Robustness**

Enhance the system's accuracy by refining the facial recognition algorithms, especially in challenging conditions such as low lighting or varying angles.

Explore the integration of advanced machine learning techniques, possibly leveraging deep learning architectures, to continually improve the system's ability to recognize faces accurately

- **Integration with Other Technologies**

Explore integration with other bio-metric technologies to create multi-modal recognition

systems for heightened security and accuracy. Consider incorporating IoT (Internet of Things) devices to create a more interconnected attendance management system, allowing for a seamless flow of information

- **Mobile and Cloud Integration**

Develop mobile applications that allow for remote monitoring and management of attendance records.

Investigate cloud-based solutions to store and process attendance data, enabling easy access, scalability, and data analytics capabilities

- **User Experience and Accessibility** Focus on improving the user interface to make the system more user-friendly for both administrators and end-users

Ensure accessibility features to accommodate individuals with diverse needs, such as people with disabilities or those with cultural differences

in facial features.

- **Scalability and Customization**

Design the system to be easily salable to accommodate organizations of different sizes and structures.

Provide customization options to meet the specific needs and preferences of different users and organizations

- **Continuous Training and Updates**

Implement a system for continuous training of the facial recognition algorithms to adapt to changes in facial appearances over time.

Regularly update the software to patch security vulnerabilities and integrate the latest advancements in facial recognition technology.

CHAPTER 14

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ANNEXURE A

**LABORATORY ASSIGNMENTS ON
PROJECT ANALYSIS OF
ALGORITHMIC DESIGN**

1. System Architecture Overview

- **Objective:**

Understand the overall architecture of the face recognition attendance system

- **Tasks:**

- (a) Study the system's architectural diagram provided in the project documentation.
- (b) Identify and describe the key components of the system, such as the user interface, database, face recognition modules, and attendance logging mechanisms.
- (c) Analyze the interactions between these components and their roles in the system's functionality.

2. Algorithm Selection and Analysis

- **Objective:**

Evaluate the Haar Cascade and LBPH algorithms used in the face recognition system.

- **Tasks:**

- (a) Research and summarize the principles and working mechanisms of the Haar Cascade algorithm.
- (b) Research and summarize the principles and working mechanisms of the LBPH (Local Binary Patterns Histograms) algorithm.
- (c) Compare and contrast the strengths and weaknesses of these algorithms in the context of face recognition accuracy, computational efficiency, and robustness to variations.

3. Database Design and Implementation

- **Objective:**

Examine the database design and implementation for storing student/client details and attendance records.

- **Tasks:**

- (a) Study the ERD (Entity-Relationship Diagram) provided for the database schema.
- (b) Analyze the tables and their relationships for storing user credentials, student/client details, and attendance records.

- (c) Evaluate the database's performance and scalability considerations for handling large datasets.

4. User Interface Evaluation

- **Objective:**
Assess the user interface design and usability of the face recognition attendance system.
- **Tasks:**
 - (a) Use the system's user interface to perform common tasks such as user registration, face sample collection, and attendance marking.
 - (b) Evaluate the interface's intuitiveness, responsiveness, and overall user experience.
 - (c) Propose any recommendations for improving the user interface based on your evaluation

I	D	E	A
Improve Attendance accuracy	User engagement through mobile app	Staff on best practices	System response time through optimization
Deliver Real-time attendance tracking	Enhanced user experience	Training sessions for employees	Automated reporting tools
Evaluate	Data analytics for attendance patterns	User feedback on system usability	Training effectiveness metrics
Ignore Unreliable face recognition algorithms	Complex manual attendance processes	Eliminate Outdated technology solutions	Avoid Non-compliance with data privacy regulations

Table A.1: IDEA Matrix

5. System Integration and Testing

- **Objective:**
Test the integration of system components and analyze the testing approach.
- **Tasks:**

- (a) Conduct integration testing by simulating user interactions across different modules (e.g., registration, face recognition, attendance marking).
- (b) Develop test cases for critical system functionalities, including error handling, data validation, and performance under load.
- (c) Analyze the testing results and propose any enhancements or fixes based on identified issues.

ANNEXURE B

**LABORATORY ASSIGNMENTS ON
PROJECT QUALITY AND
RELIABILITY TESTING OF
PROJECT DESIGN**

TC ID	TC Name	Description	Expected	Actual	Result
TC 01	Open Application	Open application on PC.	Application should get opened.	Application is getting opened	Pass
TC 02	Registration page	Click on Registration page	Application should show Registration page	Application is showing Registration form to fill.	Pass
TC 03	Registration verification	Verify the registration Details of user	Application should show Successful registration of user	Application is showing Successful registration of user.	Pass
TC 04	Login with valid credentials	Enter a valid username and password. Click the login button.	User is successfully logged in and redirected to the appropriate page	User is successfully logged in and redirected to the appropriate page	Pass
TC 05	Student or Client Details Saving	Enter valid student/client details (e.g., name, ID, contact information). Save the details to the database.	Student/client details should be stored accurately in the database with unique identifiers.	Details saved successfully, and unique IDs assigned in the database	Pass
TC 06	Face Sample Collection	Select a client from the database to collect face samples. Capture multiple face	Face samples should be captured and associated with the respective student	Face samples captured successfully and linked to the correct student/client IDs	Pass

Table B.1: Test Cases

TC ID	TC Name	Description	Expected	Actual	Result
TC 07	Training with Haar Cascade Algorithm	Trigger training using the Haar Cascade algorithm for a selected student/client. Verify training completion and accuracy metrics	Training should complete without errors, and accuracy metrics should indicate successful learning	Training completed successfully, and accuracy metrics meet expected standards	Pass
TC 08	Training with LBPH Algorithm	Initiate training using the LBPH algorithm for a selected student/client. Validate training results and accuracy.	LBPH training should yield accurate recognition results and meet defined accuracy thresholds	LBPH training successful, and recognition accuracy meets expectations.	Pass
TC 09	Face Recognition and Attendance Marking	Present a registered student/client face for recognition. Verify if the system correctly recognizes the face and marks attendance.	System should accurately recognize faces and mark attendance for recognized individuals	Face recognition successful, and attendance marked for recognized individuals	Pass
TC 10	Attendance Logging - Excel Output	Mark attendance for multiple individuals with recognized faces. Check the generated Excel attendance log for accuracy	Excel log should accurately reflect the attendance marked by the system..	Excel log generated accurately, reflecting marked attendance	Pass

Table B.2: Test Cases

ANNEXURE C
PROJECT PLANNER

Sr no	Lesion Topic	Time Allotted	Start date	End date
1.	Domain Selection	7 Days	24-Jul-2023	30-Jul-2023
2.	Paper Selection	7 Days	31-Jul-2023	6-Aug-2023
3.	Requirement Gathering	8 Days	7-Aug-2023	14-Aug-2023
4.	Literature Survey	7 Days	15-Aug-2023	21-Aug-2023
5.	Existing Architecture	9 Days	22-Aug-2023	30-Aug-2023
6.	Propose Architecture	12 Days	31-Aug-2023	11-Sep-2023
7.	Existing Algorithm	10 Days	12-Sep-2023	21-Sep-2023
8.	Propose Algorithm	11 Days	22-Sep-2023	2-Oct-2023
9.	Mathematical Model	9 Days	3-Oct-2023	11-Oct-2023
10.	Analysis	7 Days	12-Oct-2023	18-Oct-2023
11.	Planning	9 Days	19-Oct-2023	27-Oct-2023
12.	Design	12 Days	30-Oct-2023	10-Nov-2023
13.	User model	12 Days	13-Nov-2023	24-Nov-2023
14.	Server Model	9 Days	25-Nov-2023	3-Dec-2023
15.	Database model	19 Days	4-Dec-2023	22-Dec-2023
16.	Coding	60 Days	23-Dec-2023	20-Feb-2024
17.	Testing Integration	20 Days	21-Feb-2024	11-Mar-2024
18	Testing	18 Days	12-Mar-2024	28-Mar-2024

Figure C.1: Plan Timeline

ANNEXURE D

REVIEWERS COMMENTS OF

PAPER SUBMITTED

1. Paper Title: Survey Paper on Face Recognition Attendance System
2. Name of the Conference/Journal where paper submitted :2024 International Journal of Scientific Research in Engineering and Management (IJSREM)
3. Paper accepted/rejected : Accepted
4. Review comments by reviewer : N/A
5. Corrective actions if any : N/A

Survey paper on Face Recognition Based Attendance System

Sujit Yadav¹, Om Jain², Vivek Sangle³, Arjala Patil⁴, Prof. B. M. Gawale⁵

Abstract - The integration of face recognition technology into attendance management systems has emerged as a transformative solution for monitoring and optimizing attendance processes in educational institutions and organizations. This report presents an in-depth analysis of a project dedicated to the development and implementation of a Face Recognition Attendance System (FRAS) and examines its efficiency and security in various real-world scenarios. The report begins by discussing the significance of attendance tracking systems in educational and corporate environments, highlighting the shortcomings of traditional methods and the potential benefits of face recognition technology. It explores the underlying principles of face recognition, focusing on the algorithms and techniques used to capture and process facial data. The project methodology and implementation details are outlined, covering the hardware and software components used to create the FRAS. Special emphasis is placed on the challenges encountered during the system's development and the strategies employed to overcome them. A comprehensive evaluation of the system's performance and efficiency is presented, including accuracy, speed, and scalability, with comparisons to traditional attendance systems. Moreover, the report addresses security concerns associated with the FRAS, such as data protection, privacy, and vulnerability to spoofing. The report concludes with a discussion of the practical implications and future prospects of implementing face recognition technology in attendance systems. It highlights the system's potential to streamline administrative tasks, reduce errors, and enhance security, while also emphasizing the need for ongoing research and development to address emerging challenges. Overall, this report provides a valuable insight into the evolving landscape of attendance management and the role that face recognition technology can play in revolutionizing these processes, balancing efficiency with security considerations.

1. INTRODUCTION:

1.1 MOTIVATION:

A face recognition attendance system offers several advantages and motivations for its implementation in various settings. Here are some key motivations for using face recognition technology in attendance systems:

Efficiency and Accuracy:

Time-saving: Face recognition systems can quickly and accurately identify individuals, reducing the time it takes to record attendance compared to traditional methods like manual entry or card scanning.

Reduced errors: Automated systems are less prone to errors associated with manual data entry, leading to more accurate attendance records.

Security:

Prevention of proxy attendance: Face recognition helps prevent instances of proxy attendance, where someone else might attempt to clock in on behalf of another individual.

Enhanced security: Facial features are unique to individuals, making it difficult for unauthorized individuals to gain access.

Contactless Technology:

Hygiene and health considerations: Especially in the context of global health concerns, contactless systems reduce the risk of spreading germs compared to traditional methods involving physical touch or proximity.

Automation and Integration:

Integration with other systems: Face recognition attendance systems can be seamlessly integrated with



other software and databases, streamlining administrative processes and providing real-time data.

Automation of attendance tracking: Eliminates the need for manual data entry and paperwork, reducing administrative burden and potential errors.

Cost-Effective:

Reduced administrative costs: Automation of attendance tracking can lead to cost savings by reducing the need for manual record-keeping and associated administrative tasks.

Scalability:

Suitable for large groups: Face recognition attendance systems are scalable and can efficiently handle large groups of people, making them suitable for various settings, including educational institutions, workplaces, and events.

Technological Advancements:

Advancements in facial recognition technology: Continuous improvements in facial recognition algorithms and hardware contribute to enhanced accuracy and reliability over time.

User Convenience:

User-friendly: Face recognition systems are generally user-friendly, requiring minimal effort from individuals to register their attendance.

No need for physical tokens: Eliminates the need for physical ID cards or keycards, providing a more convenient experience for users.

1.2 Problem Definition:

Traditional attendance tracking methods in educational institutions, such as manual sign-in sheets and roll call, are inefficient, inaccurate, and prone to fraud. These methods can waste time, distract from instruction, and make it difficult to track student progress accurately.

Face recognition attendance systems (FRAS) offer a number of advantages over traditional attendance

tracking methods. FRAS are accurate, efficient, and convenient. They can also help to improve security and reduce cheating. However, FRAS are a relatively new technology, and there are still some challenges associated with their implementation in educational institutions, including

Accuracy and Reliability:

The face recognition algorithm must be accurate and reliable, ensuring minimal false positives and false negatives.

The system should be able to handle variations in lighting conditions, facial expressions, and minor changes in appearance over time.

Real-time Processing:

The system should operate in real-time, providing quick and efficient recognition without significant delays

It must be able to process a large number of faces in a short time, making it suitable for attendance tracking in various scenarios such as classrooms, offices, or events.

User-Friendly Interface:

The system should have an intuitive and user-friendly interface for both administrators and end-users.

Administrators should be able to easily manage the system, add or remove users, and view attendance records.

End-users (individuals being recognized) should experience a seamless and non-intrusive attendance process.

Security and Privacy:

The system must prioritize the security and privacy of the stored facial data and attendance records.

It should comply with relevant privacy regulations and standards to ensure the protection of personal information.

Scalability:

The system should be designed to scale, accommodating a growing number of users and devices.

It should be easily deployable in various environments, ranging from small classrooms to large organizations.

Integration with Existing Systems:

The system should be designed to integrate with existing attendance management systems or databases.

Compatibility with different hardware and software configurations should be considered for seamless implementation.

Reporting and Analytics:

Provide a reporting and analytics feature that allows administrators to generate attendance reports, track trends, and analyze attendance data over time.

2. LITERATURE SURVEY:**Fast Fractal Image Encoding Using an Improved Search Scheme:**

As fractal image encoding algorithms can yield high-resolution reconstructed images at very high Compression ratio, and therefore, have a great potential for improving the efficiency of image storage and image transmission. However, the baseline fractal encoding algorithm requires a great deal of time to complete the best matching search between the range and domain blocks, which greatly limits practical applications of the algorithm. In order to solve this problem, a necessary condition of the best matching search based on an image feature is proposed in this paper. The proposed method can reduce the search space significantly and excludes the most inappropriate domain blocks for each range block before carrying out the best matching search. Experimental results show that the proposed algorithm can produce good quality reconstructed images and requires much less time than the baseline encoding algorithm. Specifically, the new algorithm can speed up encoding by about 85 times with a loss of just 3 dB in the peak signal to noise ratio (PSNR), and yields compression ratios close to 34.

Smart Image Sensor with Integrated Low Complexity Image Processing for Wireless Endoscope Capsules

Smart image sensor was developed which integrates a digital pixel image sensor array with an image processor, designed for wireless endoscope capsules. The camera-on-a-chip architecture and its on-chip functionality facilitate the design of the packaging and power consumption of the integrated capsule. The power reduction techniques were carried out at both the architectural and circuit level. Gray coding and power gating in the sensor array to eliminate almost 50% of the switch activity on the data bus and more than 99% of the power dissipation in each pixel at a transmitting rate of 2 frames per second. Filtering and compression in the processor reduces the data transmission by more than 2/3. A parallel fully pipelined architecture with a dedicated clock management scheme was implemented in the JPEG-LS engine to reduce the power consumption by 15.7%. The smart sensor has been implemented in 0.18 μm CMOS technology.

Content Based Image Retrieval using Color and Texture

The increased need of content based image retrieval technique can be found in a number of different domains such as Data Mining, Education, Medical the content based image retrieval, using features like texture and color, called WBCHIR (Wavelet Based Color Histogram Image Retrieval). The texture and color features are extracted through wavelet transformation and color histogram and the combination of these features is robust to scaling and translation of objects in an image. The proposed system has demonstrated a promising and faster retrieval method on a WANG image database containing 1000 general-purpose color images. The performance has been evaluated by comparing with the existing systems in the literature.

Content Based Image Retrieval Methods Using Graphical Image Retrieval Algorithm

This document gives a brief description of a system developed for retrieving images similar to a query image from a large set of distinct images. It follows an image segmentation based approach to extract the different features present in an image. These features are stored in vectors called feature vectors and compared to the feature vectors of query image and

thus, the image database is sorted in decreasing order of similarity. Different from traditional dimensionality reduction algorithms such as Principal Component Analysis (PCA) and Linear Discriminate Analysis (LDA), which effectively see only the global Euclidean structure, GIRA is designed for discovering the local Manifold structure. Therefore, GIRA is likely to be more suitable for image retrieval, where nearest neighbour search is usually involved. After projecting the images into a lower dimensional subspace, the relevant images get closer to the query image; thus, the retrieval performance can be enhanced.

A Novel Fractal Image Compression Scheme with Block Classification and Sorting Based on Pearson's Correlation Coefficient

Fractal image compression (FIC) is an image coding Technology based on the local similarity of image structure. It is widely used in many fields such as image retrieval, image denoising, image authentication, and encryption. FIC, however, suffers from the high computational complexity in encoding. Although many schemes are published to speed up encoding, they do not easily satisfy the encoding time or the reconstructed image quality requirements. In this paper, a new FIC scheme is proposed based on the fact that the affine similarity between two blocks in FIC is equivalent to the absolute value of Pearson's correlation coefficient (APCC) between them. First, all blocks in the range and domain pools are chosen and classified using an APCC-based block classification method to increase the matching probability. Second, by sorting the domain blocks with respect to APCCs between these domain blocks and a preset block in each class, the matching domain block for a range block can be searched in the selected domain set in which these APCCs are closer to APCC between the range block and the preset block. Experimental results show that the proposed scheme can significantly speed up the encoding process in FIC while preserving the reconstructed image quality well.

A Nonlinear Mapping Approach to Stain Normalization in Digital Histopathology Images Using Image-Specific Color

Deconvolution

Histopathology diagnosis is based on visual examination of the morphology of histological sections under a microscope. With the increasing popularity of digital slide scanners, decision support systems based on the analysis of digital pathology images are in high demand. However, computerized decision support systems are fraught with problems that stem from color variations in tissue appearance due to variation in tissue preparation, variation in stain reactivity from different manufacturers/batches, user or protocol variation, and the use of scanners from different manufacturers. In this paper, we present a novel approach to stain normalization in histopathology images. The method is based on nonlinear mapping of a source image to a target image using a representation derived from color deconvolution. Color deconvolution is a method to obtain stain concentration values when the stain matrix, describing how the color is affected by the stain concentration, is given. Rather than relying on standard stain matrices, which may be inappropriate for a given image, we propose the use of a color-based classifier that incorporates a novel stain color descriptor to calculate image-specific stain matrix. In order to demonstrate the efficacy of the proposed stain matrix estimation and stain normalization methods, they are applied to the problem of tumor segmentation in breast histopathology images. The experimental results suggest that the paradigm of color normalization, as a pre-processing step, can significantly help histological image analysis algorithms to demonstrate stable performance which is insensitive to imaging conditions in general and scanner variations in particular.

CID2013: A Database for Evaluating No-Reference Image Quality Assessment Algorithms

This paper presents a new database, CID2013, to address the issue of using no-reference (NR) image quality assessment algorithms on images with multiple distortions. Current NR algorithms struggle to handle images with many concurrent distortion types, such as real photographic images captured by different digital cameras. The database consists of six image sets; on average, 30 subjects have evaluated 12–14 devices depicting eight

different scenes for a total of 79 different cameras, 480 images, and 188 subjects (67% female). The subjective evaluation method was a hybrid absolute category rating-pair comparison developed for the study and presented in this paper. This method utilizes a slideshow of all images within a scene to allow the test images to work as references to each other. In addition to mean opinion score value, the images are also rated using sharpness, graininess, lightness, and color saturation scales. The CID2013 database contains images used in the experiments with the full subjective data plus extensive background information from the subjects. The database is made freely available for the research community.

Confidentiality-Preserving Image Search: A Comparative Study Between Homomorphic Encryption and Distance-Preserving Randomization

Recent years have seen increasing popularity of storing and managing personal multimedia data using online services. Preserving confidentiality of online personal data while offering efficient functionalities thus becomes an important and pressing research issue. In this paper, we study the problem of content-based search of image data archived online while preserving content confidentiality. The problem has different settings from those typically considered in the secure computation literature, as it deals with data in rank-ordered search, and has a different security-efficiency requirement. Secure computation techniques, such as homomorphic encryption, can potentially be used in this application, at a cost of high computational and communication complexity. Alternatively, efficient techniques based on randomizing visual feature and search indexes have been proposed recently to enable similarity comparison between encrypted images. This paper focuses on comparing these two major paradigms of techniques, namely, homomorphic encryption based techniques and feature/index randomization-based techniques, for confidentiality-preserving image search. We develop novel and systematic metrics to quantitatively evaluate security strength in this unique type of data and applications. We compare these two paradigms of techniques in terms of their

search performance, security strength, and computational efficiency. The insights obtained through this paper and comparison will help design practical algorithms appropriate for privacy-aware cloud multimedia systems.

Real-Time RGB-D Camera Relocalization via Randomized Ferns for Key frame Encoding

Recovery from tracking failure is essential in any simultaneous localization and tracking system. In this context, we explore an efficient key frame-based relocalization method based on frame encoding using randomized ferns. The method enables automatic discovery of key frames through online harvesting in tracking mode, and fast retrieval of pose candidates in the case when tracking is lost. Frame encoding is achieved by applying simple binary feature tests which are stored in the nodes of an ensemble of randomized ferns. The concatenation of small block codes generated by each fern yields a global compact representation of camera frames. Based on those representations we define the frame dissimilarity as the block-wise hamming distance (BlockHD). Dissimilarities between an incoming query frame and a large set of key frames can be efficiently evaluated by simply traversing the nodes of the ferns and counting image co-occurrences in corresponding code tables. In tracking mode, those dissimilarities decide whether a frame/pose pair is considered as a novel key frame. For tracking recovery, poses of the most similar key frames are retrieved and used for initialization of the tracking algorithm. The integration of our relocalization method into a hand-held KinectFusion system allows seamless continuation of mapping even when tracking is frequently lost.

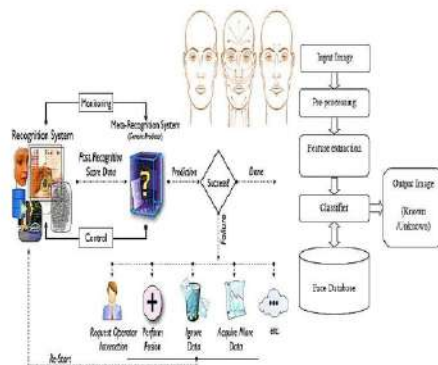
Multimedia Information Retrieval Based on Late Semantic Fusion Approaches: Experiments on a Wikipedia Image Collection

Main goal of this work is to show the improvement of using a textual pre-filtering combined with an image re-ranking in a Multimedia Information

Retrieval task. The defined three step based retrieval processes and a well- selected combination of visual and textual techniques help the developed Multimedia Information Retrieval System to overcome the *semantic gap* in a given query. In the paper, five different late semantic fusion approaches are discussed and experimented in a realistic scenario for multimedia retrieval like the one provided by the publicly available Image CLEF Wikipedia Collection.

3. SYSTEM ARCHITECTURE:

Face detection (using Haar cascade): The Haar cascade algorithm is applied to detect faces in the webcam images.



Store images in dataset: Detected faces are stored in a dataset for future reference.

Create YAML file: A YAML file is generated, containing information about the detected faces, including bounding boxes and coordinates.

Face recognition: The system compares the detected faces with those in the dataset to identify the person in the image.

Make box around face: A bounding box is drawn around the recognized face in the image.

Take multiple images: The system captures multiple

webcam images of the identified person.

Mark attendance: The attendance of the person is recorded as part of the system's functionality.

4. Conclusion:

Face recognition attendance systems are a modern and efficient way to track attendance in a variety of settings. They offer a number of advantages over traditional attendance systems, such as accuracy, convenience, security, automation, and integration.

Face recognition attendance systems are becoming increasingly popular as the technology becomes more accurate and affordable. These systems offer a number of advantages over traditional attendance systems, making them a valuable tool for organizations of all sizes. While face recognition attendance systems do have some limitations, such as accuracy bias, privacy concerns, cost, and complexity, the benefits outweigh the risks for many organizations.



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"Face Recognition Based Attendance System"

Sujit Yadav¹, Om Jain², Vivek Sangle³, Arjala Patil⁴, Prof. B. M. Gawale⁵

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Loknete Gopinathji Munde Institute of Engineering Education & Research, Nashik, India

Abstract :- The integration of face recognition technology into attendance management systems has emerged as a transformative solution for monitoring and optimizing attendance processes in educational institutions and organizations. This report presents an in-depth analysis of a project dedicated to the development and implementation of a Face Recognition Attendance System (FRAS) and examines its efficiency and security in various real-world scenarios. The report begins by discussing the significance of attendance tracking systems in educational and corporate environments, highlighting the shortcomings of traditional methods and the potential benefits of face recognition technology. It explores the underlying principles of face recognition, focusing on the algorithms and techniques used to capture and process facial data. The project methodology and implementation details are outlined, covering the hardware and software components used to create the FRAS. Special emphasis is placed on the challenges encountered during the system's development and the strategies employed to overcome them. A comprehensive evaluation of the system's performance and efficiency is presented, including accuracy, speed, and scalability, with comparisons to traditional attendance systems. Moreover, the report addresses security concerns associated with the FRAS, such as data protection, privacy, and vulnerability to spoofing. The report concludes with a discussion of the practical implications and future prospects of implementing face recognition technology in attendance systems. It highlights the system's potential to streamline administrative tasks, reduce errors, and enhance security, while also emphasizing the need for ongoing research and development to address emerging challenges. Overall, this report provides a valuable insight into the evolving landscape of attendance management and the role that face recognition technology can play in revolutionizing these processes, balancing efficiency with security considerations.

Key Words:

FRAS : Face recognition attendance systems
SDLC: Software Development Life Cycle

1. INTRODUCTION :- This report is dedicated to a project of paramount relevance—a comprehensive exploration of the "Face Recognition Attendance System" (FRAS), a groundbreaking technological advancement that has the potential to revolutionize the way attendance is managed across educational institutions, corporate settings, and beyond. Through this project, we aim to delve into the core principles, development, and practical implications of employing face

recognition technology to address the perennial challenges of attendance tracking.

In the ever-evolving landscape of education and workforce management, the traditional practice of recording attendance through manual methods, such as paper registers or cardbased systems, has proven to be increasingly inadequate. These age-old approaches are not only time-consuming but also susceptible to inaccuracies, proxy attendance, and other forms of manipulation. In an era characterized by technological advancements and an unyielding need for efficient, secure, and contact-less processes, it is imperative to explore innovative solutions to address these challenges.

This report is dedicated to a project of paramount relevance—a comprehensive exploration of the "Face Recognition Attendance System" (FRAS), a groundbreaking technological advancement that has the potential to revolutionize the way attendance is managed across educational institutions, corporate settings, and beyond. Through this project, we aim to delve into the core principles, development, and practical implications of employing face recognition technology to address the perennial challenges of attendance tracking.

2. Literature Survey :-

1.Face Recognition based Class Management and Attendance System-

proposed a system of using different algorithms like haar cascade along with another one, LBPH which is mostly utilize for the object and the image detection along with the record of the attendance through this system

2.A real time face recognition system based on improved LBPH algorithm-

The Local Binary Pattern Histogram (LBPH) algorithm is a simple and efficient face recognition algorithm. It works by converting the image of the face to a grayscale image and then dividing the image into a grid of small cells. For each cell, the algorithm calculates the LBPH value for the cell. The LBPH value is a histogram of the differences between the pixel values in the cell and the pixel values in the neighboring cells. The improved LBPH algorithm is a modified version of the LBPH

algorithm that is designed to improve the accuracy of face recognition in challenging conditions, such as low light and facial expressions. The improved LBPH algorithm uses a number of techniques to improve accuracy.

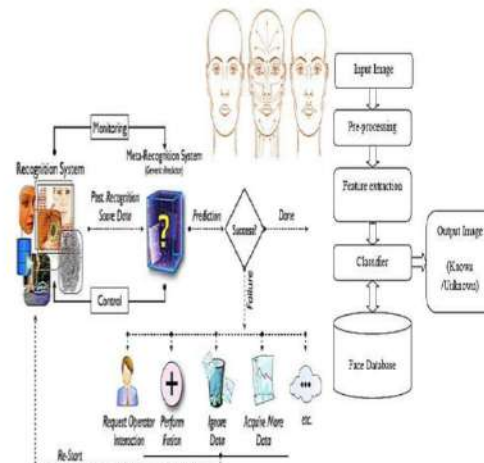
3. Real time eye state detection system using harscade classifier & circular hough transform similarity-

Real-time eye state detection systems are used to detect whether a person's eyes are open or closed. These systems have a wide range of applications, such as driver fatigue detection, human-computer interaction, and facial expression analysis. One approach to real-time eye state detection is to use a Haar Cascade classifier and a Circular Hough Transform. A Haar Cascade classifier is a machine learning algorithm that can be trained to detect specific features in images, such as faces and eyes. A Circular Hough Transform is an image processing algorithm that can be used to detect circles in images.

4. Face recognition for attendance recognition system-

It proposed an attendance system by face recognition using the Haar cascade algorithm. Face images are captured using a camera. To determine the location of the face it is performed by a haar algorithm for image detection. It basically draws a box around the faces of the person to whom we want to detect his face. Using different types of the algorithms they have done it. After they have stored the data in the cloud to determine the location of the face. Using Haar Cascade algorithm face detection is done. When the image of the face had been made it would now make or draw a box which is covering the whole face as an Region Of Interest (ROI). Extraction of image features was performed using the LBPH algorithm. Compare the detected faces with all the faces in the database to observe the one closest to the identified face. We have used CSV file format for storing the database. It is used to present the names and directories in the database format.

1. Architectural Design :-



The integration of face recognition technology into attendance management systems has emerged as a transformative solution for monitoring and optimizing attendance processes in educational institutions and organizations. This report presents an in-depth analysis of a project dedicated to the development and implementation of a Face Recognition Attendance System (FRAS) and examines its efficiency and security in various real-world scenarios. The report begins by discussing the significance of attendance tracking systems in educational and corporate environments, highlighting the shortcomings of traditional methods and the potential benefits of face recognition technology. It explores the underlying principles of face recognition, focusing on the algorithms and techniques used to capture and process facial data. The project methodology and implementation details are outlined, covering the hardware and software components used to create the FRAS. Special emphasis is placed on the challenges encountered during the system's development and the strategies employed to overcome them. A comprehensive evaluation of the system's performance and efficiency is presented, including accuracy, speed, and scalability, with comparisons to traditional attendance systems. Moreover, the report addresses security concerns associated with the FRAS, such as data protection, privacy, and vulnerability to spoofing. The report concludes with a discussion of the practical implications and future prospects of implementing face recognition technology in attendance systems. It highlights the system's potential to streamline administrative tasks, reduce errors, and enhance security, while also emphasizing the need for ongoing research and development to address emerging challenges. Overall, this report provides a valuable insight into the evolving landscape of attendance management and the role that face recognition

technology can play in revolutionizing these processes, balancing efficiency with security considerations.

2. Outcomes / discussions :-

Using 'confidence' we can see if algorithm recognizes the image correctly. We may deduce that the algorithm recognizes properly from confidence level. If the confidence level is too low, the image will be labelled as unknown and keep in unknown image folder.

Person	Confidence
Sujit	80%
Om	76%
Vivek	78%
Arjala	75%



3. Future Scope :-

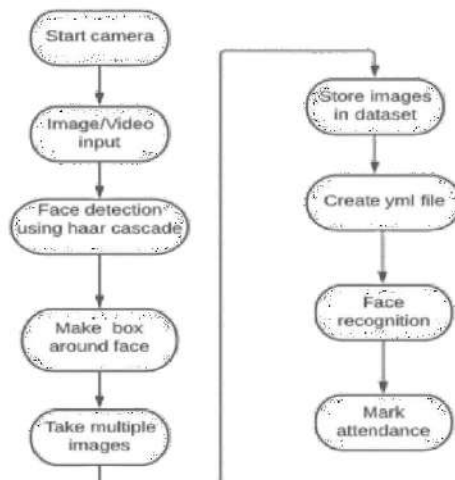
- Improving Accuracy & robustness**-Enhance the system's accuracy by refining the facial recognition algorithms, especially in challenging conditions such as low lighting or varying angles. Explore the integration of advanced machine learning techniques, possibly leveraging deep learning architectures, to continually improve the system's ability to recognize faces accurately.
- Real time monitoring and alert**- Develop real-time monitoring features to allow for immediate response to attendance discrepancies or unauthorized access. Implement alert systems that can notify administrators or security personnel in the case of anomalies, such as unrecognized faces or multiple registrations.

3. **Integration with another technology**- Implement real-time feedback mechanisms to gather continuous input from students on their learning experiences. Use data analytics to analyze feedback and make prompt improvements to the integrated portal system. Prioritize inclusive design principles to ensure that the integrated portal system is accessible to all students, including those with disabilities.
4. **Mobile & cloud Integration**- Develop mobile applications that allow for remote monitoring and management of attendance records. Investigate cloud-based solutions to store and process attendance data, enabling easy access, scalability, and data analytics capabilities.

4. Methodology :-

A. Flowchart:

Face Detection:



It finds and extract faces in images so that face recognition algorithm can use them. The Haar Cascade approach was used to recognise faces in image. This is an object detection technique which will detect face in images.

Face Recognition:

The algorithm of face recognition is important for discovering features that best describe the image once we have retrieved facial images, cropped recognize human faces is called Local Binary Pattern Histogram. It is well known for its performance and ability to distinguish a person's face from both front and side.

Function	Task Done
VideoCapture()	For starting the camera
cvtColor()	Input image is converted into a specified format.
detectMultiScale()	Detect different-sized items in the provided image.
Cv2.imwrite()	To save the images into dataset.
Cv2.face.LBPHFaceRecognizerCreate()	Used to load recognizer.
Import cv2	To import the OpenCV module.

1. Requirement gathering & analysis:

Identification of Stakeholder Needs: Involves identifying and gathering the needs of stakeholders, such as school administration, teachers, and students.

Requirements Analysis: Ensures completeness, consistency, and feasibility of gathered requirements.

2. System Design:

Architectural Design: The system architect designs the face recognition attendance system based on the gathered requirements.

Components Definition: Includes defining hardware and software components.

System Architecture: Specifies the overall system architecture.

Database Schema: Defines the structure of the database.

3. Implementation:

Code Development: The system is developed according to design specifications.

Configuration: Configures hardware and software components.

Integration: Integrates the system with the existing IT infrastructure.



4. Testing:

Thorough Testing: The system undergoes unit testing, integration testing, system testing, and acceptance testing. Implement mechanisms for continuous monitoring and evaluation of the portal system's performance. Use key performance indicators (KPIs) to assess the impact on student engagement, satisfaction, and academic outcomes.

5. Deployment:

Verification: Tested and verified system is deployed to the production environment.

Installation: Installs the software on target servers.

Configuration: Configures the network.

User Training: Provides training to users.

6. Maintenance & support:

Ongoing Maintenance: Provides ongoing maintenance for the system.

Defect Fixing: Addresses any defects found.

Feature Addition: Adds new features and functionality.

Technical Support: Provides technical support to users.

CONCLUSION :-

In conclusion, the implementation of face recognition attendance system offers a myraids of benefits that significantly streamlined attendance tracking process in various environments, ranging from educational institutes to corporate settings. By leveraging advanced facial recognition technology, such systems provides enhanced accuracy, efficiency and association compared to traditional methods. Face recognition attendance systems are a modern and efficient way to track attendance in a variety of settings. They offer a number of advantages over traditional attendance systems, such as accuracy, convenience, security, automation, and integration. Face recognition attendance systems are becoming increasingly popular as the technology becomes more accurate and affordable. These systems offer a number of advantages over traditional attendance systems, making them a valuable tool for organizations of all sizes. While face recognition attendance systems do have some limitations, such as accuracy bias, privacy concerns, cost, and complexity, the benefits outweigh the risks for many organizations.

ACKNOWLEDGMENT :-

We wish to express our deepest appreciation to our revered mentors and professors, with special gratitude towards Prof. G.M. Gawale, for their unwavering guidance in our academic pursuits and project endeavours. Our heartfelt thanks also extend to the dedicated team at the Computer Engineering Department for their consistent support. We are indebted to Dr. K. V. Chandratre, the Principal of Loknete Gopinathji Munde Institute of Engineering Education & Research, for his encouragement and authorization that enabled the successful completion of this project. The invaluable assistance provided by the support staff of our department is sincerely acknowledged. Our profound gratitude goes out to our parents, friends, and all those who stood by us throughout this project, offering their unwavering support and encouragement.

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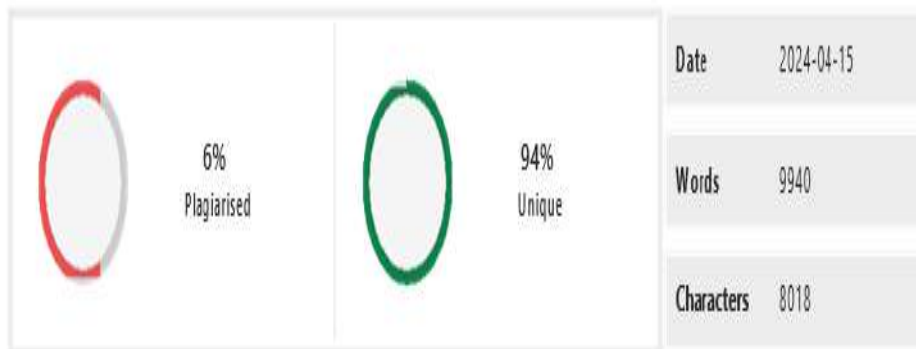

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ANNEXURE E
PLAGIARISM REPORT



PLAGIARISM SCAN REPORT



Content Checked For Plagiarism

ANNEXURE F
TERM-II PROJECT LABORATORY
ASSIGNMENTS



Krantiveer Vasantao Narayanrao Naik Shikshan Prasarak Sanstha

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Canada Corner, Sharanpur Road, Nashik 422 002
Approved by AICTE, Accredited 'B' Grade by NAAC



Computer Engineering Department

Academic Year: 2023-24

Class:BE

Subject: Project Lab

Sem :II

Assignment No: 01

Title: Project workstation selection and review design necessary corrective action taking into consideration the feedback report of term I assessment.

Due Date:

Date Submitted:

Guide Name: Prof. B.M.GAWALE

Guide Remark:.....

Student Name(Roll No.)	Performance	Understanding	Submission	Total
SUJIT KANHAIYALAL YADAV	3	3	2	8
OM RAJENDRA JAIN	3	3	2	8
VIVEK SUNIL SANGALE	3	2	2	7
ARJALA SAJENDRA PATIL	3	2	2	7

Project Coordinator: Prof. D. D. Sharma

Head of Department: Prof. R. M. Shaikh

1. Which algorithm you have used for implementation?

Ans: Haar Cascade Algorithm:

Detection: The Haar Cascade algorithm is used for face detection. It works by using a set of pre-trained classifiers to detect objects in images based on their features. These features are represented as Haar-like features, which are simple rectangular patterns that are used to compute the difference between light and dark areas in an image.

Training: Initially, the Haar Cascade algorithm requires training with positive and negative images. Positive images contain the object to be detected (in this case, faces), while negative images do not contain the object. During training, the algorithm learns to distinguish between these images based on the Haar-like features.

Integration: In your project, the OpenCV library is commonly used for implementing the Haar Cascade algorithm. OpenCV provides pre-trained Haar Cascade classifiers for face detection, making it easier to integrate into your system.

Usage: Once the Haar Cascade classifier is trained and integrated into your system, it can be used to detect faces in real-time images or video streams. The classifier scans the image using a sliding window technique, searching for regions that match the learned features. When a face is detected, its coordinates are returned, allowing further processing such as face recognition..

LBPH Algorithm (Local Binary Patterns Histograms):

Feature Extraction: LBPH is a method used for face recognition. It extracts features from facial images based on local binary patterns (LBP). LBP is a texture descriptor that encodes the local structure of an image by comparing each pixel's intensity with its neighbors. Histograms: In LBPH, histograms of LBP patterns are created for each face in the training dataset. These histograms represent the distribution of local patterns in different regions of the face.

Recognition: During recognition, the LBPH algorithm compares the LBP histogram of a new input face with the histograms of faces in the training set. It calculates the similarity between histograms using distance measures such as Euclidean distance or Chi-Square distance.

Integration: Similar to the Haar Cascade algorithm, OpenCV provides functions for implementing the LBPH algorithm for face recognition. These functions handle the extraction of LBP features, histogram creation, and comparison for recognition.

Usage: In your project, the LBPH algorithm would be used after face detection (using Haar Cascade) to recognize and match detected faces with known faces in the database. This allows for attendance tracking or identity verification based on recognized faces.

2. Your Project or Application can be used in real life? Justify it.

Ans: Yes

Schools & Universities: Face recognition can be used in educational institutions to track student attendance. This can streamline attendance management for teachers and administrators, reduce paperwork, and ensure accurate records.

Events & Conferences: Face recognition can be used at events and conferences for registration, access control, and tracking attendee participation. This can improve the overall event experience and provide valuable data for event organizers.

Corporate Attendance Tracking: In corporate settings, face recognition systems can replace traditional attendance methods like swipe cards or manual sign-ins. Employees can simply walk in front of a camera, and the system can automatically record their attendance based on facial recognition.

Time & Attendance Management: Face recognition systems can be used in various industries such as retail, healthcare, and hospitality for time and attendance management. This helps in monitoring employee attendance, calculating work hours, and generating accurate payroll reports.

3. Future scope of the project.

Ans:

Enhanced Accuracy: Continued research and development in face recognition algorithms will lead to improved accuracy and reliability. This includes better recognition of faces in varying lighting conditions, angles, and facial expressions.

Integration with IoT & Hardware: Face recognition attendance systems can be integrated with Internet of Things (IoT) devices and smart infrastructure for seamless connectivity and automation.

4. What are the goals of the project?

Ans:

Automated Attendance Tracking: The primary goal is to automate the attendance marking process using facial recognition. This eliminates the need for manual attendance registers or biometric devices, making the system more efficient and reducing human errors.

Efficiency: One of the primary goals is to improve the efficiency of attendance tracking processes. Face recognition systems can automate attendance marking, reducing the time and effort required compared to manual methods.

Accuracy: Another important goal is to achieve high accuracy in identifying individuals based on their facial features. This helps in minimizing errors and ensuring that attendance records are reliable.

5 What is Front end and Back end of project

Ans:

Front end : Tkinter GUI ToolKit

- Tkinter is a standard Python library for creating graphical user interfaces (GUIs).
- It provides tools for creating windows, buttons, menus, and other GUI elements.

Library/ Frameworks: OpenCv

- OpenCV (Open Source Computer Vision Library) is a popular open-source library for computer vision and image processing tasks.
- OpenCV is extensively used in applications requiring image and video analysis, including face recognition systems

Database : MySQL

- MySQL is used to store and manage data related to user information, attendance records, and other system data.

6 Explain the Algorithms used in the project.

Ans:

Step 1: Start .

Step 2: The first step is to detect images.

Step 3: Perform face alignment to normalize orientation and size of detected face.

Step 4: Relevant features are extracted from aligned faces

Step 5: Compare features of detected face with features of known face stored in database.

Step 6: The system records attendance of recognized individual.

Step 7: The system maintains database of known faces along with their corresponding attendance record.

Haar Cascade Face Detection Algorithm

Step 1: Load the pre-trained Haar Cascade face detection model

Step 2: Initialize the webcam or image source for capturing frames.

Step 3: Repeat for each frame:

- Read the frame from the webcam or image source.
- Convert the frame to gray scale for efficient processing.
- Use the Haar Cascade model to detect faces in the gray scale frame.

Step 4: For each detected face:

- Draw a bounding box around the face on the original color frame.
- Extract the face region from the frame for further processing.

Step 5: Display the processed frame with detected faces.

Step 6: If required, save or export the processed frames with bounding boxes.

LBPH Face Recognition Algorithm

Step 1: Load the pre-trained LBPH face recognition model.

Step 2: Initialize the database or data structure to store user profiles with face templates.

Step 3: Initialize the webcam or image source for capturing frames.

Step 4: Repeat for each frame:

- a. Read the frame from the webcam or image source.
- b. Convert the frame to grayscale for efficient processing.
- c. Use the LBPH model to recognize faces in the grayscale frame.

Step 5: For each recognized face:

- d. Compare the face template with stored user profiles in the database.
- e. If a match is found with sufficient confidence:

(a) Mark attendance for the corresponding user.
(b) Display the user's name or ID on the frame.

Step 6: If no match is found or confidence is low:
Display "Unknown" or handle unrecognized faces based on
System requirements.

Step 7: Display the processed frame with recognized faces and attendance status

Step 8: If required, save or export attendance records and processed frames.



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Computer Engineering Department

Academic Year: 2023-24

Class:BE

Subject: Project Lab

Sem :II

Assignment No: 02

Title: Project environment creation along with setup and installation report preparations.

Due Date:

Date Submitted:

Guide Name: Prof. B.M.GAWALE

Guide Remark:.....

Student Name (Roll No.)	Performance	Understanding	Submission	Total
SUJIT KANHAIYALAL YADAV	3	3	2	8
OM RAJENDRA JAIN	3	3	2	8
VIVEK SUNIL SANGALE	3	3	2	8
ARJALA SAJENDRA PATIL	3	3	2	8

Project Coordinator: Prof. D. D. Sharma

Head of Department: Prof. R. M. Shaikh

Project Environment Setup:

1. **Install Visual Studio 2020:** Begin by installing Visual Studio 2020 on your system. This integrated development environment (IDE) is essential for compiling and running the FRAS (Face Recognition Attendance System) project, which is developed using the python language.
2. Install MySQL Workbench an MySQL database for Database and other data storing purpose.
3. Install Python libraries like Tkinter, Pillow & OpenCV for GUI (Graphical User Interface) purpose.
4. **Clone Source Code:** Clone the source code repository of the FRAS (Face Recognition Attendance System) project from the designated repository or source control system. Ensure that you have the necessary permissions to access the repository.
5. **Open Project in Visual Studio:** Open Visual Studio 2020 and load the project solution file from the cloned repository. This step allows you to access and modify the project files within the IDE.
6. **Configure Database Connection:** If the project utilizes a database, configure the database connection settings in the project configuration files Specify the database server, credentials, and other necessary parameters.
7. **Build and Test Project:** Build the project solution in Visual Studio to compile the code and resolve dependencies. Run the project in debug mode to ensure that it launches successfully and functions as expected. Conduct thorough testing to verify the integrity and functionality of the project.

Installation Setup:

1. Installation:

Prerequisites Check: Before installation, ensure that Visual Studio 2020 is installed on the target system.

2. Download Source Code: Obtain the source code of the FRAS(Face Recognition Attendance System) project from the given repository or source control system.

3. Open Project in Visual Studio: Launch Visual Studio 2020 and open the project.

4. Build Solution: Build the solution in Visual Studio to compile the project and resolve any dependencies.

5. Configure Database Connection: If the project uses a database, configure the database connection settings in the project configuration files.

6. Run Project: Run the project in debug mode within Visual Studio to verify that it launches successfully and functions as expected. Setup Database. If the project requires a database, ensure that the necessary database schema and initial data are set up.



Krantiveer Vasantao Narayanrao Naik Shikshan Prasarak Sanstha

LOKNETE GOPINATHJI MUNDE
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Approved by AICTE, Accredited 'B' Grade by NAAC



Computer Engineering Department

Academic Year: 2023-24

Class: BE

Subject: Project Lab

Sem : II

Assignment No: 03

Title: Programming of the project functions, interfaces and GUI (if any) as per 1st Term termwork submission using corrective actions recommended in Term-I assessment of Termwork.

Due Date:

Date Submitted:

Guide Name: Prof. B. M. Gawale.....

Guide Remark:.....

Student Name (Roll No.)	Performance	Understanding	Submission	Total
Sujit K. Yadav	4	3	2	9
Om Rajendra Jain	4	3	2	9
Vivek Sunil Sangale	3	3	2	8
Arjala Sajendra Patil	3	3	2	8

Project Coordinator: Prof. D. D. Sharma.....

Head of Department: Prof. R. M. Shaikh.....

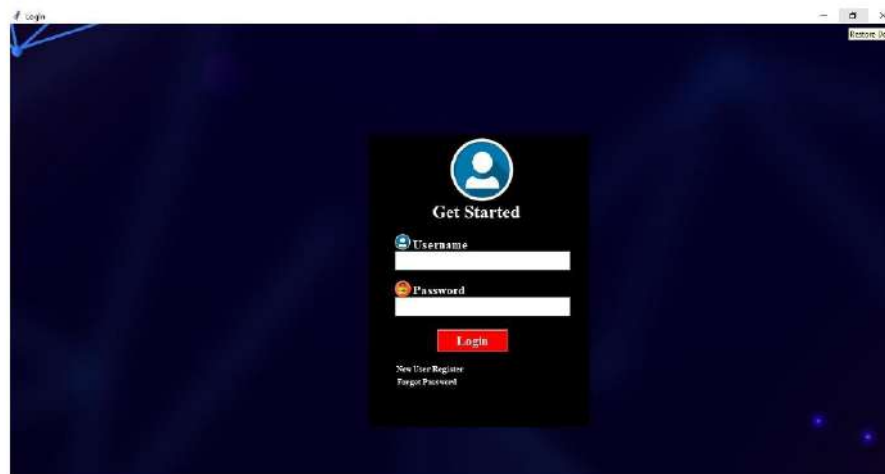
GUI

Registration Page



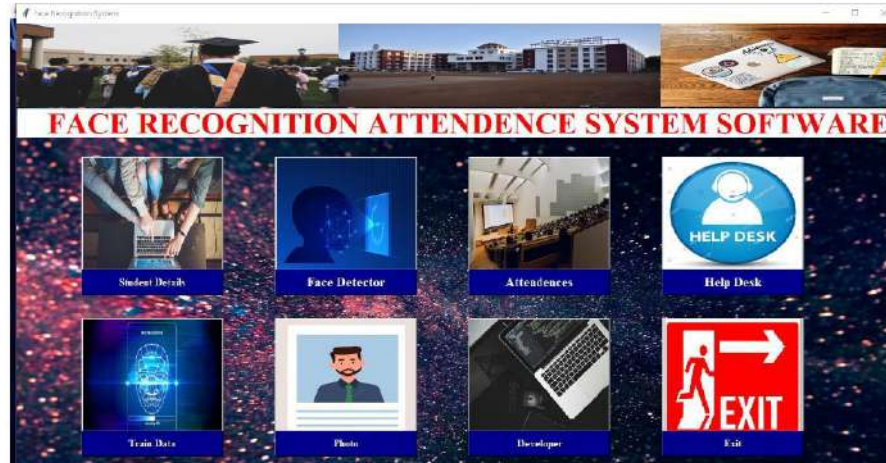
The Registration Page GUI features a dark blue background with a network-like pattern of white dots and lines. On the left, there is a white rectangular area containing an image of a laptop displaying the registration form. To the right of this image, the word "Registration" is written in green. Below the title, there are two columns of input fields: "First Name", "Last Name", "Contact No", "Email", "Select Security Questions" (with a dropdown arrow), "Security Answer", "Password", and "Confirm Password". A checkbox labeled "I Agree The Terms & condition" is located below the "Select Security Questions" field. At the bottom, there are two buttons: a purple "REGISTER" button and a green "LOGIN" button with a white arrow icon.

Login Page



The Login Page GUI has a dark blue background with a network-like pattern of white dots and lines. In the center, there is a black rectangular box with a white user icon at the top. Below the icon, the text "Get Started" is displayed. Underneath, there are two input fields: "Username" with a blue user icon and "Password" with an orange key icon. A red "Login" button is positioned below the password field. At the bottom of the black box, there are two links: "New User Register" and "Forgot Password".

Main Page



Student Detail Page



Sample Collecting Page for face Detection

MANAGEMENT SYSTEM

Search System

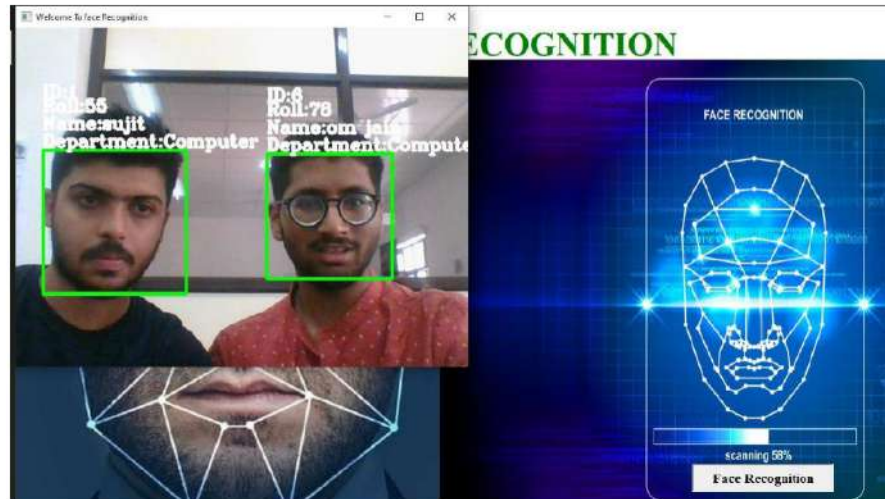
Search By:

Department	Course	Year	Semester	Studentid	Name
Computer	SE	2023-24	Semester-1	1	ajit
Computer	BE	2023-24	Semester-2	10	sky
IT	TE	2021-22	Semester-2	11	sachin
IT	SA	2022-23	Semester-1	12	akash
Computer	PL	2022-23	Semester-2	2	nikh
IT	TE	2021-22	Semester-2	3	rony
IT	SA	2023-23	Semester-1	4	calvin
Civil	TE	2021-22	Semester-2	5	tony
Computer	BE	2023-24	Semester-2	6	ow jia
Computer	BE	Select Year	Semester-2	7	dargesh
Civil	TE	2021-22	Semester-2	8	an
Civil	BE	2020-21	Semester-2	9	an

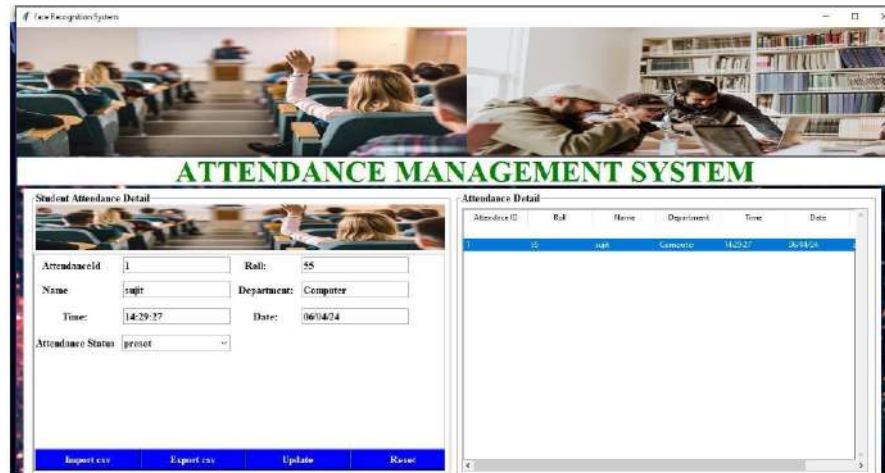
Training Data



Face Detection



Attendance Management Page



Registration Page

```
from tkinter import* from tkinter
import ttk from PIL import
Image, ImageTk from tkinter
import messagebox import
mysql.connector import re

class Register: def __init__(self,root):
self.root = root
self.root.title("Register")
self.root.geometry("1650x900+0+0")

# ===== Variable for text data =====
self.var_fname=StringVar() self.var_lname=StringVar()
self.var_contact=StringVar() self.var_email=StringVar()
self.var_securityQ=StringVar() self.var_securityA=StringVar()
self.var_pass=StringVar() self.var_confpass=StringVar()

# background image
self.bg=ImageTk.PhotoImage(file=r"D:\z\trial3\image\re.jpg") bg_lb1
= Label(self.root, image=self.bg)
bg_lb1.place(x=0, y=0,relwidth=1, relheight=1)

# left image of background self.bg1 =
ImageTk.PhotoImage(file=r"D:\z\trial3\image\bg1.jpg") left_lb1 =
Label(self.root,image=self.bg1) left_lb1.place(x=50, y=100,
width=470, height=550)
```



```

# =====main frame=====
frame = Frame(self.root,bg="white")
frame.place(x=520, y=100,width=800,height=550)

register_lb1 = Label(frame,text= "Registration",font=('times new roman',20,"bold"),fg="darkgreen",bg="white")
register_lb1.place(x=20,y=20)

# ===== label entry =====
# row -1

fname=Label(frame,text="First Name" ,font=("times new roman",15,"bold"),bg="white")
fname.place(x=50, y=100)

fname_entry=tk.Entry(frame,textvariable=self.var_fname,font=("times new roman",15,"bold"))
fname_entry.place(x=50, y=130,width=250)

l_name=Label(frame,text="Last Name" ,font=("times new roman",15,"bold"),bg="white",fg="black")
l_name.place(x=370, y=100)

self.txt_lname=tk.Entry(frame,textvariable=self.var_lname,font=("times new roman",15))
self.txt_lname.place(x=370, y=130,width=250)

# ===== row 2 =====

contact =Label(frame,text="Contact No",font=("times new roman",15,"bold"),bg="white",fg="black")
contact.place(x=50, y=170)

self.txt_contact=tk.Entry(frame, textvariable=self.var_contact,font=("times new roman",15))
self.txt_contact.place(x=50, y=200,width=250)

email=Label(frame,text="Email",font=("times new roman",15,"bold"),bg="white",fg="black")
email.place(x=370, y=170)

```

```

self.txt_email=tk.Entry(frame,textvariable=self.var_email,font=("times new roman",15))
self.txt_email.place(x=370, y=200,width=250)

# ===== row 3 =====

security_Q=Label(frame, text="Select Security Question: ",font=("times new roman",15,"bold"),bg="white",fg="black")
security_Q.place(x=50, y=240)

self.combo_securiy_Q=tk.Combobox(frame,textvariable=self.var_securityQ,font=("times new
roman",15,"bold"),state='readonly') self.combo_securiy_Q['values']=( "Select", "Your Birth Place", "Your
Girlfriend name", "Your Pet Name") self.combo_securiy_Q.place(x=50, y=270, width=250)
self.combo_securiy_Q.current(0)

security_A = Label(frame,text="Security Answer", font=("times new roman", 15,"bold"), bg="white",fg="black")
security_A.place(x=370, y=240)

self.txt_security=tk.Entry(frame,textvariable=self.var_securityA,font=("times new roman",15))
self.txt_security.place(x=370, y=270,width=250)

# ----- row4 -----

pswd= Label(frame,text="Password",font=( "times new roman",15,"bold"),bg="white",fg="black")
pswd.place(x=50,y=310)

self.txt_pswd=tk.Entry(frame,textvariable=self.var_pass,font=("times new roman",15))
self.txt_pswd.place(x=50,y=340,width=250)

confirm_pswd=Label(frame, text="Confirm Password",font=( "times new roman",15,"bold"),bg="white",fg="black")
confirm_pswd.place(x=370, y=310)

self.txt_confirm_pswd=tk.Entry(frame,textvariable=self.var_confpass,font=("times new roman",15))
self.txt_confirm_pswd.place(x=370, y=340,width=250)

# ===== checkbutton =====

self.var_check=IntVar() # check value variable checkbtn=Checkbutton(frame,variable=self.var_check,
text = "I Agree The Term & comdition",font=("times new roman",12,"bold"),onvalue=1,offvalue=0 )
checkbtn.place(x=50 , y = 380)

```

```

if self.var_fname.get() == "" or self.var_email.get() == "" or self.var_securityQ.get() == "Select":
    messagebox.showerror("Error", "All fields are required.", parent=self.root)
elif self.var_pass.get() != self.var_confpass.get():
    messagebox.showerror("Error", "Password and Confirm Password do not match.", parent=self.root)
elif self.var_check.get() == 0:
    messagebox.showerror("Error", "Please agree with the terms & conditions.", parent=self.root)
elif not self.validate_email():
    pass # Validation error message already
    handled in validate_email function
elif not self.validate_contact():
    pass # Validation error message already handled in validate_contact function
else:
    conn = mysql.connector.connect(host="localhost", user="root", password="4381", database="login")
    my_cursor = conn.cursor()
    query = ("select * from register where email=%s")
    value = (self.var_email.get(),)
    my_cursor.execute(query, value)
    row = my_cursor.fetchone()
    if row is not None:
        messagebox.showerror("Error", "User already exists. Please use another email.",
        parent=self.root)
    else:
        my_cursor.execute("insert into register values(%s,%s,%s,%s,%s,%s,%s)", (
        self.var_fname.get(), self.var_lname.get(),
        self.var_contact.get(), self.var_email.get(),
        self.var_securityQ.get(), self.var_securityA.get(),
        self.var_pass.get()
        ))
        conn.commit()
        conn.close()
    messagebox.showinfo("Success", "Successfully Saved", parent=self.root)

if __name__ == "__main__":
    root=Tk()
    app =
    Register(root)
    root.mainloop()

```

Login Page

```
import tkinter from tkinter import* import
tkinter as tk from tkinter import ttk from PIL
import Image, ImageTk from tkinter import
messagebox import mysql.connector from PIL
import Image, ImageTk from time import
strftime from datetime import datetime import
os from student import Student from train
import Train from face_recognition import
Face_Recognition from attendance import
Attendance from developer import Developer
from help import Help

def main(): win=Tk()
app=Login_Window(win)
win.mainloop()

class Login_Window:
    def __init__(self, root): self.root =
root self.root.title("Login")
self.root.geometry("1560x800+0+0")
# self.root.geometry("1400x700+0+0") self.bg=ImageTk.PhotoImage(file=r"D:\z\trial3\image\lo.jpg")

lbl_bg=Label(self.root,image=self.bg)
lbl_bg.place(x=0,y=0,relwidth=1,relheight=1)

frame = Frame(self.root,bg="black")
frame.place(x=550 , y=170,width=340,height=450)
```

```

img1=Image.open(r'D:\z\trial3\image\icon.png')
img1=img1.resize((100, 100) , Image.ANTIALIAS)
self.photoimage1=ImageTk.PhotoImage(img1)    lblimg1 =
Label(image=self.photoimage1,bg= "black",borderwidth=0)
lblimg1 .place(x=675,y=175,width=100,height=100)

get_str=Label(frame,text="Get Started",font=("times new roman",20,"bold"),fg="white",bg= "black")
get_str.place(x=95,y=100)

# label making    username=lbl=Label(frame,text="Username",font=("times new
roman",15,"bold"),fg="white",bg= "black")    username.place(x=65, y= 155)

self.txtuser=ttk.Entry(frame,font=("times new roman",15,"bold"))
self.txtuser.place(x=40, y=180, width=270)

# password code    password=lbl=Label(frame,text="Password",font=("times new
roman",15,"bold"),fg="white",bg= "black")    password.place(x=65, y= 225)

self.txtpass=ttk.Entry(frame,font=("times new roman",15,"bold"))
self.txtpass.place(x=40, y=250, width=270)

# ===== ICON Image =====
img2=Image.open(r'D:\z\trial3\image\icon.png')
img2=img2.resize((25, 25) , Image.ANTIALIAS)
self.photoimage2=ImageTk.PhotoImage(img2)    lblimg2 =
Label(image=self.photoimage2,bg= "black",borderwidth=0)
lblimg2 .place(x=590,y=323,width=25,height=25)

```

```

        self.txt_security=tk.Entry(self.root2,font=("times new roman",15))
self.txt_security.place(x=50, y=180,width=250)

        new_password = Label(self.root2,text="New Password", font=("times new roman", 15,"bold"),
bg="white",fg="black")
        new_password.place(x=50, y=220)

        self.txt_newpass=tk.Entry(self.root2,font=("times new roman",15))
self.txt_newpass.place(x=50, y=250,width=250)

        btn=Button(self.root2,text="Reset",command=self.reset_pass,font=("times new
roman",15,"bold"),fg="white",bg="green")
        btn.place(x=100, y=290)

# registration
class Register:
    def
    __init__(self,root):
        self.root = root
self.root.title("Register")
self.root.geometry("1650x900+0+0")

# ===== Variable for text data =====
self.var_fname=StringVar()
self.var_lname=StringVar()

self.var_contact=StringVar()
self.var_email=StringVar()
self.var_securityQ=StringVar()
self.var_securityA=StringVar()
self.var_pass=StringVar()
self.var_conpass=StringVar()

```

```

# background image
self.bg=ImageTk.PhotoImage(file=r"D:\z\trial3\image\rc.jpg")
bg_lb1 = Label(self.root, image=self.bg)    bg_lb1.place(x=0,
y=0,relwidth=1, relheight=1)

# left image of background    self.bg1 =
ImageTk.PhotoImage(file=r"D:\z\trial3\image\bg1.jpg")    left_lb1 =
Label(self.root,image=self.bg1)    left_lb1.place(x=50 , y=100,
width=470, height=550)

# =====main frame=====
frame = Frame(self.root,bg="white")
frame.place(x=520, y=100,width=800,height=550)

register_lb1 = Label(frame,text= "Registration",font=("times new roman",20,"bold"),fg="darkgreen",bg="white")
register_lb1.place(x=20,y=20)

# ===== label entry =====

# row -1
fname=Label(frame,text="First Name" ,font=("times new roman",15,"bold"),bg="white")
fname.place(x=50, y=100)

fname_entry=tk.Entry(frame,textvariable=self.var_fname,font=("times new roman",15,"bold"))
fname_entry.place(x=50, y=130,width=250)

```

```

l_name=Label(frame,text="Last Name" ,font=("times new roman",15,"bold"),bg="white",fg="black")
l_name.place(x=370, y=100)

self.txt_lname=tk.Entry(frame,textvariable=self.var_lname,font=("times new roman",15))
self.txt_lname.place(x=370, y=130,width=250)

# ===== row 2 =====

contact =Label(frame,text="Contact No",font=("times new roman",15,"bold"),bg="white",fg="black")
contact.place(x=50, y=170)

self.txt_contact=tk.Entry(frame, textvariable=self.var_contact,font=("times new roman",15))
self.txt_contact.place(x=50, y=200,width=250)

email=Label(frame,text="Email", font=("times new roman",15,"bold"),bg="white",fg="black")
email.place(x=370, y=170)

self.txt_email=tk.Entry(frame,textvariable=self.var_email,font=("times new roman",15))
self.txt_email.place(x=370, y=200,width=250)

# ===== row 3 =====

security_Q=Label(frame, text="Select Security Question: ",font=("times new roman",15,"bold"),bg="white",fg="black")
security_Q.place(x=50, y=240)

self.combo_securiy_Q=tk.Combobox(frame,textvariable=self.var_securityQ,font=("times new
roman",15,"bold"),state='readonly') self.combo_securiy_Q['values']=( "Select" , "Your Birth Place","Your Girlfriend
name", "Your Pet Name") self.combo_securiy_Q.place(x=50, y=270, width=250) self.combo_securiy_Q.current(0)

```



```

security_A = Label(frame,text="Security Answer", font=("times new roman", 15,"bold"), bg="white",fg="black")
security_A.place(x=370, y=240)

self.txt_security=tk.Entry(frame,textvariable=self.var_securityA,font=("times new roman",15))
self.txt_security.place(x=370, y=270,width=250)

# =====row4=====

pswd= Label(frame,text="Password",font=("times new roman",15,"bold"),bg="white",fg="black")
pswd.place(x=50,y=310)

self.txt_pswd=tk.Entry(frame,textvariable=self.var_pass,font=("times new roman",15))
self.txt_pswd.place(x=50,y=340,width=250)

confirm_pswd=Label(frame, text="Confirm Password",font=("times new roman",15,"bold"),bg="white",fg="black")
confirm_pswd.place(x=370, y=310)

self.txt_confirm_pswd=tk.Entry(frame,textvariable=self.var_confpass,font=("times new roman",15))
self.txt_confirm_pswd.place(x=370, y=340,width=250)

# ===== checkbutton =====

self.var_check=IntVar() # check value variable checkbtn=Checkbutton(frame,variable=self.var_check,
text = "I Agree The Term & comdition",font=("times new roman",12,"bold"),onvalue=1,offvalue=0 )
checkbtn.place(x=50 , y = 380)

# ===== button ===== img=Image.open(r"D:\z\trial3\image\bgReg.JPG")
img=img.resize((200,55),Image.ANTIALIAS) self.photoimage=ImageTk.PhotoImage(img)
b1=Button(frame,image=self.photoimage,command=self.register_data,borderwidth=0,cursor="hand2",font=("tims new
roman",15,"bold")) b1.place(x=10,y=420,width=200)

# b1.place(x=10, y=10, width=200)

```

```

# img1=Image.open(r"D:\z\trial3\image\login.jpg")    img1=Image.open(r"D:\z\trial3\image\lgin2.jpg")
img1=img1.resize((200,55),Image.ANTIALIAS)    self.photoimage1=ImageTk.PhotoImage(img1)    b1=Button(frame,
command=self.return_login,image=self.photoimage1,borderwidth=0,cursor="hand2",font=("times new roman",15,"bold"))
b1.place(x=330,y=420,width=200)

# ===== function deceleration =====

def register_data(self):    if self.var_fname.get()==" or self.var_email.get()==" or
self.var_securityQ.get()=="Select":

    messagebox.showerror("Error", "All fields are required.",parent=self.root)
elif self.var_pass.get()!=self.var_conpass.get():

    messagebox.showerror("Error", "Password and Confirm Password does not match.",parent=self.root)
elif self.var_check.get()==0:

    messagebox.showerror("Error", "Please agree with the terms & conditions.",parent=self.root)
else:

    # messagebox.showinfo("Success", "Welcome Client")
conn=mysql.connector.connect(host="localhost",user="root",password="4381",database="login")
my_cursor=conn.cursor()    query=("select * from register where email=%s")
value=(self.var_email.get(),)    my_cursor.execute(query,value)
row=my_cursor.fetchone()    if row!=None:    messagebox.showerror("Error","User
already exist", "please from another Email",parent=self.root)
else:

    my_cursor.execute("insert into register values(%s,%s,%s,%s,%s,%s,%s)",(
                                self.var_fname.get(),
self.var_lname.get(),
self.var_contact.get(),
self.var_email.get(),
self.var_securityQ.get(),
self.var_securityA.get(),
self.var_pass.get()

```

```

    ))

    conn.commit()    conn.close()

messagebox.showinfo("Success","Successfully Saved",parent=self.root)

# ===== destroy login Button =====

def return_login(self):
self.root.destroy()

# ===== main code ===== class
Face_Recognition_System:
def __init__(self,root):
    self.root = root    self.root.geometry("1400x700+0+0")    # the
size of Screen display    self.root.title("face Recognition System")

    # image part    #first images    img1 =
Image.open(r"D:\face2\images\2.jpg") img1 =
img1.resize((500,130),Image.ANTIALIAS)
self.photoimg = ImageTk.PhotoImage(img1)

    f_lb1 = Label(self.root,image=self.photoimg)
f_lb1.place(x=0,y=0, width=500,height=130)

    # Second Images    img2 =
Image.open(r"D:\face2\images\1.jpeg")    img2 =
img2.resize((500,130),Image.ANTIALIAS)
self.photoimg2= ImageTk.PhotoImage(img2)

    f_lb1 = Label(self.root,image=self.photoimg2)
f_lb1.place(x=500,y=0, width=500,height=130)

```

```

# Third Images      img3 =
Image.open(r"D:\face2\images\3.jpg")      img3 =
img3.resize((500,130),Image.ANTIALIAS)
self.photoimg3= ImageTk.PhotoImage(img3)

f_lb1 = Label(self.root,image=self.photoimg3)
f_lb1.place(x=1000,y=0, width=500,height=130)

# BackGround Images      img4 =
Image.open(r"D:\face2\images\4.jpg")      img4 =
img4.resize((1400,700),Image.ANTIALIAS)
self.photoimg4= ImageTk.PhotoImage(img4)

bg_img = Label(self.root,image=self.photoimg4)
bg_img.place(x=0,y=130, width=1400,height=700)      title_lb1 =
Label(bg_img,text="FACE RECOGNITION ATTENDENCE
SYSTEM SOFTWARE",font=("times new
roman",35,"bold"),bg="white",fg="red")
title_lb1.place(x=0,y=0,width=1400,height=45)

#===== time =====
def time():
    string = strftime("%H:%M:%S %p")      lbl.config(text=string)      lbl.after(1000,time)
lbl=Label(title_lb1,font=('times new roman',14,'bold'),background='white',foreground='blue')
lbl.place(x=30,y=50,width=110,height=50)

time()

# student button      img5 = Image.open(r"D:\face2\images\7.jpg")      img5 =
img5.resize((220,190),Image.ANTIALIAS)      self.photoimg5= ImageTk.PhotoImage(img5)      b1 =

```

```

Button(bg_img,image=self.photoimg5,command=self.student_details,cursor="hand2") #change done
b1.place(x=100,y=75,width=220,height=190)    b1_1 = Button(bg_img,text="Student
Details",command=self.student_details,cursor="hand2",font=("times new
roman",12,"bold"),bg="darkblue",fg="white")    b1_1.place(x=100,y=250,width=220,height=40)

    # Detect face button    img6 = Image.open(r"D:\face2\images\8.jpg")    img6 =
img6.resize((220,190),Image.ANTIALIAS)    self.photoimg6= ImageTk.PhotoImage(img6)    b1 =
Button(bg_img,image=self.photoimg6,cursor="hand2",command=self.face_data)    b1.place(x
-400,y-75,width-220,height-190)    b1_1 = Button(bg_img,text="Face
Detector",cursor="hand2",command=self.face_data,font=("times new
roman",15,"bold"),bg="darkblue",fg="white")    b1_1.place(x=400,y=250,width=220,height=40)

    # Attendance face button    img7 =
Image.open(r"D:\face2\images\5.jpg")    img7 =
img7.resize((220,190),Image.ANTIALIAS)
self.photoimg7= ImageTk.PhotoImage(img7)

    b1 = Button(bg_img,image=self.photoimg7)
b1.place(x=700,y=75,width=220,height=190)

    b1_1 = Button(bg_img,text="Attendences",cursor="hand2",command=self.attendance_data,font=("times new
roman",15,"bold"),bg="darkblue",fg="white")    b1_1.place(x=700,y=250,width=220,height=40)

    # # Help Desk button

    # img8 = Image.open(r"D:\face2\images\8.jpg")

    # img8 = img8.resize((220,190),Image.ANTIALIAS)

    # self.photoimg8= ImageTk.PhotoImage(img8)

    # b1 = Button(bg_img,image=self.photoimg8)

    # b1.place(x=1000,y=75,width=220,height=190)

    # b1_1 = Button(bg_img,text="Help Desk",cursor="hand2",font=("times new roman",15,"bold"),bg="darkblue",fg="white")

```

```

        if self.iExit > 0:
            self.root.destroy()
        else:
            return

# ----- Function Button -----
def student_details(self):
    self.new_window = Toplevel(self.root)
    self.app=Student(self.new_window)

# ----- training Data -----
def train_data(self):
    self.new_window = Toplevel(self.root)
    self.app=Train(self.new_window)

# ----- face recongnition Data -----
def face_data(self):
    self.new_window = Toplevel(self.root)
    self.app=Face_Recognition(self.new_window)

# ----- attendance display -----
def attendance_data(self):
    self.new_window = Toplevel(self.root)
    self.app=Attendance(self.new_window)

def developer_data(self):
    self.new_window = Toplevel(self.root)
    self.app=Developer(self.new_window)

# ----- help display -----
def help_data(self):
    self.new_window = Toplevel(self.root)
    self.app=Help(self.new_window) if __name__ == "__main__":
        main()

```



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Computer Engineering Department

Academic Year: 2023-24

Class: BE

Subject: Project Lab

Sem : II

Assignment No: 04

Title: Test tool selection and testing of various test cases for the project performed and generate various testing result charts, graphs etc. including reliability testing

Due Date:

Date Submitted:

Guide Name: Prof. B.M. GAWALE

Guide Remark:

Student Name(Roll No.)	Performance	Understanding	Submission	Total
SUJIT KANHAIYALAL YADAV	4	3	2	9
OM RAJENDRA JAIN	4	3	2	9
VIVEK SUNIL SANGALE	3	3	2	8
ARJALA SAJENDRA PATIL	3	3	2	8

Project Coordinator: Prof. D. D. Sharma

Head of Department: Prof. R. M. Shaikh

Testing Tools: Selenium

The testing tools for a face recognition attendance system, it is crucial to ensure accuracy, efficiency, and reliability. Various facial recognition libraries and frameworks such as OpenCV, Dlib, Face_recognition, provide robust algorithms for face detection and recognition. Performance testing involves benchmarking tools to measure processing time and accuracy, and stress testing to simulate high loads and assess system scalability. Data preparation and annotation tools like Labelling and SuperAnnotate help label images for object detection and recognition tasks. Validation and verification using confusion matrices, ROC and AUC curves, and metrics like precision, recall, and F1 score are essential to evaluate system accuracy. Utilizing evaluation datasets ensures comprehensive testing. Integration testing with unit tests and end-to-end tests verifies that individual components and the entire workflow function correctly. Security testing, including penetration testing and liveness detection, ensures the system's resilience against unauthorized access and spoofing attacks. Usability testing, through user feedback and assessing ease of use, ensures the system is user-friendly and intuitive. These combined efforts help create a robust face recognition attendance system.

Testing:

1. Unit System:

Unit Testing is a type of software testing where individual units or components of software are tested. The purpose is to validate that each unit of the software code performs as expected. Unit Testing is done during the development (coding phase) of an application by the developers. Unit testing is used to check the execution path of the module, function, and procedure of the system. The test is conducted with the help of normal data and abnormal data. This testing includes different factors like statement coverage, branch coverage, loop processing, abnormality, circulation, etc. With the help of this Unit testing system check that all the statement in the code is executed or not so it avoids the dead code statement. It checks all the branches and execution paths of the code. It ensures that all the internal methods of the program are executed and properly integrated with the program.

2. Black-box Testing:

It is a software testing method where the tester evaluates the functionality of an application without peering into its internal structures or workings. This type of testing focuses on the inputs and outputs of the system, ensuring that it behaves as expected according to the specifications.

For a face recognition attendance system (FRAS), black box testing would involve validating the system's overall functionality, performance, usability, and compliance with requirements. Below are the key areas to focus on and some example test cases.

1. Functional Testing:
2. Performance Testing
3. Usability Testing
4. Security Testing

3. **Integration Testing:**

In integrated testing, all the modules are checked together to ensure that all the modules are executing together according to the program specification. Once all the modules have been tested individually, the most legitimate question that can be asked is when all the modules are working properly, and why there is a need for integrated testing. The answer is, though all modules are working properly, problems may occur while interfacing individual modules. Data can lose across an interface one module can have an adverse on another

4. **System Testing:**

System testing for a face recognition attendance system involves evaluating the system as a whole to ensure it meets the specified requirements, functions correctly, and delivers the expected outcomes. • **Functional Testing:** Verify that the system's functions, such as face detection, recognition, attendance marking, and database operations, work as intended. • **Non-functional Testing:** Evaluate the system's nonfunctional aspects, including performance, security, usability, reliability, and compliance.

5. **Functional Testing:**

Enrollement process testing:

- Verify that users can enroll their faces successfully.
- Test the system's ability to capture and store face templates accurately.
- Check if the system provides feedback on successful or failed enrollments.

Recognition Accuracy Testing:

- Test the system's accuracy in recognizing enrolled faces.
- Verify if the system can handle variations in lighting, angles, and facial expressions.

Attendance Marking Testing:

- Verify that the system correctly marks attendance when a recognized face is detected.
- Test scenarios such as multiple faces in the frame or no faces detected.
- Ensure that attendance records are accurate and timestamped correctly.

System Integration Testing:

- Verify data synchronization and accuracy between the face recognition system

User Interface Testing:

- Test the usability of the system's interface for administrators and users.
- Verify that users can easily navigate enrollment, recognition, and attendance viewing processes.

Error Handling Testing:

- Test how the system handles errors such as face recognition failures or database connectivity issues.
- Verify that error messages are informative and help users troubleshoot issues.

Regression Testing:

- Perform regression testing after updates or changes to ensure that existing functionality still works correctly.
- Test backward compatibility with older versions of the system if applicable.

Performance Testing:

- Test the system's performance metrics such as response time, throughput, and resource utilization.

- Verify scalability by testing with a large number of users or concurrent requests

6. Acceptance testing:

Requirement Review:

- Verify adherence to privacy principles, data retention policies, and user consent requirements.

Test Plan Creation:

- Develop a detailed test plan that outlines the test cases, scenarios, and acceptance criteria for the face recognition system.
- Include input from stakeholders to ensure that the test plan aligns with their expectations and requirements.

Installation & configuration Testing:

- Verify that the system can be installed and configured correctly according to the deployment guidelines.
- Test the system's compatibility with hardware and software dependencies.

Functional Testing:

- Perform functional testing as described earlier, covering aspects such as enrollment, recognition accuracy, attendance marking, system integration, user interface, security, error handling, regression, and performance.

Usability Testing:

- Evaluate the system's usability by conducting user acceptance testing (UAT) with representative users.
- Gather feedback on the user interface, ease of use, and overall user experience.

Performance Testing:

- Conduct performance testing to ensure that the system meets performance requirements under expected loads and usage patterns.
- Measure response times, throughput, and resource utilization during peak usage scenarios.

Security Testing:

- Verify that the system's security features are effective in preventing unauthorized access and protecting sensitive data.
- Test for vulnerabilities such as spoofing attacks and unauthorized data access.

Scalability Testing:

- Evaluate the system's scalability by testing with increasing numbers of users, face templates, and concurrent requests.
- Verify that the system can handle growth without significant degradation in performance.

Compatibility Testing:

- Test the system's compatibility with different devices, browsers, operating systems, and network configurations.
- Ensure that the system works seamlessly across a variety of environment.

User Acceptance Technique:

- Conduct UAT with key stakeholders and end users to validate that the system meets their expectations and business needs.

- Address any feedback or issues identified during UAT before final acceptance.

7. Security Testing:

Authentication Security

- Test the strength of user authentication mechanisms, such as passwords or biometric verification, for accessing the system's administrative functions.
- Verify that strong password policies are enforced, including requirements for complexity and periodic changes.

Access Control

- Test role-based access control (RBAC) to ensure that users are granted appropriate permissions based on their roles.
- Verify that unauthorized users are denied access to sensitive functions and data.

Session Management:

- Test session management mechanisms to prevent session hijacking or fixation attacks.
- Verify that sessions are securely maintained, with proper timeout and logout functionalities.

Data Integrity:

- Test the integrity of stored data, including face templates and attendance records, to prevent unauthorized modifications or tampering.
- Verify that data validation and integrity checks are implemented at critical points.

Vulnerability Assessment:

- Conduct vulnerability scanning and penetration testing to identify and remediate security vulnerabilities in the system.
- Test for common vulnerabilities such as SQL injection, cross-site scripting (XSS), and insecure file uploads.

Test Cases

TC_ID	TC_NAME	OBJECTIVE	PRE_REQUISITE	DISCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	RESULT
TC_01	TC01_Open Application	Verify to open application on the device.	Application should be open on device.	Open application on PC.	Application should get opened.	Application is getting opened	Pass	1
TC_02	TC02_Registration page	Verify the fuctinality of registration page	Application should be open on device.	Click on Registration	Application should show Registration page	Application is showing Registration form to fill.	Pass	1
TC_03	TC_03_registration verification	Verify user registration	Application should be open on device.	Verify the registration Details of user	Application should show Successful registration of user	Application is showing Successful registration of user.	Pass	1
TC_04	TC_04_Login with valid credentials	Verify user can login with correct username and password	User account created	Enter a valid username and password. * Click the login button.	User is successfully logged in and redirected to the appropriate page	User is successfully logged in and redirected to the appropriate page	pass	1

TC_05	TC_05_Login with invalid username	Verify login fails with an incorrect username	User account created	Enter an invalid username and a valid password. * Click the login button.	Login fails and an error message is displayed indicating an invalid username.	Login fails and an error message is displayed indicating an invalid username.	pass	1
TC_06	TC_06_Student/Client Details Saving	Verify that student/client details are saved correctly in the MySQL database.	User is logged in, and the system is operational.	Enter valid student/client details (e.g., name, ID, contact information). Save the details to the database.	Student/client details should be stored accurately in the database with unique identifiers.	Details saved successfully, and unique IDs assigned in the database.	pass	1
TC_07	TC_07_Face Sample Collection	Test the process of collecting face samples from students/clients.	Student/client details are saved, and the camera interface is accessible.	Select a student/client from the database to collect face samples. Capture multiple face samples	Face samples should be captured and associated with the respective student/client in the database.	Face samples captured successfully and linked to the correct student/client IDs.	pass	1
TC_08	TC_08_Training with Haar Cascade Algorithm	Verify the training process using the Haar Cascade algorithm.	Face samples are collected, and the training module is operational.	Trigger training using the Haar Cascade algorithm for a selected student/client. Verify training completion and accuracy metrics.	Training should complete without errors, and accuracy metrics should indicate successful learning.	Training completed successfully, and accuracy metrics meet expected standards.	pass	1

TC_09	TC_09_Training with LBPH Algorithm	Test the training process using the LBPH (Local Binary Patterns Histograms) algorithm.	Face samples are collected, and the LBPH training module is operational.	Initiate training using the LBPH algorithm for a selected student/client. Validate training results and accuracy.	LBPH training should yield accurate recognition results and meet defined accuracy thresholds.	LBPH training successful, and recognition accuracy meets expectations.	pass	1
TC_10	TC_10_Face Sample Quality Check	Ensure that the captured face samples meet quality standards.	Captured face samples are available for testing.	Analyze the quality of captured face samples. Verify that samples meet defined quality criteria.	Face samples should be clear and suitable for accurate recognition.	Face samples meet quality standards, suitable for recognition.	Pass	1
TC_11	TC_11_Face Recognition and Attendance Marking	Test the face recognition and attendance marking process.	System is trained with Haar Cascade and LBPH algorithms, and attendance module is active	Present a registered student/client face for recognition. Verify if the system correctly recognizes the face and marks attendance.	System should accurately recognize faces and mark attendance for recognized individuals.	Face recognition successful, and attendance marked for recognized individuals.	pass	1
TC_12	TC_12_Attendance Logging - Excel Output	Verify the accuracy of attendance logging in Excel format.	Attendance is marked for recognized faces, and Excel logging module is active.	Mark attendance for multiple individuals with recognized faces. Check the generated Excel attendance log for accuracy.	Excel log should accurately reflect the attendance marked by the system.	Excel log generated accurately, reflecting marked attendance.	pass	1

TC_13	TC_13 Detect Feedback testing	Verify if application give feedback of successful registration of user	Application should be open on device.	Verify if application gives feedback	Application should give feedback of successful registration or failure	Application should give Feedback of successful user Register	Pass	1
TC_14	TC14_Recognition Accuracy testing	Verify system's accuracy in recognizing faces.	Application should be open on device.	Verify that if system recognises enrolled faces.	Application should detect Enrolled faces accurately.	Application is detecting Enrolled faces	Pass	1
TC_15	TC_15_ System variation testing	Verify system's variation in lightning, angles and facial expression.	Application should be open on device.	Verify variation in lightning, angles and facial expression.	System should handle variation in lightning, angles and facial expressions	System is able to handle variation in lightning, angles and facial expression	Pass	1
TC_16	TC_16 Attendance marking testing	Verify that application should Able to mark attendance accurately.	Applications should be open on device.	Verify if the application marks attendance correctly or not.	Should record attendance accurately.	Application correctly records attendance accurately.	Pass	1
TC_017	TC_017 Interface Testing	Verify the usability Of system's interface for users and administrators.	Application should be browsed and opened.	Verify that users can navigate through application	Usability of system's interface should be smooth for user.	User interface is simple and easy to navigate through application	Pass	1

TC_13	TC_13 Detect Feedback testing	Verify if application give feedback of successful registration of user	Application should be open on device.	Verify if application gives feedback	Application should give feedback of successful registration or failure	Application should give Feedback of successful user Register	Pass	1
TC_14	TC14_Recognition Accuracy testing	Verify system's accuracy in recognizing faces.	Application should be open on device.	Verify that if system recognises enrolled faces.	Application should detect Enrolled faces accurately.	Application is detecting Enrolled faces	Pass	1
TC_15	TC_15_ System variation testing	Verify system's variation in lightning, angles and facial expression.	Application should be open on device.	Verify variation in lightning, angles and facial expression.	System should handle variation in lightning, angles and facial expressions	System is able to handle variation in lightning, angles and facial expression	Pass	1
TC_16	TC_16 Attendance marking testing	Verify that application should Able to mark attendance accurately.	Applications should be open on device.	Verify if the application marks attendance correctly or not.	Should record attendance accurately.	Application correctly records attendance accurately.	Pass	1
TC_017	TC_017 Interface Testing	Verify the usability Of system's interface for users and administrators.	Application should be browsed and opened.	Verify that users can navigate through application	Usability of system's interface should be smooth for user.	User interface is simple and easy to navigate through application	Pass	1



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Computer Engineering Department

Academic Year: 2023-24

Class: BE

Subject: Project Lab

Sem : II

Assignment No: 05

Title: Installations and Reliability Testing Reports at the client end.

Due Date:

Date Submitted:

Guide Name: Prof. B.M.GAWALE

Guide Remark:

Student Name (Roll No.)	Performance	Understanding	Submission	Total
Sujit Yadav.	4	3	2	9
Om Jain.	4	3	2	9
Vivek Sangale.	4	3	2	9
Arjala Patil.	4	3	2	9

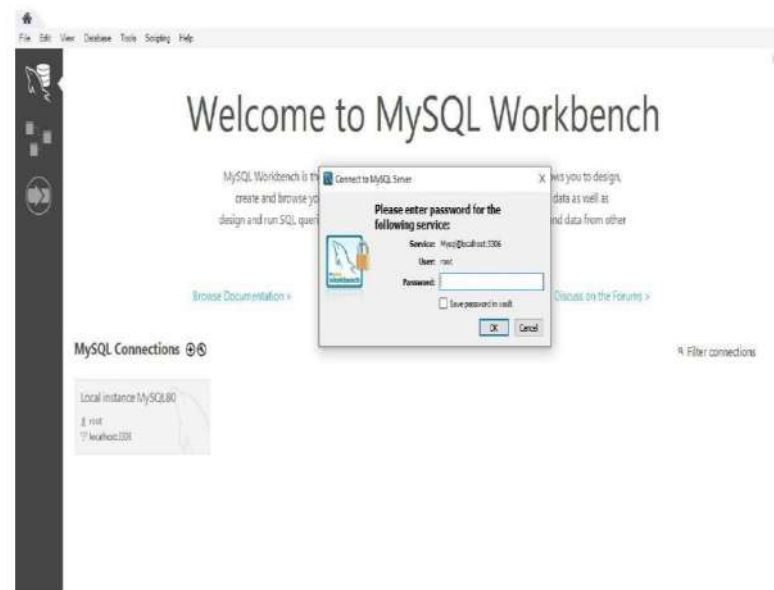
Project Coordinator: Prof. D. D. Sharma

Head of Department: Prof. R. M. Shaikh

Installation Summary:

Provide a brief overview of the installation process, including the hardware and software components deployed.

1. **MySQL WorkBench:** <https://dev.mysql.com/downloads/installer/>



2. Microsoft Visual Studio code:

<https://code.visualstudio.com/download>



Hardware Setup:

1. Outline the hardware setup process, including:
2. Camera installation and positioning
3. Connection to power source and network
4. i3 processor 10 th Generation
5. 256 GB SSD

Software Installation:

1. Detail the software installation steps, including:
2. Python 3.7.0 <https://www.python.org/downloads/>
3. Installation of OpenCV, face recognition libraries, and other dependencies
<https://opencv.org/>
4. MySQL <https://dev.mysql.com/downloads/installer/>
5. Microsoft Visual Code <https://code.visualstudio.com/download>

Testing:

Outline the testing procedures conducted to ensure the system's functionality, including:

1. Camera functionality test
2. Face recognition accuracy test
3. Attendance recording test

Face Recognition Attendance System Reliability Testing:

Reliability is a critical aspect of any face recognition attendance system, ensuring that it consistently performs its intended functions accurately and efficiently under various conditions. Our system enhances user's trust and improve efficiency. We have tested the system in various environments and backgrounds to ensure that system recognises authorised faces accurately without any error.

The system is reliable for face recognition and authentication. This system has wide applications in real world applications such as schools, offices and other Institutions where attendance tracking is crucial.

Testing:**Unit Testing:**

Unit Testing is a type of software testing where individual units or components of software are tested. The purpose is to validate that each unit of the software code performs as expected. Unit Testing is done during the development (coding phase) of an application by the developers. Unit testing is used to check the execution path of the module, function, and procedure of the system. The test is conducted with the help of normal data and abnormal data. This testing includes different factors like statement coverage, branch coverage, loop processing, abnormality, circulation, etc. With the help of this Unit testing system check that all the statement in the code is executed or not so it avoids the dead code statement. It checks all the branches and execution paths of the code. It ensures that all the internal methods of the program are executed and properly integrated with the program

Integration Testing:

Integrated testing, all the modules are checked together to ensure that all the modules are executing together according to the program specification. Once all the modules have been tested individually, the most legitimate question that can be asked is when all the modules are

working properly, and why there is a need for integrated testing. The answer is, though all modules are working properly problems may occur while interfacing individual modules. Data can lose across an interface one module can have an adverse on another

System Testing:

System testing for a face recognition attendance system involves evaluating the system as a whole to ensure it meets the specified requirements, functions correctly, and delivers the expected outcomes.

- **Functional Testing:** Verify that the system's functions, such as face detection, recognition, attendance marking, and database operations, work as intended.
- **Non-functional Testing:** Evaluate the system's non-functional aspects, including performance, security, usability, reliability, and compliance.

Testing Results:

Present the results of the reliability testing, including:

- System uptime/downtime
- Face recognition accuracy under different conditions
- Performance metrics (response time, processing speed, etc.)

Reliability Assessment:

Reliability testing for a face recognition attendance system involves evaluating the system's ability to perform consistently under various conditions. Here are some key aspects and methods to consider:

1. Accuracy Testing:

For a Face Recognition Attendance System we have measured how often system incorrectly identifies unauthorised person as authorised person and also measured how often system fails to recognise authorised person.

2. Environmental Testing:

We tested the system under various lightning conditions including low light, bright light and varying angles of light. Also we evaluated performance with different backgrounds to ensure system can identify faces in various environments.

3. Performance Testing:

In performance testing, we measured time taken for face recognition and authentication of user. System usage of user's device is also evaluated to ensure user's device operates efficiently.

4. User Experience testing:

We have evaluated that if system is user friendly or not including face recognition and authentication process.

ANNEXURE G
INFORMATION OF PROJECT
GROUP MEMBERS



- (a) Name : Sujit Kanhaiyalal Yadav
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- (h) Paper Published : Face Recognition Based Attendance System
Survey Paper on Face Recognition Attendance System



- (a) Name : OM Rajendra jain
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Survey Paper on Face Recognition Attendance System



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Survey Paper on Face Recognition Attendance System



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- (h) Paper Published :Face Recognition Based Attendance System
Survey Paper on Face Recognition Attendance System